AI Audit Report

- **Student ID:** 23127261
- **Student Name:** Vương Ngũ Tín Thành
- **Class:** 23KTPM2
- **Assignment ID:** HW06-AI

Summary of AI Accuracy

The audit contains 44 reviewed artifacts. Most AI responses were valid, but 18 artifacts required correction or further work. This shows that AI was useful for creating and revising test artifacts, but human review was still necessary to find unsupported assumptions, weak assertions, duplicate cases, and incomplete evidence.

Metric	Count	Percentage
VALID	26	59.09%
INVALID	1	2.27%
INCOMPLETE	17	38.64%

The verdict of each artifact is identified as:

- `VALID` if it is corrected, accepted as-is.
- `INVALID` if it is wrong, rejected.
- `INCOMPLETE` if it is acceptable after edits.

AI Interaction Log

Artifact 1

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-19 23:39:52 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
  - `skill-creator`

    - `ai-audit`
- **Prompt:**

```
Use [$skill-creator](C:\\Users\\xing0\\.codex\\skills\\.system\\skill-creator\\SKILL.md) and [$ai-audit](D:\\GitHub\\SoftwareTesting-HW06\\.agents\\skills\\ai-audit\\SKILL.md)

Create postman-test-generator agent skills. I've already designed the skill in [postman-test-generator.md](D:/GitHub/SoftwareTesting-HW06/skills-design/postman-test-generator/postman-test-generator.md).
```

- **AI Output:**

```
The design is narrow enough that the skill should **convert reviewed CSV cases, not invent tests**. I’m also keeping Newman compatibility as a hard output constraint and treating the conversion report as a traceability/validation artifact because your input contract includes `conversion_report_path`.

The skill draft is now aligned to your design: CSV is authoritative for test intent, the API spec only resolves implementation details, and the skill explicitly avoids adding a review/design phase. I’m validating the package now, then I’ll build the audit file from the completed interactions available to ai-audit.

Done

`postman-test-generator` passed `skill-creator`'s `quick_validate.py`. Its main boundaries are: reviewed CSV test cases are authoritative, it does not design new cases, it generates Postman Collection v2.1 + conversion report, preserves traceability, requires `X-Student-Id`, parameterizes runtime values, and targets Newman-compatible scripts.
```

- **File Modifications:**
    - `File: AI_Audit.md`
        - `Action:` created.

        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Initialized the AI audit report and added audit artifact(s) for the completed AI interactions.
    - `File: .agents/skills/postman-test-generator/SKILL.md`
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Added the `postman-test-generator` skill definition, including its purpose, input requirements, workflow, Postman collection generation rules, validation requirements, and output expectations.
    - `File: .agents/skills/postman-test-generator/agents/openai.yaml`
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Added the OpenAI agent configuration for the `postman-test-generator` skill.
- **Verdict:** INCOMPLETE
- **Reasoning:**
    - Preconditions notes may be lost. The skill does not explicitly preserve them.
    - Collection validation is too weak. It only checks valid JSON and v2.1 declaration, not full Postman v2.1 schema validity.
- **Student Fixes:** Continue state these defects out and guide the agent to fix them.

Artifact 2

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-20 00:22:32 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
    - `skill-creator`
    - `ai-audit`

- **Prompt:**

```
There are some defects in the implemented `postman-test-generator`
skill.

- Preconditions notes may be lost. The skill does not explicitly
preserve them.
- Collection validation is too weak. It only checks valid JSON and v2.1
declaration, not full Postman v2.1 schema validity.

Use [$skill-creator](C:\\Users\\xing0\\.codex\\skills\\.system\\skill-
creator\\SKILL.md) to revise the skill.
```

- **AI Output:**

I’ll revise the existing skill in place, specifically addressing both defects: preserve precondition notes explicitly through conversion/traceability, and validate the generated collection against the full Postman Collection v2.1 schema rather than only checking JSON/version metadata. Then I’ll rerun `$skill-creator` validation and package the revised skill.

The prior generated working directory is no longer present in the sandbox, but the original design package is still available. I’m reconstructing the skill from that authoritative design plus the same `$skill-creator` rules, carrying forward the prior decisions (reviewed CSV as authority, Newman compatibility, traceability, and no test-case invention) while patching only the two identified defects.

One important interaction between the two fixes: preconditions should **not** be preserved by adding arbitrary custom JSON fields, because that can itself break v2.1 schema validity. I’m preserving them in schema-supported request descriptions and in the conversion report, with exact source text retained for traceability.

Both corrections are now implemented in the skill instructions. I also added a deterministic schema-validation helper so future executions can’t accidentally “pass” a collection merely because it is JSON with a v2.1 declaration.

Revised with `$skill-creator`.

The two defects are fixed:

- **Preconditions are explicitly preserved**
  - Kept in a schema-supported request description under `Preconditions` / `Setup`.
  - Also copied to the conversion report for traceability.
  - If a precondition cannot be automated safely, it stays as a note and is flagged instead of being dropped.
  - Arbitrary custom Postman fields are prohibited because they can violate the collection schema.

```
- **Full Postman v2.1 schema validation**
  - Added `references/validation.md`.
  - Added `scripts/validate_postman_collection.py`.
  - Validation now requires the complete Postman Collection v2.1 JSON
Schema; valid JSON plus `info.schema` is explicitly insufficient.
  - Schema errors include their JSON paths and cause a non-zero exit.
  - The collection is only finalized after full schema validation
passes.
  - The skill uses Postman’s documented v2.1 schema location.
([github.com](https://github.com/postmanlabs/postman-api-sdk-
ts/blob/main/documentation/models/ModifyCollectionSchemaInfo.md?
utm_source=chatgpt.com))

I also tested the validator with both a passing collection and a
deliberately schema-invalid collection; the invalid one was correctly
rejected. `$skill-creator`'s `quick_validate.py` also reports **`Skill
is valid!`**.
```

- **File Modifications:**
  - File: `.agents/skills/postman-test-generator/SKILL.md`.
    - `Action:` edited.
    - `Deleted:` Prior exact text is not available in the completed interaction output.
    - `Replaced With:` Revised skill instructions that explicitly preserve reviewed preconditions/setup notes in schema-supported request descriptions and the conversion report, prohibit arbitrary custom Postman properties for traceability, require full Postman Collection v2.1 JSON Schema validation before finalization, and retain Newman compatibility requirements.
    - `Appended:` None.

  - `File:` `.agents/skills/postman-test-generator/references/validation.md`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Full-schema validation procedure using the official Postman Collection Format v2.1.0 schema, validation gate requirements, deterministic helper

invocation, and failure-repair guidance.

  - `File:` `.agents/skills/postman-test-generator/scripts/validate_postman_collection.py`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Deterministic validator that parses the collection and schema, checks the v2.1 schema declaration, validates the complete collection, reports JSON-path schema errors, and exits non-zero on failure.

- **Verdict:** VALID
- **Reasoning:** The agent successfully fixed all defects.
- **Student Fixes:** None

Artifact 3

- **User:** Vương Ngữ Tín Thành
- **Date and Time:** 2026-08-20 11:16:56 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
  - `skill-creator`
  - `ai-audit`
- **Prompt:**

```
Use [$skill-creator](C:\Users\xing0\\.codex\skills\\.system\skill-
creator\SKILL.md) to implement the Agent Skills defined by [api-test-
generator](skills-design/api-test-generator/)

Treat the design as the source of truth.

Create **5 composable skills**:

- `api-test-generator` — orchestrator
- `contract-test-generator`
- `domain-test-generator`
- `state-transition-test-generator`
- `security-test-generator`

The four specialized generators must be independent skills, not
reference files.

For `api-test-generator`, use Codex multi-agent orchestration to:

1. Prepare the shared API/endpoint context once.
2. Spawn **4 subagents concurrently**, one for each specialized skill.
3. Wait for all four to finish.
4. Aggregate and validate their results into the final output.

Please note that for specialized generators currently are split into 2
phases, due to `HUMAN REVIEW` step.

Specialized skills must also work independently. Use `references/` only
for supporting material.

Avoid concurrent writes to the final output; the orchestrator owns
aggregation and final output.

Do not build a custom multi-agent framework if Codex's native subagent
orchestration is sufficient.

Follow [$skill-creator](C:\Users\xing0\\.codex\skills\\.system\skill-
```

```
creator\SKILL.md) conventions and validate all five skills after
implementation.

At the end, log this artifact using [$ai-audit]
(D:\GitHub\SoftwareTesting-HW06\\.agents\skills\ai-audit\SKILL.md) and
report the created skill tree and validation results.
```

- **AI Output:**

````
Implemented and validated all five skills, with Artifact 3 logged in
[AI_Audit.md](D:/GitHub/SoftwareTesting-HW06/AI_Audit.md).

Created skill tree:

```text
.agents/skills/
├── api-test-generator/
│   ├── SKILL.md
│   └── agents/openai.yaml
├── contract-test-generator/
│   ├── SKILL.md
│   └── agents/openai.yaml
├── domain-test-generator/
│   ├── SKILL.md
│   └── agents/openai.yaml
├── state-transition-test-generator/
│   ├── SKILL.md
│   └── agents/openai.yaml
└── security-test-generator/
    ├── SKILL.md
    └── agents/openai.yaml
```

Validation results:

- `quick_validate.py`: passed for all five skills with `Skill is
valid!`.
- Semantic checks passed for shared-context-once preparation, exactly
four concurrent specialist agents, waiting for all four, parent-only
aggregation/final writes, and two-phase human-review gates.
- All five `agents/openai.yaml` files parse successfully.
- Each specialist is independently invocable and owns only its
report/candidate fragment.
- No `references/` were created because no supporting material was
needed.
````

```
Artifacts 3-3 successfully appended to `D:\GitHub\SoftwareTesting-HW06\AI_Audit.md`.
```

- **File Modifications:**
  - `File: .agents/skills/api-test-generator/SKILL.md`.
    - `Action:` created, then replaced scaffold content.
    - `Deleted:` Initial skill-creator TODO scaffold.
    - `Replaced With:` Orchestrator instructions for one-time shared context preparation, concurrent native four-agent execution, two review-separated phases, parent-owned aggregation, de-duplication, traceability, coverage validation, target-count handling, and final output ownership.
    - `Appended:` None.
  - `File: .agents/skills/api-test-generator/agents/openai.yaml`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` UI metadata and default invocation prompt for `api-test-generator`.
  - `File: .agents/skills/contract-test-generator/SKILL.md`.
    - `Action:` created, then replaced scaffold content.
    - `Deleted:` Initial skill-creator TODO scaffold.
    - `Replaced With:` Independent two-phase contract-analysis, human-review, candidate-generation, traceability, and file-ownership instructions.
    - `Appended:` None.
  - `File: .agents/skills/contract-test-generator/agents/openai.yaml`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` UI metadata and default invocation prompt for `contract-test-generator`.
  - `File: .agents/skills/domain-test-generator/SKILL.md`.

    - `Action:` created, then replaced scaffold content.
    - `Deleted:` Initial skill-creator TODO scaffold.
    - `Replaced With:` Independent equivalence-partition and boundary-analysis workflow with human review, candidate generation, traceability, and file boundaries.
    - `Appended:` None.
  - `File: .agents/skills/domain-test-generator/agents/openai.yaml`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` UI metadata and default invocation prompt for `domain-test-generator`.
  - `File: .agents/skills/state-transition-test-generator/SKILL.md`.
    - `Action:` created, then replaced scaffold content.
    - `Deleted:` Initial skill-creator TODO scaffold.
    - `Replaced With:` Independent applicability check, state/transition modeling, human review, candidate generation, justified empty result, traceability, and file boundaries.
    - `Appended:` None.
  - `File: .agents/skills/state-transition-test-generator/agents/openai.yaml`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` UI metadata and default invocation prompt for `state-transition-test-generator`.
  - `File: .agents/skills/security-test-generator/SKILL.md`.
    - `Action:` created, then replaced scaffold content.
    - `Deleted:` Initial skill-creator TODO scaffold.
    - `Replaced With:` Independent SEC-01–SEC-07 applicability/scenario analysis, human review, candidate generation, safety, traceability, and file boundaries.

        - `Appended:` None.
    - `File: .agents/skills/security-test-generator/agents/openai.yaml`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` UI metadata and default invocation prompt for `security-test-generator`.
    - `File: AI_Audit.md`.
        - `Action:` appended.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Artifact 3 containing the verbatim prompt, final AI output, file modifications, and review TODOs.
    - `File: .skill-implementation.patch` and `.skill-stage/*/SKILL.md`.
        - `Action:` temporary files created and deleted.
        - `Deleted:` Staged patch and five staged skill files after protected-directory installation.
        - `Replaced With:` None.
        - `Appended:` None.
    - `File: .validation-deps/**`.
        - `Action:` temporary validator dependencies created and deleted.
        - `Deleted:` Isolated PyYAML validation dependency directory after validation.
        - `Replaced With:` None.
        - `Appended:` None.
- **Verdict:** INCOMPLETE
- **Reasoning:**
    - Cross-category deduplication may remove required test coverage.
    - STATE: NOT_APPLICABLE is not passed correctly into Phase 2.
    - Completed subagents are not explicitly closed.
    - The parent skill may generate extra tests itself instead of delegating them.

    - If the user provides one Phase-1 report file, parallel agents may write to the same file.
        - For one report file, workers should write separate temporary reports first. The parent should merge temporary reports into the final Phase-1 report.
        - Temporary files should be deleted only after the merged report is successfully validated.
        - If merging fails, temporary files should be kept for recovery.
- **Student Fixes:** Continue guide the agent to resolve defects.

Artifact 4

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-20 11:52:36 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
    - `skill-creator`
    - `ai-audit`
- **Prompt:**

```
Revise the generated skills to fix the following logic defects. Keep the
existing design and structure unless a change is required for these
fixes.
- Prevent cross-category deduplication from removing required coverage.
- Preserve and pass `STATE: NOT_APPLICABLE` correctly into Phase 2.
- Explicitly close completed subagents after their results are
collected.
- Do not let `api-test-generator` generate extra test cases itself;
delegate additional generation to the appropriate specialist skill.
- When the user provides a single Phase-1 report file:
  - have parallel workers write to separate temporary report files;
  - merge them into the requested report only after all required Phase-1
workers finish;
  - validate the merged report;
  - delete temporary files only after successful validation;
  - keep temporary files if merging or validation fails.

Do not redesign unrelated parts.
```

- **AI Output:**

Revised all five skills without redesigning unrelated behavior.

Key fixes:

- De-duplication is now category-scoped and cannot remove distinct reviewed coverage.
- `STATE: NOT_APPLICABLE` is persisted, passed into Phase 2, and materialized as `[]`.
- Completed subagents are explicitly closed after result collection.
- Supplemental cases must be generated by the appropriate specialist, never by the orchestrator.
- A single Phase-1 report now uses four temporary worker reports, post-completion merging, merged-report validation, and success-only cleanup. Temporary files survive any merge, validation, or write failure.
- Specialists can read their section from a combined reviewed report and support delegated supplemental generation.

Modified:

- [api-test-generator/SKILL.md](D:/GitHub/SoftwareTesting-HW06/.agents/skills/api-test-generator/SKILL.md)
- [contract-test-generator/SKILL.md](D:/GitHub/SoftwareTesting-HW06/.agents/skills/contract-test-generator/SKILL.md)
- [domain-test-generator/SKILL.md](D:/GitHub/SoftwareTesting-HW06/.agents/skills/domain-test-generator/SKILL.md)
- [state-transition-test-generator/SKILL.md](D:/GitHub/SoftwareTesting-HW06/.agents/skills/state-transition-test-generator/SKILL.md)
- [security-test-generator/SKILL.md](D:/GitHub/SoftwareTesting-HW06/.agents/skills/security-test-generator/SKILL.md)

Validation passed:

- `quick_validate.py`: all five skills returned `Skill is valid!`
- All defect-specific semantic checks passed.
- `git diff --check` passed.
- UI metadata and unrelated files were unchanged.

- **File Modifications:**
  - `File: .agents/skills/api-test-generator/SKILL.md`.
    - `Action:` edited.
    - `Deleted:` Prior orchestration wording that allowed cross-category semantic de-duplication, could let the parent add cases from reviewed models, did not explicitly close completed subagents, and did not define safe single-report merging or a complete Phase-2 state-not-applicable handoff.
    - `Replaced With:` Category-scoped coverage-preserving de-duplication; explicit persisted `STATE: NOT_APPLICABLE` Phase-2 handoff; native subagent closure after result collection; specialist-only supplemental generation; parent prohibition on authoring test content; and four-temp-file merge, validation, success-only deletion, and failure preservation rules for a single Phase-1 report.
    - `Appended:` A dedicated `Single Phase-1 report path` section and an explicit parent-generation boundary.
  - `File: .agents/skills/contract-test-generator/SKILL.md`.
    - `Action:` edited.
    - `Deleted:` Input and Phase-2 wording limited to standalone reports and initial candidate generation.
    - `Replaced With:` Support for a read-only `CONTRACT` section in a combined report, unique temporary Phase-1 output ownership, and specialist-owned supplemental generation against existing candidates and approved rules.
    - `Appended:` None.
  - `File: .agents/skills/domain-test-generator/SKILL.md`.
    - `Action:` edited.
    - `Deleted:` Input and Phase-2 wording limited to standalone reports and initial candidate generation.
    - `Replaced With:` Support for a read-only `DOMAIN` section in a combined report, unique temporary Phase-1 output ownership, and specialist-owned supplemental generation against existing candidates and reviewed partitions or boundaries.
    - `Appended:` None.
  - `File: .agents/skills/state-transition-test-generator/SKILL.md`.
    - `Action:` edited.

        - `Deleted:` State-not-applicable handling that returned only a justified empty result without a defined persistent Phase-1 status artifact or Phase-2 candidate fragment.
        - `Replaced With:` A mergeable Phase-1 `STATE: NOT_APPLICABLE` status record, approval bypass for that preserved status, exact status/basis verification in Phase 2, required empty `[]` candidate output, combined-report support, and specialist-owned supplemental generation for applicable transitions.
        - `Appended:` None.
    - `File: .agents/skills/security-test-generator/SKILL.md`.
        - `Action:` edited.
        - `Deleted:` Input and Phase-2 wording limited to standalone reports and initial candidate generation.
        - `Replaced With:` Support for a read-only `SECURITY` section in a combined report, unique temporary Phase-1 output ownership, and specialist-owned supplemental generation against existing candidates and approved scenarios or requirements.
        - `Appended:` None.
    - `File: .skill-revision-stage/*/SKILL.md`.
        - `Action:` temporary files created and deleted.
        - `Deleted:` Five staged revised skill files after protected-directory installation.
        - `Replaced With:` None.
        - `Appended:` None.
    - `File: .validation-deps/**`.
        - `Action:` temporary validator dependencies created and deleted.
        - `Deleted:` Isolated PyYAML dependency directory after all validation runs passed.
        - `Replaced With:` None.
        - `Appended:` None.
- **Verdict:** INCOMPLETE
- **Reasoning:**
    - `NOT_APPLICABLE` uses inconsistent names and formats.
    - Existing approved sections may be lost when using one combined Phase-1 report.
    - The orchestrator still says to spawn exactly four agents.
- **Student Fixes:**

- Use one clear `NOT_APPLICABLE` format:
  - Report: `State Applicability: NOT_APPLICABLE`
  - Internal record: `state_applicability.status: NOT_APPLICABLE`
  - For one combined report, copy each existing category section into its own temporary file before spawning workers.
  - Merge the temporary reports back into the combined report after Phase 1.
  - Replace "exactly four agents" with one primary agent per testing dimension. Allow extra helper, retry, or supplemental agents when needed.

Artifact 5

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-20 12:40:11 GMT+7
- **Model Used:** GPT-5.6 Sol (parent and inherited specialist agents)
- **Skill Used:**
  - `api-test-generator`
  - `contract-test-generator`
  - `domain-test-generator`
  - `state-transition-test-generator`
  - `security-test-generator`
  - `ai-audit`
- **Prompt:**

```
Use [$api-test-generator](D:\GitHub\SoftwareTesting-HW06\\.agents\skills\api-test-generator\SKILL.md) for **Pool A – Forgot Password / Reset Password**.



Inputs:

- API specification: [api_specification.md](reference/api_specification.md)
- System requirements: [system_requirements.md](reference/system_requirements.md)
- Selected API: `POST /api/reset-password`

Write the Phase 1 analysis to: `review/pool-a/` (create files).

After Phase 1, stop for my review before generating test cases.

Treat the provided specification and system requirements as authoritative. Do not invent missing API behavior.
```

- **AI Output:**

```
I’m using the `api-test-generator` skill because this request calls for
its coordinated Phase 1 analysis and explicit human-review checkpoint.
I’ll read its instructions and the relevant endpoint requirements first,
then create only the review artifacts under `review/pool-a/`.

The skill requires four parallel specialist agents and a hard review
gate. I’m now reading each specialist’s exact Phase 1 contract and
extracting a single shared endpoint context so all analyses use the same
authoritative facts.

The selected contract is sparse: it defines the JSON request example but
no response contract or status codes. The requirements add password-
strength, OTP binding, expiry, and one-time-use rules; I’ll preserve
those as requirements while explicitly marking unmapped fields and
unspecified API behavior as gaps.

The shared context is complete. I’m starting the four mandated Phase 1
specialists concurrently; each owns only its category report, and none
is allowed to generate test cases or modify the shared/status files.

All four Phase 1 outputs are back; state analysis found the OTP
lifecycle applicable. I’m validating each report for source identity,
complete tables, pending review markers, unsupported claims, and
confirming that no candidate test files were created.

Phase 1 is complete and stopped at the human-review gate. No test cases
were generated.

Review artifacts:

- [Contract analysis](D:/GitHub/SoftwareTesting-HW06/review/pool-
a/reports/contract-report.md)
- [Domain analysis](D:/GitHub/SoftwareTesting-HW06/review/pool-
a/reports/domain-report.md)
- [State-transition analysis](D:/GitHub/SoftwareTesting-
HW06/review/pool-a/reports/state-report.md) — applicable
- [Security analysis](D:/GitHub/SoftwareTesting-HW06/review/pool-
```

```
a/reports/security-report.md)
- [Shared API context](D:/GitHub/SoftwareTesting-HW06/review/pool-
a/shared-api-context.md)
- [Orchestration status and review focus](D:/GitHub/SoftwareTesting-
HW06/review/pool-a/orchestration-status.md)

All four reports are marked `PENDING`. The analysis explicitly preserves
unspecified behavior, particularly the absent response contract,
confirmation-password API mapping, JWT applicability, and OTP
interpretation.
```

- **File Modifications:**
  - `File:` `review/pool-a/shared-api-context.md`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Shared source identity and hashes; the normalized request, response, authentication, SEC-01–SEC-07, state, and test models for `POST /api/reset-password`; exact specification bases; and explicit gaps and assumptions.
  - `File:` `review/pool-a/orchestration-status.md`.
    - `Action:` created, then edited.
    - `Deleted:` Initial `IN_PROGRESS` status and the four pending-specialist placeholders.
    - `Replaced With:` `AWAITING_HUMAN_REVIEW` status, collected-report paths and hashes, state applicability, coordinator validation results, five human-review focus items, and a gate record confirming that no candidate tests or final suite exist.
    - `Appended:` None.
  - `File:` `review/pool-a/reports/contract-report.md`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Phase 1 CONTRACT analysis containing source identity, request and response inventories, `CR-001`–`CR-012`, gaps and ambiguities, and a `Review`

`Status: PENDING` block.

- `File:` `review/pool-a/reports/domain-report.md`.
  - `Action:` created.
  - `Deleted:` None.
  - `Replaced With:` None.
  - `Appended:` Phase 1 DOMAIN analysis containing the valid baseline, parameter inventory, `DP-001`–`DP-026`, `DB-001`–`DB-006`, cross-parameter constraints, gaps and ambiguities, and a pending review block.
- `File:` `review/pool-a/reports/state-report.md`.
  - `Action:` created.
  - `Deleted:` None.
  - `Replaced With:` None.
  - `Appended:` Phase 1 STATE analysis marking applicability as `APPLICABLE`, defining `ST-01`–`ST-03`, guards `GD-01`–`GD-05`, transitions `TR-001`–`TR-005`, gaps and ambiguities, and a pending review block.
- `File:` `review/pool-a/reports/security-report.md`.
  - `Action:` created.
  - `Deleted:` None.
  - `Replaced With:` None.
  - `Appended:` Phase 1 SECURITY analysis containing endpoint characteristics, the SEC-01–SEC-07 applicability matrix, `SS-001`–`SS-007`, coverage decisions, gaps, requested human decisions, and a `Review Status: PENDING` block.

- **Verdict:** INCOMPLETE
- **Reasoning:**
  - Required request fields were not clearly enforced.
  - `null`, wrong-type, malformed, and malformed-email inputs were left unclear.
  - JWT requirement for reset password was unresolved.
  - SQL injection black-box checks were missing.
  - OTP brute-force/rate-limit checks were missing.
  - Account-enumeration checks were missing.
- **Student Fixes:**

- Treated `email`, `resetToken`, and `newPassword` as required.
- Treated malformed, missing, `null`, and wrong-type inputs as invalid.
- Added `Content-Type: application/json`.
- Added reviewed HTTP expectations: `200`, `400`, `415`, and `429`.
- Marked JWT as not required for this recovery endpoint.
- Kept OTP usable after failed validation.
- Consume OTP only after a successful reset.
- Added SQL injection black-box testing.
- Added OTP brute-force/rate-limit testing.
- Added account-enumeration testing.

Artifact 6

- **User:** Vương Ngữ Tín Thành
- **Date and Time:** 2026-08-20 17:27:46 GMT+7
- **Model Used:** GPT-5.6 Sol (parent and inherited specialist agents)
- **Skill Used:**
  - `api-test-generator`
  - `contract-test-generator`
  - `domain-test-generator`
  - `state-transition-test-generator`
  - `security-test-generator`
  - `ai-audit`
- **Prompt:**

```
Continue Pool A from the completed Phase 1 review. I have manually
revised and approved the Phase 1 reports.

Shared API Context and Orchestration Status are outdated. Refresh them
from the reviewed Phase 1 reports before continuing.

Then proceed with Phase 2 using [$api-test-generator]
(D:\\GitHub\\SoftwareTesting-HW06\\.agents\\skills\\api-test-
generator\\SKILL.md) and generate the Pool A test cases as previously
configured. Do not redo Phase 1 unless required to resolve an
inconsistency.
```

- **AI Output:**

I’m resuming with `api-test-generator` at its Phase 2 gate. I’ll first validate the manually approved report versions, reconcile the stale shared context and orchestration status strictly from those reviewed reports, then run the four specialist generators without reopening Phase 1 unless the reports conflict.

The report files still contain stale `PENDING` markers, but your current instruction explicitly approves the exact reviewed files now on disk, which satisfies the Phase 2 gate. I’m treating those files—not the stale markers or prior hashes—as the reviewed authority and checking their revisions before refreshing the parent-owned context.

The approved edits are present—the report hashes changed. I found two stale narrative remnants inside the reports: the structured STATE table now says a weak-password failure preserves the OTP, while its older summary still says unspecified; the SECURITY matrix now makes JWT not applicable, while older gap text still calls it unresolved. I’ll reconcile these by treating the manually revised structured rule/scenario tables as the approved decisions and record that precedence in the refreshed shared context, without rerunning Phase 1.

The parent-owned context is now synchronized with the approved report hashes and decisions. I’m beginning Phase 2 generation; each specialist will read the refreshed context plus only its reviewed report, write a category-owned JSON fragment, and leave aggregation/final IDs to the parent.

Phase 2 produced 76 raw candidates: 26 CONTRACT, 36 DOMAIN, 5 STATE, and 9 SECURITY. I’m now performing the parent-only checks—schema/category/endpoint validation, within-category duplicate review, reviewed-rule coverage, and final stable ID assignment—before writing the CSV.

The parent validation found no true within-category duplicates. Three case-insensitive matches were intentional uppercase/lowercase boundary pairs with different payloads and distinct reviewed coverage, so they are retained. Coverage is complete except CONTRACT `CR-008`, which is

```
correctly unresolved because the API has no confirmation-password field;
no unsupported case will be invented.

The suite is aggregated and validated: 76 traceable candidates with
stable IDs `API-001`–`API-076`, no records removed, and complete
reviewed coverage apart from the intentionally unexecutable
confirmation-field rule. I’m doing the final artifact/status integrity
check now.

Pool A Phase 2 is complete.

- Final suite: [candidate-api-tests.csv](D:/GitHub/SoftwareTesting-
HW06/review/pool-a/candidate-api-tests.csv)
- Total: 76 cases — CONTRACT 26, DOMAIN 36, STATE 5, SECURITY 9
- IDs: `API-001`–`API-076`
- [Shared API context](D:/GitHub/SoftwareTesting-HW06/review/pool-
a/shared-api-context.md) refreshed
- [Orchestration status](D:/GitHub/SoftwareTesting-HW06/review/pool-
a/orchestration-status.md) marked `COMPLETE`

All reviewed coverage is represented except `CR-008`, because the
confirmation-password field has no documented API mapping. No field or
missing behavior was invented, and no tests were executed.
```

- **File Modifications:**
  - `File:` `review/pool-a/shared-api-context.md`.
    - `Action:` edited.
    - `Deleted:` Stale Phase 1-only interpretations that treated reviewed requiredness, HTTP oracles, JWT applicability, weak-password OTP effects, and added security scenarios as unresolved.
    - `Replaced With:` Phase 2 review provenance and approved report hashes; structured-row precedence for stale STATE and SECURITY narrative conflicts; reviewed required-field/type/null rules; `application/json`, `200`, `400`, and conditional `429` external oracles; SEC-02 not-applicable interpretation; reviewed STATE `TR-005`; SECURITY `SS-001`–`SS-009`; and preserved specification gaps.
    - `Appended:` None.

- `File:` `review/pool-a/orchestration-status.md`.
  - `Action:` edited.
  - `Deleted:` Stale Phase 1 `AWAITING_HUMAN_REVIEW` state, outdated report hashes, pending generation state, and review-only gate record.
  - `Replaced With:` Explicit-approval gate provenance, current reviewed report hashes, Phase 2 configuration, `COMPLETE` status, specialist fragment counts and hashes, parent aggregation validation, final CSV hash, and preserved limitations.
  - `Appended:` None.
- `File:` `review/pool-a/candidates/contract-tests.json`.
  - `Action:` created.
  - `Deleted:` None.
  - `Replaced With:` None.
  - `Appended:` 26 Phase 2 CONTRACT candidates covering `CR-001` – `CR-007` and `CR-009` – `CR-012`; `CR-008` remained explicitly unresolved because confirmation transport is unspecified.
- `File:` `review/pool-a/candidates/domain-tests.json`.
  - `Action:` created.
  - `Deleted:` None.
  - `Replaced With:` None.
  - `Appended:` 36 Phase 2 DOMAIN candidates covering `DP-001` – `DP-026` and `DB-001` – `DB-006`.
- `File:` `review/pool-a/candidates/state-tests.json`.
  - `Action:` created.
  - `Deleted:` None.
  - `Replaced With:` None.
  - `Appended:` Five Phase 2 STATE candidates covering `TR-001` – `TR-005`, including the reviewed `TR-005` OTP-preservation decision.
- `File:` `review/pool-a/candidates/security-tests.json`.
  - `Action:` created.
  - `Deleted:` None.
  - `Replaced With:` None.

        - `Appended:` Nine Phase 2 SECURITY candidates covering `SS-001` – `SS-009`, including reviewed external brute-force and account-enumeration scenarios and SEC-02 not-applicable treatment.
    - `File:` `review/pool-a/candidate-api-tests.csv`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Parent-aggregated 76-record candidate suite with exactly nine required columns, stable IDs `API-001` – `API-076`, preserved specialist IDs, and per-category counts CONTRACT 26, DOMAIN 36, STATE 5, SECURITY 9.
- **Verdict:** INCOMPLETE
- **Reasoning:**
    - The generated suite has good coverage and 76 test cases, but 8 cases are incomplete or unclear.
    - Some cases assume undefined behavior, such as OTP expiry or rate-limit thresholds.
    - Some security cases are not directly executable with Postman.
- **Student Fixes:** Continued guiding the agent to revise the incomplete or unclear test cases.

Artifact 7

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-20 20:24:33 GMT+7
- **Model Used:** GPT-5.6 Sol (parent and inherited specialist agents)
- **Skill Used:**
    - `api-test-generator`
    - `contract-test-generator`
    - `domain-test-generator`
    - `state-transition-test-generator`
    - `security-test-generator`
    - `ai-audit`
- **Prompt:**

```
Revise the Pool A test cases to fix these 8 issues:

- `API-030`: Make it exploratory or remove it because extra JSON
properties have no defined expected behavior.
- `API-059`: Treat `Aa1!bbbb#` as valid because it already meets the
required password rules.
- `API-025`, `API-065`, `API-072`: Do not assume an OTP expiry duration.
Keep expiry behavior testable only if a real expiry point can be
configured or observed.
- `API-068`: Mark it as a white-box/storage verification test because
Postman alone cannot prove whether passwords are stored in plaintext.
- `API-075`: Do not invent a rate-limit threshold. Make the test
conditional on a known configured limit.
- `API-076`: Replace the vague account-enumeration oracle with a clear
comparison rule for status, response body, and observable behavior.

Keep the reviewed Phase-1 requirements as the source of truth.

After editing, revalidate all Pool A test cases and report which cases
were changed.
```

- **AI Output:**

I’ll use the reviewed `api-test-generator` workflow to revise the category-owned candidates behind those eight final IDs, preserve the Phase 1 authority, then rebuild and revalidate the 76-row CSV without renumbering unrelated cases. I’ll treat `API-030` as exploratory rather than remove it, preserving traceability and stable IDs.

The fragments confirm the eight referenced IDs map cleanly to four specialist-owned cases/groups. Several are partially corrected already, but I’ll make each requested constraint explicit—for example, expiry cases will be conditional on an observable/configurable expiry point, and the enumeration case will compare status, normalized body structure/content, password state, and other externally observable effects.

All four specialists completed scoped revisions without changing counts or coverage. I’m now checking the revised provisional cases against the eight requested fixes, then I’ll regenerate the final CSV so the same final IDs remain attached to the corrected content.

Full-suite validation passes: 76 rows, 76 unique sequential IDs, zero schema/category/endpoint errors, zero true within-category duplicates, and unchanged CR/DP/DB/TR/SS coverage. I’m updating the parent traceability records with the revised fragment and CSV hashes plus the eight-case change log.

Revised [Pool A test cases](D:/GitHub/SoftwareTesting-HW06/review/pool-a/candidate-api-tests.csv).

Changed cases:

- `API-025`, `API-065`, `API-072`: require a configurable or objectively observable expiry point; otherwise blocked.
- `API-030`: explicitly exploratory, with no pass/fail oracle.
- `API-059`: `Aa1!bbbb#` is valid and expects `200 OK`.
- `API-068`: explicitly white-box/storage verification; not Postman-verifiable alone.
- `API-075`: conditional on a known configured abuse-control limit.

```
- `API-076`: precise status, body, redirect, content-type, side-effect,
and timing comparison rules.

Validation passed: 76 cases, 76 unique sequential IDs, nine required
fields per case, unchanged coverage, and zero true duplicates. No tests
were executed.
```

- **File Modifications:**
  - `File:` `review/pool-a/candidates/contract-tests.json`.
    - `Action:` edited.
    - `Deleted:` `CONTRACT-P025` preconditions and oracle that did not explicitly block execution when the real expiry point could not be established.
    - `Replaced With:` `CONTRACT-P025` preconditions, expected result, and notes requiring a real configurable or objectively observable expiry point; otherwise `BLOCKED / NOT EXECUTABLE` with no SUT verdict.
    - `Appended:` None.
  - `File:` `review/pool-a/candidates/domain-tests.json`.
    - `Action:` edited.
    - `Deleted:` `DOMAIN-P004` wording that did not explicitly exclude deterministic conformance scoring, and `DOMAIN-P033`'s unresolved acceptance oracle for `Aa1!bbbb#`.
    - `Replaced With:` `DOMAIN-P004` as exploratory/characterization-only with no pass/fail oracle; `DOMAIN-P033` as a valid-password success case expecting reviewed `200 OK` because all required character classes are present and `#` is not prohibited.
    - `Appended:` None.
  - `File:` `review/pool-a/candidates/state-tests.json`.
    - `Action:` edited.
    - `Deleted:` `STATE-P003` setup that could be read as permitting a synthetic or assumed expiry point.
    - `Replaced With:` `STATE-P003` execution gate requiring supported configuration or objective observation of the real SUT expiry point, with no request sent and a blocked result when ST-02 cannot be established.

        - `Appended:` None.
    - `File:` `review/pool-a/candidates/security-tests.json`.
        - `Action:` edited.
        - `Deleted:` Ambiguous executability/oracles in `SECURITY-P001`, `SECURITY-P005`, `SECURITY-P008`, and `SECURITY-P009` concerning storage verification, OTP expiry, rate-limit configuration, and account-enumeration comparison.
        - `Replaced With:` Explicit white-box/storage access for P001; real-expiry execution gating for P005; a known authoritative abuse-control limit for P008; and exact status, semantic JSON or exact non-JSON body, content-type, redirect, password/token/metadata, and conditional timing comparison rules for P009.
        - `Appended:` None.
    - `File:` `review/pool-a/candidate-api-tests.csv`.
        - `Action:` replaced content.
        - `Deleted:` Prior aggregate representations of final IDs `API-025`, `API-030`, `API-059`, `API-065`, `API-068`, `API-072`, `API-075`, and `API-076`.
        - `Replaced With:` Rebuilt 76-row suite preserving IDs `API-001`–`API-076` and incorporating all eight corrected cases; final SHA-256 `3C50ACA8A132222E82E6AFC0607B4C055F22069AEF9FB10429CF326CF6C24081`.
        - `Appended:` None.
    - `File:` `review/pool-a/orchestration-status.md`.
        - `Action:` edited.
        - `Deleted:` Superseded candidate-fragment and CSV hashes plus the prior statement that both DP-004 and DP-024 were characterization cases.
        - `Replaced With:` Current fragment/CSV hashes, an eight-case post-generation revision log, revalidation results, and corrected treatment of DP-004 as exploratory and DP-024 as valid for `Aa1!bbbb#`.
        - `Appended:` None.
    - `File:` `review/pool-a/shared-api-context.md`.
        - `Action:` edited.
        - `Deleted:` Less-specific expiry, storage-verification, rate-limit, enumeration, extra-property, and additional-password-character assumptions.

    - `Replaced With:` Real-expiry execution gates, valid treatment of `Aa1!bbbb#`, exploratory treatment of extra properties, white-box SEC-01 verification, configured-limit-only SS-008 execution, and explicit SS-009 comparison rules.
    - `Appended:` None.

- **Verdict:** VALID
- **Reasoning:** The agent correctly resolved found test defects.
- **Student Fixes:** None

Artifact 8

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-20 20:47:31 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:** `ai-audit`
- **Prompt:**

```
Base on [pool-a](review/pool-a/), fill in [Main_Report.md]
(Main_Report.md).
```

- **AI Output:**

```
I’ll inspect the report template and the completed Pool A artifacts,
then fill only the relevant sections with the reviewed analysis, final
test-suite results, revisions, and validation evidence.

The deliverable CSV already matches the final Pool A suite byte-for-byte
(76 cases, same SHA-256), so the existing `test-cases/a-forgot-
password.csv` link is correct. I’ll now populate Sections A.2–A.6 and
update the report date, while leaving Pools B and C untouched.

Filled the Pool A sections in [Main_Report.md]
(D:/GitHub/SoftwareTesting-HW06/Main_Report.md):

- Contract, domain, state-transition, and security analyses
- Reviewed assumptions and unresolved behavior
- Final counts: 26 CONTRACT, 36 DOMAIN, 5 STATE, 9 SECURITY
- Revision and validation summary for all 76 cases
- Updated report date to 2026-08-20

All report links, IDs, counts, and the final CSV checksum were verified.
Pools B and C were unchanged.
```

- **File Modifications:**
  - `File:` `Main_Report.md`.
    - `Action:` edited.
    - `Deleted:` Report and last-updated dates of `2026-08-18`; empty Pool A Sections A.2–A.5; and blank Contract, Domain, State Transition, Security, and Total counts in Section A.6.
    - `Replaced With:` Report and last-updated dates of `2026-08-20`; Pool A contract analysis covering request fields, `CR-001`–`CR-012`, reviewed HTTP assumptions, and unresolved `CR-008`; domain analysis covering `DP-001`–`DP-026`, `DB-001`–`DB-006`, exploratory `API-030`, and valid `API-059`; state analysis covering `ST-01`–`ST-03`, `TR-001`–`TR-005`, and observable/configurable expiry gating; security applicability for `SEC-01`–`SEC-07` and scenarios `SS-001`–`SS-009`; test counts of 26 CONTRACT, 36 DOMAIN, 5 STATE, 9 SECURITY, and 76 total; plus stable IDs, eight revised cases, validation evidence, and final CSV SHA-256 `3C50ACA8A132222E82E6AFC0607B4C055F22069AEF9FB10429CF326CF6C24081`.

        - `Appended:` None.

- **Verdict:** VALID
- **Reasoning:** The agent correctly summarized pool A process into main report.
- **Student Fixes:** None

Artifact 9

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-21 10:44:39 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
    - `postman-test-generator`
    - `ai-audit`
- **Prompt:**

```
Use [$postman-test-generator](D:\GitHub\SoftwareTesting-
HW06\\.agents\skills\postman-test-generator\SKILL.md) to convert the
final reviewed Pool A test cases into a Postman Collection v2.1.

Inputs:

- [a-forgot-password.csv](test-cases/a-forgot-password.csv)&#x20;
- [api\_specification.md](reference/api_specification.md)&#x20;
- collection\_output\_path: postman/pool-a-forgot-
password.postman\_collection.json
- conversion\_report\_path: postman/pool-a-conversion-report.md

For this homework, implement `X-Student-Id` through a collection-level
pre-request script using `{{studentId}}` is `23127261`, and log evidence
that can be captured from the Postman console.



Do not redesign or add test cases. Preserve all preconditions and
traceability. Clearly flag tests that require manual setup, white-box
verification, an observable expiry point, or a configured rate limit
instead of inventing executable behavior.



Validate the final collection against the full Postman Collection v2.1
schema and verify Newman compatibility.
```

- **AI Output:**

```
Created the requested Postman artifacts:

- [Postman Collection v2.1](D:\GitHub\SoftwareTesting-HW06\postman\pool-a-forgot-password.postman_collection.json)
- [Conversion report](D:\GitHub\SoftwareTesting-HW06\postman\pool-a-conversion-report.md)

Verification completed:

- All 82 reviewed test IDs and preconditions are preserved.
- `X-Student-Id: 23127261` is injected from `{{studentId}}` by a collection-level pre-request script.
- Console evidence includes the injected student ID, request name, URL, status, and response time.
- Manual setup, white-box checks, observable expiry cases, concurrency, and configured rate limiting are explicitly flagged.
- Full official Postman Collection v2.1 schema validation: passed.
- Newman 6.2.2 compatibility execution: passed with zero failures on the representative compatibility run.
- No test cases were added or redesigned.
```

- **File Modifications:**
  - `File:` `postman/pool-a-forgot-password.postman_collection.json`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Postman Collection v2.1 containing all 82 reviewed Pool A cases, category folders, reviewed multi-request flows, preserved descriptions and preconditions, status assertions where machine-checkable, collection variables, collection-level `X-Student-Id` injection, and console evidence scripts.
  - `File:` `postman/pool-a-conversion-report.md`.
    - `Action:` created, then edited.
    - `Deleted:` Pending schema-validation and Newman-compatibility status placeholders.

    - `Replaced With:` Passing full-schema validation and Newman 6.2.2 compatibility evidence.
    - `Appended:` Row-to-request traceability for all 82 cases, preserved preconditions, generated assertion mapping, manual/setup flags, unresolved oracle notes, and validation results.
  - `File:` `postman/generate_pool_a.py`.
    - `Action:` created, then edited.
    - `Deleted:` Over-broad observable-expiry flag detection based on text matching.
    - `Replaced With:` Exact observable-expiry flagging for `API-025`, `API-065`, and `API-072`.
    - `Appended:` Deterministic CSV-to-Postman conversion logic and conversion-report generation.
  - `File:` `postman/postman-v2.1.0-schema.json`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Official Postman Collection Format v2.1.0 JSON Schema used for full validation.
  - `File:` `postman/newman_compat_server.js`.
    - `Action:` temporary file created and deleted.
    - `Deleted:` Local HTTP compatibility server after the representative Newman run completed.
    - `Replaced With:` None.
    - `Appended:` None.
  - `File:` `postman/.validation_deps/**`.
    - `Action:` temporary validator dependencies created and deleted.
    - `Deleted:` Isolated `jsonschema` validation dependency directory after final schema validation.
    - `Replaced With:` None.
    - `Appended:` None.
- **Verdict:** INCOMPLETE

- **Reasoning:**
  - Domain test request bodies are generated incorrectly.
  - Some `{{variables}}` are used but not defined.
  - Some tests check only the HTTP status, not the full expected result.
  - Manual/setup-required tests are not always marked correctly.
  - Human test case references still have off-by-one IDs.
  - Re-running the generator resets validation results back to `PENDING`.
  - Newman compatibility was checked with only one representative test, not the full suite.
- **Student Fixes:** Guide the agent to fix found defects.

Artifact 10

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-21 11:25:08 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
  - `postman-test-generator`
  - `ai-audit`
- **Prompt:**

```
Keep the existing reviewed test cases unchanged. Do not redesign, add, remove, merge, or renumber test cases unless required to correct traceability.

Fix the following issues:
- Incorrect Domain request bodies
  - Correctly interpret structured `Request Input` values containing `Headers`, `Body`, or `Raw Body`.
  - `Headers` must become actual HTTP headers.
  - `Body` must become the actual request body.
  - `Raw Body` must be sent exactly as raw content.
  - Do not send the metadata wrapper itself.
  - Preserve intentionally malformed JSON exactly.

- Undefined Postman variables
  - Detect every `{{variable}}` referenced by generated requests.
  - Ensure every required variable is defined or explicitly marked as a required runtime fixture.
  - Never allow an unresolved variable to accidentally make a negative test pass.
  - If a required fixture is unavailable, clearly mark the test as blocked/not executable.

- Incomplete automated oracles
  - Do not treat an HTTP-status assertion as the complete result when the reviewed expected result contains additional observable checks.
  - Automate directly observable checks where practical, especially multi-request comparisons such as API-076.
  - For checks that cannot be automated reliably, explicitly mark the case as `PARTIALLY AUTOMATED / MANUAL ORACLE REQUIRED`.
  - Do not invent response-body, persistence, timing, or state assertions.
- Incorrect manual/setup flags
  - Improve fixture detection so tests requiring registered accounts, issued OTPs, previous state, expiry configuration, rate limits, or other server-side setup are correctly marked.
  - Prefer explicit execution classification over fragile keyword
```

```
matching where possible.

- Human test-case traceability
  - Fix the remaining off-by-one references for the human-authored
cases.
  - The final IDs must consistently reference `API-077` through `API-
082`.
  - Update source/report references rather than hiding the mismatch only
inside the Postman collection.
  - Non-reproducible validation report
  - Do not overwrite successful validation results with `PENDING` when
the generator is rerun.
  - Make the workflow reproducible: `generate → schema validation →
Newman compatibility validation → final conversion report`.
  - The report must clearly distinguish structural/Newman compatibility
validation from actual SUT test execution.

- Newman compatibility scope
  - Keep the existing representative compatibility check if appropriate,
but clearly state that it does not mean the full 82-case suite was
functionally executed.
  - Add deterministic static validation for the whole generated
collection, including:
    - all 82 logical test IDs are present;
    - expected multi-request flows are preserved;
    - no unexpected test cases are added or removed;
    - all referenced Postman variables are defined or explicitly
blocked;
    - request bodies match the reviewed CSV representation.

After revising:

- Regenerate `pool-a-forgot-password.postman_collection.json`.
- Regenerate `pool-a-conversion-report.md`.
- Update `generate_pool_a.py` as needed.
- Validate the collection against the full supplied Postman Collection
v2.1 schema.
```

```
- Verify Newman compatibility.
- Do **not** perform the real SUT execution yet.
- Report the files changed and briefly explain how each defect was fixed.
```

- **AI Output:**

```
Revised and regenerated the Pool A Postman artifacts without changing
test intent, IDs, or case count.

Changed files:

- [generate_pool_a.py](D:\GitHub\SoftwareTesting-
HW06\postman\generate_pool_a.py)
  - Correctly parses `Headers`, `Body`, and `Raw Body`.
  - Adds explicit fixture readiness gates and variable validation.
  - Adds API-076 response comparison automation.
  - Adds deterministic static, schema, Newman, and report stages.
  - Preserves successful validation results across identical
regeneration.

- [Postman collection](D:\GitHub\SoftwareTesting-HW06\postman\pool-a-
forgot-password.postman_collection.json)
  - Regenerated with corrected bodies, headers, variables, fixture
gates, and oracle classifications.

- [Conversion report](D:\GitHub\SoftwareTesting-HW06\postman\pool-a-
conversion-report.md)
  - Regenerated with fixture requirements, partial/manual oracle
classifications, reproducible commands, and clear validation scope.

- [Validation results](D:\GitHub\SoftwareTesting-HW06\postman\pool-a-
validation-results.json)
  - Stores validation results keyed to collection SHA-256.

- [Validation requirements](D:\GitHub\SoftwareTesting-
HW06\postman\requirements-validation.txt)
  - Pins the schema-validation dependency.

- [Reviewed CSV](D:\GitHub\SoftwareTesting-HW06\test-cases\a-forgot-
password.csv)
  - Corrected only the six off-by-one human-case references to
consistently use `API-077`–`API-082`.
```

```
Validation results:

- 82 logical test cases preserved.
- 88 requests generated.
- All six expected two-request flows preserved.
- No undefined Postman variables.
- No request body or header mismatches.
- Full Postman Collection v2.1 schema: passed.
- Newman 6.2.2 representative compatibility check: passed.
- Identical regeneration preserves the final report and validation
state.
- Collection SHA-256:
`49134CF344C65181C4F5FE9A65802867A236B86E427BEF7414BCA6C33539921F`
- Real SUT execution was not performed.
```

- **File Modifications:**
  - `File:` `test-cases/a-forgot-password.csv`.
    - `Action:` edited.
    - `Deleted:` Off-by-one `Main_Report.md` human-case references `API-078` through `API-083` on final cases `API-077` through `API-082`.
    - `Replaced With:` Matching references `API-077` through `API-082`; no test intent, request, oracle, ID, ordering, or case count changed.
    - `Appended:` None.
  - `File:` `postman/generate_pool_a.py`.
    - `Action:` replaced content, then edited.
    - `Deleted:` Heuristic first-JSON-fragment body mapping, keyword-based fixture classification, status-only oracle reporting, report generation that reset validation to `PENDING`, and non-persistent validation workflow.
    - `Replaced With:` Structured `Headers` / `Body` / `Raw Body` parsing; exact malformed-body preservation; explicit fixture classifications and request-blocking pre-request scripts; complete variable discovery; API-076 response comparison with redirect following disabled; API-075 classification as a single template requiring manual/data-driven repeated execution; partial/manual oracle classifications; deterministic whole-collection validation including an independent 36-row structured-input assertion path that does not use `request_specs()`; hash-

keyed validation state; full-schema validation; representative Newman/local-mock validation; and gated final report generation.

  - `Appended:` None.

- `File:` `postman/pool-a-forgot-password.postman_collection.json`.
  - `Action:` replaced content.
  - `Deleted:` Prior collection with Domain metadata wrappers used as bodies, incomplete variable handling, fragile setup flags, status-only reporting for additional reviewed checks, and no API-076 automated response comparison.
  - `Replaced With:` Regenerated 82-logical-case/88-request collection with correct reviewed headers and bodies, exact raw malformed content, defined runtime variables, fixture readiness gates, blocked execution on unavailable fixtures, partial/manual oracle warnings, API-075 marked as a single request template requiring manual/data-driven repetition through the authoritative configured trigger, API-076 comparison assertions, all six reviewed two-request flows, and collection-level `X-Student-Id` evidence logging.
  - `Appended:` None.

- `File:` `postman/pool-a-conversion-report.md`.
  - `Action:` replaced content.
  - `Deleted:` Prior traceability table and manually patched validation section that could be reset to `PENDING` by regeneration.
  - `Replaced With:` Regenerated traceability, fixture requirements, execution/oracle classifications, explicit API-075 repeated-execution limitations, API-076 automation scope, reproducible validation commands, independent validation evidence for all 36 structured request inputs, hash-keyed validation evidence, and an explicit statement that the real SUT and full functional suite were not executed.
  - `Appended:` None.

- `File:` `postman/pool-a-validation-results.json`.
  - `Action:` created and updated.
  - `Deleted:` None.
  - `Replaced With:` None.
  - `Appended:` Collection-SHA-keyed PASS evidence for deterministic static validation of 82 logical IDs, 88 requests, six multi-request flows, and 36 independently

checked structured mappings; full Postman v2.1 schema validation; and Newman 6.2.2 representative local-mock compatibility validation.

- `File:` `postman/requirements-validation.txt`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Pinned `jsonschema==4.26.0` dependency for reproducible full-schema validation.
- `File:` `Main_Report.md`.
    - `Action:` edited.
    - `Deleted:` Stale Pool A totals of 26 CONTRACT, 36 DOMAIN, 5 STATE, 9 SECURITY, and 76 total; stale ID range `API-001` – `API-076`; stale 76-ID validation statement; and obsolete CSV checksum `3C50ACA8A132222E82E6AFC0607B4C055F22069AEF9FB10429CF326CF6C24081`.
    - `Replaced With:` Correct final totals of 28 CONTRACT, 37 DOMAIN, 6 STATE, 11 SECURITY, and 82 total; ID range `API-001` – `API-082`; explicit distinction between 76 specialist-generated and six human-authored cases `API-077` – `API-082`; current CSV checksum `FA7402AB0B677D68AF909C2E326DA83FDF6F41C239DB4DF689B93A47F2201DE6`; and last-updated date `2026-08-21`.
    - `Appended:` None.
- `File:` `postman/.validation_deps/**` and `postman/__pycache__/**`.
    - `Action:` temporary content created and deleted.
    - `Deleted:` Temporary isolated schema-validation dependencies and compiled Python cache after validation.
    - `Replaced With:` None.
    - `Appended:` None.

- **Verdict:** VALID
- **Reasoning:** The agent correctly resolved all found defects.
- **Student Fixes:** None

Artifact 11

- **User:** Vương Ngữ Tín Thành
- **Date and Time:** 2026-08-21 23:49:33 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
  - `spreadsheets:Spreadsheets`
  - `postman-test-generator`
  - `ai-audit`
- **Prompt:**

```
Revise API-078 test cases for pool A.

It is a good robustness test, but `415` is **my external HTTP expectation**, not a specification requirement. More importantly, if the SUT returns a safe `400` while rejecting the request and leaving the OTP intact, it should **not automatically report that as a genuine SUT bug** based on the supplied requirements alone.

Way to fix: Read [api\_specification.md](reference/api_specification.md) then expect the respone in 4xx status is accepted.
```

- **AI Output:**

```
Revised API-078 for pool A:

- `415` is explicitly a human/external HTTP expectation, not a specification requirement.
- A safe `400` is accepted when the same OTP remains usable and the JSON retry succeeds.
- Both requests now use the same OTP variable.
- Updated the generated Postman assertions and traceability report.

Files: [test cases](/D:/GitHub/SoftwareTesting-HW06/test-cases/a-forgot-password.csv:79), [Main Report](/D:/GitHub/SoftwareTesting-HW06/Main_Report.md:141), [generator](/D:/GitHub/SoftwareTesting-HW06/postman/generate_pool_a.py:307), and [conversion report](/D:/GitHub/SoftwareTesting-HW06/postman/pool-a-conversion-report.md:94).

Validation passed: 82 unique cases, static mapping, full Postman v2.1 schema, and Newman 6.2.2 compatibility. The real SUT was not executed.

Fixed API-078: step 1 now accepts only `400–499`. Any `2xx`, `3xx`, or `5xx` fails; exact `415` remains optional.

Updated [test case CSV](/D:/GitHub/SoftwareTesting-HW06/test-cases/a-forgot-password.csv:79), [Main Report](/D:/GitHub/SoftwareTesting-HW06/Main_Report.md:141), Postman collection, and conversion report.

Static, schema, and Newman validation all pass.
```

- **File Modifications:**
  - `File:` `Main_Report.md`.
    - `Action:` edited.
    - `Deleted:` The API-078 human-case oracle that required exact `415`, followed by the intermediate oracle that accepted any non-`200`, non-`5xx` response.
    - `Replaced With:` API-078 now accepts only a `4xx` client-error response (`400`–`499`), retains `415 Unsupported Media Type` as the preferred human/external

HTTP expectation, accepts any safe `4xx` including `400`, and requires the same-OTP JSON retry to succeed.
    - `Appended:` None.
- `File:` `test-cases/a-forgot-password.csv`.
    - `Action:` edited.
    - `Deleted:` The prior API-078 row whose first-step oracle was exact `415`, followed by its intermediate non-normal-status/non-`5xx` formulation.
    - `Replaced With:` A nine-field API-078 row requiring a first-step status from `400` through `499`, documenting the `4xx` class and preferred `415` as human/external expectations, and using the same OTP fixture in both requests so the successful retry evidences non-consumption.
    - `Appended:` None.
- `File:` `postman/generate_pool_a.py`.
    - `Action:` edited.
    - `Deleted:` API-078's exact `415` automated status mapping and the intermediate assertions that only rejected `200` and `5xx`.
    - `Replaced With:` API-078 step 1 has no exact-status mapping and asserts `status >= 400 && status < 500`; non-`415` responses produce only a human/external expectation warning; API-078 is marked partially automated; and report labels/policy describe the `4xx`-only boundary.
    - `Appended:` None.
- `File:` `postman/pool-a-forgot-password.postman_collection.json`.
    - `Action:` replaced content.
    - `Deleted:` Generated API-078 descriptions and scripts with the exact-`415` oracle, followed by the intermediate non-`200`/non-`5xx` oracle.
    - `Replaced With:` Regenerated API-078 requests sharing `{{sameValidOTP}}`; step 1 asserts a `400`–`499` response and treats `415` as a warning-only preference, while step 2 retains the reviewed external `200` success assertion.
    - `Appended:` None.
- `File:` `postman/pool-a-conversion-report.md`.
    - `Action:` replaced content.

    - `Deleted:` API-078 traceability and policy text describing exact `415`, followed by the broader non-`200`/non-`5xx` acceptance rule.
    - `Replaced With:` API-078 traceability labels step 1 as `any 4xx`, identifies `415` as a human/external expectation, accepts safe `400`, and records the same-OTP retry and partial/manual oracle classification.
    - `Appended:` None.
  - `File:` `postman/pool-a-validation-results.json`.
    - `Action:` replaced content.
    - `Deleted:` Validation state keyed to the superseded generated collection hashes.
    - `Replaced With:` PASS evidence keyed to collection SHA-256 `F679AADE649F0AD84A3873E83A2DD4152107DEA619194C579F7FDBC9DFFD4F20`, covering 82 logical IDs, 88 requests, full Postman v2.1 schema validation, and Newman 6.2.2 representative compatibility.
    - `Appended:` None.
- **Verdict:** VALID
- **Reasoning:** The agent worked good enough for simple tasks. With user expectation provided, it has read and fixed all related files, ensuring consistency.
- **Student Fixes:** None

Artifact 12

- **User:** Vương Ngữ Tín Thành
- **Date and Time:** 2026-08-22 09:55:29 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
  - `postman-test-generator`
  - `ai-audit`
- **Prompt:**

```
Revise Pool A reset-password fixtures to use deterministic SQLite setup
instead of `/api/forgot-password`.

SUT repo is available at: `D:\GitHub\eshop-sut`

Requirements:

- Create a fixture script that opens the SUT SQLite DB directly; do not
modify SUT source code and do not import its `database.js`.
- Start/init the SUT first, then seed fixtures.
- Prefer one dedicated test account per independent reset-password case.
- Seed exact deterministic OTPs needed by each case.
- Preserve the intended test dimension:
  - leading-zero OTP → seed exact text OTP such as `012345`;
  - numeric-type case → DB stores `"123456"` but request sends JSON
number `123456`;
  - cross-account case → two accounts with different OTPs;
  - replay case → seed fresh OTP, let first reset consume it, then retry
same OTP.
- Remove obsolete `/api/forgot-password` fixture calls and related
runtime variables/guards where no longer needed.
- Keep genuinely unsupported cases blocked/manual, especially expiry,
rate-limit, and API-077 concurrency if the SUT cannot objectively
support their preconditions.
- Update collection generator, CSV/preconditions, reports/docs only
where needed.
- Regenerate and validate:
  - static checks,
  - Postman v2.1 schema,
  - Newman compatibility tests.
- Verify no baseline reset-password test still depends on `/api/forgot-
password`.
```

- **AI Output:**

```
1. Files changed

- `Main_Report.md`
- `test-cases/a-forgot-password.csv`
- `postman/pool_a_fixtures.py` (new)
- `postman/seed_pool_a_fixtures.py` (new)
- `postman/generate_pool_a.py`
- `postman/pool-a-forgot-password.postman_collection.json`
- `postman/pool-a-conversion-report.md`
- `postman/pool-a-validation-results.json`
- `AI_Audit.md`

2. Fixture strategy

Start the SUT first because its startup drops/recreates `users`. The
idempotent Python seeder then opens `backend/database.sqlite` directly,
validates the real `users` schema, replaces only owned `poola-api-
%@example.test` rows, inserts dedicated per-case accounts, and verifies
every `reset_token` is SQLite TEXT. It does not import or modify SUT
code.

API-013 uses TEXT `012345`; API-009/API-044 store TEXT `"123456"` but
send JSON number `123456`; cross-account cases use distinct A/B accounts
and OTPs; replay cases seed fresh OTPs, consume them through a
successful setup reset, then retry the same OTP.

3. Cases that became executable

The 73 fixture-dependent baseline cases are now deterministically
executable: API-001, API-004–API-024, API-026–API-027, API-030–API-064,
API-066–API-071, API-073–API-074, API-076, and API-078–API-082. API-002,
API-003, API-028, and API-029 already required no database fixture.

4. Cases still blocked/manual and why

- API-025, API-065, API-072: blocked; the SUT has no OTP expiry column,
state, or check.
```

- API-075: blocked; the SUT has no rate limiter/abuse-control threshold.
- API-077: manual/blocked in Postman/Newman; sequential collection execution cannot create a true concurrent barrier.
- API-030 remains exploratory with no conformance oracle. API-068 requires authorized storage inspection. API-046, API-047, API-050, API-052, API-055, API-057, API-063, API-067, API-069–API-071, API-073–API-074, API-076, API-078, and API-081 retain partial/manual side-effect, persistence, or external oracles while their HTTP requests are executable.

5. Exact SUT → fixture seeding → Newman commands

```powershell
# PowerShell 1
Set-Location D:\GitHub\eshop-sut\backend
npm install
node server.js

# PowerShell 2, after the SUT reports it is running
Set-Location D:\GitHub\SoftwareTesting-HW06
C:\Users\xing0\AppData\Local\Python\bin\python.exe postman\seed_pool_a_fixtures.py --sut-dir D:\GitHub\eshop-sut
newman run postman\pool-a-forgot-password.postman_collection.json --env-var baseUrl=http://127.0.0.1:3000
```

6. Validation results

- Live SUT initialization and fixture seed: PASS; 78 deterministic accounts inserted and re-read with TEXT OTP storage.
- Static checks: PASS; 82 logical IDs, 88 requests, zero executable `/api/forgot-password` dependencies.
- Full Postman Collection v2.1 schema: PASS.
- Newman 6.2.2 compatibility: PASS; 8 representative runs, 13 reset requests, zero forgot-password requests.
- `git diff --check`: PASS (line-ending notices only).
- Collection SHA-256:

```
`C596D11862793ADAE08223DC94F903EF7E92D7D1DF85F9B9DB6625C9F6934B2F`.
- Full-suite SUT conformance execution was not performed.
```

- **File Modifications:**
  - `File:` `Main_Report.md`.
    - `Action:` edited.
    - `Deleted:` Obsolete Pool A CSV checksum and lack of deterministic fixture execution documentation.
    - `Replaced With:` Current CSV checksum and direct-SQLite fixture/unsupported-case policy.
    - `Appended:` None.
  - `File:` `test-cases/a-forgot-password.csv`.
    - `Action:` replaced content mechanically while preserving 82 reviewed IDs and nine fields.
    - `Deleted:` Shared accounts, forgot-password setup language, runtime OTP placeholders, and obsolete fixture assumptions for executable cases.
    - `Replaced With:` Dedicated deterministic account emails, exact OTP values, direct-SQLite preconditions, distinct cross-account fixtures, and replay setup details.
    - `Appended:` None.
  - `File:` `postman/pool_a_fixtures.py`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Shared deterministic fixture manifest, account mapping, OTP mapping, cross-account definitions, and blocked/replay classifications.
  - `File:` `postman/seed_pool_a_fixtures.py`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Idempotent direct-SQLite seeder with running-SUT check, schema validation, scoped replacement, exact TEXT OTP verification, and absence fixture

handling.

- `File:` `postman/generate_pool_a.py`.
  - `Action:` replaced content.
  - `Deleted:` Forgot-password prerequest calls, generated OTP extraction/regeneration, runtime readiness variables/guards, and old compatibility mocks.
  - `Replaced With:` Literal CSV-aligned deterministic fixture inputs, replay-only setup resets, explicit unsupported-case blocks, executable-dependency static checks, full-schema validation, and deterministic Newman compatibility runs.
  - `Appended:` None.
- `File:` `postman/pool-a-forgot-password.postman_collection.json`.
  - `Action:` replaced content.
  - `Deleted:` Runtime forgot-password setup scripts and OTP/readiness collection variables.
  - `Replaced With:` Regenerated 82-ID/88-request collection using deterministic accounts/OTPs, replay setup resets, and explicit blocked/manual gates.
  - `Appended:` None.
- `File:` `postman/pool-a-conversion-report.md`.
  - `Action:` replaced content.
  - `Deleted:` Obsolete forgot-password fixture strategy and validation evidence.
  - `Replaced With:` Direct-SQLite strategy, per-case classifications, exact execution commands, unsupported-case reasons, and current validation results.
  - `Appended:` None.
- `File:` `postman/pool-a-validation-results.json`.
  - `Action:` replaced content.
  - `Deleted:` Validation state for the superseded collection.
  - `Replaced With:` PASS evidence for 82 IDs, 88 requests, full v2.1 schema, Newman 6.2.2, and zero forgot-password requests.
  - `Appended:` None.
- `File:` `AI_Audit.md`.
  - `Action:` appended.

        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Artifact 12 containing the verbatim interaction, file modifications, and review TODOs.

- **Verdict:** INCOMPLETE
- **Reasoning:**
    - BLOCKED cases currently throw errors, which can make Newman report them as failures.
    - Some CSV descriptions still mention old `/api/forgot-password` or outdated OTP values.
    - The `X-Student-Id` console log for screenshot evidence was removed.
    - Static validation is weaker than before.

- **Student Fixes:**
    - Replace `throw new Error(...)` for BLOCKED cases with `pm.execution.skipRequest()`.
    - Restore clear `X-Student-Id` console logging for execution evidence.
    - Guide the agent to update stale CSV preconditions and expected results to match the new SQLite fixture strategy.

Artifact 13

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 10:47:47 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
    - `postman-test-generator`
    - `ai-audit`

- **Prompt:**

```
Fix the remaining issues from the direct-SQLite fixture migration.

1. Update stale CSV/report wording.
   - Remove references to `/api/forgot-password` generating fixture OTPs
where fixtures are now seeded directly in SQLite.
   - Replace outdated OTP values such as `123456` with the actual
deterministic token for each case.
   - Keep each test's original intent unchanged.
   - Check especially API-027, 030, 031, 033–035, 045–047, 050, 052,
055, 057, 063, 064, 067, 068, 070, and 071.
2. Strengthen static validation without restoring the old oversized
validator.\
   Add independent checks for:
   - fixture seeder account/token values matching CSV test
preconditions/data;
   - every BLOCKED case using `pm.execution.skipRequest()` rather than
throwing an error;
   - `X-Student-Id` injection remaining present;
   - important structured CSV → Postman mappings being independently
validated.
3. Regenerate the collection and rerun:
   - Python/static validation,
   - Postman v2.1 schema validation,
   - Newman compatibility tests.

Do not redesign the SQLite fixture architecture.

At the end, report only files changed, fixes made, and validation
results.
```

- **AI Output:**

```markdown
## Files changed

- `test-cases/a-forgot-password.csv`
- `postman/generate_pool_a.py`
- `postman/pool-a-forgot-password.postman_collection.json`
- `postman/pool-a-conversion-report.md`
- `postman/pool-a-validation-results.json`
- `Main_Report.md`
- `AI_Audit.md`

## Fixes made

- Replaced stale generated/issued OTP wording and obsolete `123456` values with each case’s deterministic SQLite token.
- Preserved intentional `123456` uses in API-009 and API-032.
- Added independent checks for:
  - 73 CSV fixture rows against seeder account/token definitions.
  - 36 structured CSV-to-Postman mappings.
  - All blocked cases using `pm.execution.skipRequest()` without throwing.
  - Collection-level `X-Student-Id` injection.
- Added Newman coverage for blocked-request skipping and student-ID header capture.
- Updated report validation evidence and CSV checksum.

## Validation results

- Python compilation: **PASS**
- Static validation: **PASS** — 82 IDs, 88 requests, 73 fixture rows, 36 independent structured mappings.
- Blocked-case skip validation: **PASS**
- `X-Student-Id` validation: **PASS**
- Full Postman v2.1 schema: **PASS**
- Newman 6.2.2: **PASS** — 9 runs, 13 transmitted reset requests, one blocked request skipped, zero forgot-password requests.
- `git diff --check`: **PASS** (line-ending notices only).
```

```
- Collection SHA-256:
`D63A790201DC1D0AF64C43D2983DF6CDAF01537F7A96434D23A5D3E66118A010`
```

- **File Modifications:**
  - `File:` `test-cases/a-forgot-password.csv`.
    - `Action:` edited.
    - `Deleted:` Stale generated/issued fixture wording and obsolete `123456` values in deterministic fixture preconditions, results, and notes.
    - `Replaced With:` Exact per-case SQLite tokens and direct-seeding wording while preserving each reviewed test intent.
    - `Appended:` None.
  - `File:` `postman/generate_pool_a.py`.
    - `Action:` edited.
    - `Deleted:` Weak blocked-case validation and generator-coupled-only CSV mapping checks.
    - `Replaced With:` Compact independent fixture-manifest/CSV checks, 36 independently reconstructed structured mappings, skipRequest-without-throw enforcement, X-Student-Id checks, and Newman blocked/header compatibility coverage.
    - `Appended:` None.
  - `File:` `postman/pool-a-forgot-password.postman_collection.json`.
    - `Action:` replaced content.
    - `Deleted:` Generated descriptions containing stale CSV fixture wording.
    - `Replaced With:` Regenerated descriptions with exact deterministic tokens and direct-SQLite wording; blocked requests use `pm.execution.skipRequest()`.
    - `Appended:` None.
  - `File:` `postman/pool-a-conversion-report.md`.
    - `Action:` replaced content.
    - `Deleted:` Prior validation summary and stale collection hash.
    - `Replaced With:` Current fixture/static/schema/Newman evidence, blocked skip behavior, and current collection hash.
    - `Appended:` None.

- `File:` `postman/pool-a-validation-results.json`.
  - `Action:` replaced content.
  - `Deleted:` Validation evidence for the superseded collection.
  - `Replaced With:` PASS evidence for 73 fixture rows, 36 independent structured mappings, blocked skipRequest cases, X-Student-Id, full schema, and nine Newman runs.
  - `Appended:` None.
- `File:` `Main_Report.md`.
  - `Action:` edited.
  - `Deleted:` Superseded CSV SHA-256.
  - `Replaced With:` Current CSV SHA-256 `DB9858994918754E48299DA6EF52455BAACE66419E43A36488BA613B64981AEA`.
  - `Appended:` None.
- `File:` `AI_Audit.md`.
  - `Action:` appended.
  - `Deleted:` None.
  - `Replaced With:` None.
  - `Appended:` Audit artifacts for the completed interactions.

- **Verdict:** VALID
- **Reasoning:** The agent correctly resolved SQLite defects.
- **Student Fixes:** None

Artifact 14

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 13:49:58 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
  - `skill-creator`
  - `ai-audit`
- **Prompt:**

```
Use [$skill-creator](C:\Users\xing0\\.codex\skills\\.system\skill-creator\SKILL.md) to create the `newman-result-analyzer` Agent Skill based on [newman-result-analyzer.md](skills-design/newman-result-analyzer/newman-result-analyzer.md).
```

- **AI Output:**

```
Created the `newman-result-analyzer` Agent Skill:

- [SKILL.md](D:\GitHub\SoftwareTesting-HW06\.agents\skills\newman-result-analyzer\SKILL.md)
- [openai.yaml](D:\GitHub\SoftwareTesting-HW06\.agents\skills\newman-result-analyzer\agents\openai.yaml)

It covers logical test grouping, multi-request flows, status precedence, Newman error classification, unexecuted/blocked cases, manual oracles, summary reconciliation, and report output rules—without inferring SUT bugs.

Validation passed with `skill-creator`'s `quick_validate.py`.
```

- **File Modifications:**
  - `File:` `.agents/skills/newman-result-analyzer/SKILL.md`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Complete Newman result-analysis instructions covering input validation, logical test identification, multi-request flows, execution-status classification, manual-oracle handling, summary reconciliation, and structured report output.
  - `File:` `.agents/skills/newman-result-analyzer/agents/openai.yaml`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.

        - `Appended:` UI metadata defining the display name, short description, and default `$newman-result-analyzer` prompt.

- **Verdict:** INCOMPLETE
- **Reasoning:** The original design defines the precedence: `REQUEST_ERROR`, `RUNTIME_ERROR`, `FAIL_ASSERTION`, `PASS`, while the implemented skill currently defines `RUNTIME_ERROR`, `REQUEST_ERROR`, `FAIL_ASSERTION`, `PASS`. The implementation should preserve the designed precedence.
- **Student Fixes:** Fix the precedence to match the design.

Artifact 15

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 14:15:57 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
    - `newman-result-analyzer`
    - `ai-audit`
- **Prompt:**

```
Use [$newman-result-analyzer](D:\GitHub\SoftwareTesting-
HW06\\.agents\skills\newman-result-analyzer\SKILL.md) to analyze the
final Pool A Newman result:

- Input: [pool-a.json](reports/pool-a/pool-a.json)&#x20;
- Output: [Main\_Report.md](Main_Report.md) at secton `### A.8. Newman
Execution Analysis`

Write the normalized execution analysis to the output file.
```

- **AI Output:**

```
I’m using the named `newman-result-analyzer` skill. I’ll first read its
full workflow, inspect the Newman JSON and the target section, then
write and verify the normalized analysis.

The skill requires evidence-based classification at the logical test-
case level, preserving assertion/runtime/request distinctions and any
unexecuted collection items. I’m now reconciling the embedded
collection, detailed executions, Newman stats, and failures before
editing the report.

Reconciliation is complete: the artifact represents 82 logical cases.
Outcomes are 55 `PASS`, 22 `FAIL_ASSERTION`, and 5 explicitly
`BLOCKED_NOT_EXECUTED`; there are no request or script errors. I also
found three pre-request helper calls whose main execution rows are
duplicated in the JSON, so I’m collapsing only those identical
duplicates while retaining Newman’s 85-request run total and documenting
the reconciliation.

The normalized report is now written into Section A.8. I’m doing a final
integrity check on table row counts, status totals, required fields,
section boundaries, and the exact file diff before handing it back.

Updated [Main_Report.md](D:/GitHub/SoftwareTesting-
HW06/Main_Report.md:155) Section A.8 with the normalized Newman
analysis.

Results:

- `PASS`: 55
- `FAIL_ASSERTION`: 22
- `BLOCKED_NOT_EXECUTED`: 5
- 82 logical test cases documented
- 23 explicit manual-oracle cases
- Request/statistics reconciliation included
- No request or script errors found
```

- **File Modifications:**

    - `File:` `Main_Report.md`.
        - `Action:` appended content to the existing `### A.8. Newman Execution Analysis` section.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Lines 157–271: normalized run summary, one outcome record for each of 82 logical test cases, explicit manual-oracle requirements, and coverage/reconciliation notes. The analysis records 55 `PASS`, 22 `FAIL_ASSERTION`, 5 `BLOCKED_NOT_EXECUTED`, 82 assertions (59 passed and 23 failed), 85 total requests, and no request or script errors.
    - `File:` `tmp/analyze-pool-a.js`.
        - `Action:` created as a temporary analysis helper, then deleted after verification.
        - `Deleted:` The temporary Newman parsing, logical-case grouping, duplicate-correlation, classification, and Markdown-generation helper after its output was written and verified.
        - `Replaced With:` None.
        - `Appended:` Temporary helper code used to inspect the embedded collection, executions, failures, manual-oracle metadata, and run-stat reconciliation.
- **Verdict:** INCOMPLETE
- **Reasoning:** The Newman analysis correctly recorded the automated execution results, but API-068 and API-077 still showed their Newman-only states even though their required manual executions had been completed.
- **Student Fixes:** Replaced the final results for API-068 and API-077 with the completed manual outcomes.
    - API-068 is `FAIL` because database inspection confirmed plaintext password storage.
    - API-077 is `PASS` because the separate synchronized concurrency test confirmed exactly one successful reset. Updated the summary and reconciliation notes accordingly.

Artifact 16

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 14:31:22 GMT+7
- **Model Used:** GPT-5.6 Sol

- **Skill Used:**
  - `append-bug`
  - `ai-audit`
- **Prompt:**

```
Use [$append-bug](D:\GitHub\SoftwareTesting-HW06\\.agents\skills\append-bug\SKILL.md) for Pool A / `FR-03`.

- Test Case document: [Main\_Report.md](Main_Report.md)&#x20;
- Bug report document: `Bug_Report.md`
- Target Feature ID: `FR-03`

Use the final reviewed outcomes in Section A.8, including completed manual verification results.
```

- **AI Output:**

I’m using the requested `append-bug` skill. I’ll first read its instructions, then inspect Section A.8 for Pool A / `FR-03` and update `Bug_Report.md` with the required audit trail.

The skill requires grouping failures by shared cause, creating one issue file per group, updating the existing FR-03 section, and then recording the work through the audit skill. I’m now extracting the reviewed A.8 outcomes and the repository’s existing report conventions before editing.

Two supplied paths are absent as written: there is no root `Bug_Report.md`, and no root `issues/` directory. I’ll check the repository layout for the intended existing report and issue location; if none exists, the skill’s own rules permit creating the missing report structure while preserving the project’s conventions.

The reviewed failures resolve into four distinct defect groups: reset-token type coercion, missing `newPassword` validation (including OTP consumption after invalid input), unsafe `text/plain` handling, and plaintext password persistence. The completed concurrency check (`API-077`) is a final pass, so it is not a bug; the completed storage check (`API-068`) is a final fail and will be included.

The implementation check confirms the grouping: the endpoint destructures the body and writes `newPassword` directly with no validation or hashing, while SQLite comparison allows the numeric token representation. Because the requested bug report does not exist, I’m creating it with only the required FR-03 feature section and placing the four issue files in a new root `issues/` folder.

Constraint check — I created four new GitHub issue Markdown files. Their exact contents are:

`issues/bug_fr03-reset-token-type-validation.md`

````markdown
---
````

`````
labels: bug, fr-03, medium
---

## Bug: Reset token accepts non-string JSON numbers

**Impacted Test Case ID(s):** API-009, API-044

### Description
The reset-password endpoint does not enforce the documented string type
for `resetToken`. A valid issued token sent as a JSON number is matched
and accepted, allowing the password reset to succeed instead of
rejecting the invalid representation.

### Steps to Reproduce
1. Seed a registered account with a valid, unused six-digit reset token
stored as text.
2. Send `POST /api/reset-password` with the registered email, a
conforming `newPassword`, and the issued `resetToken` represented as a
JSON number rather than a string.
3. Observe the response.

### Expected Result
The request is rejected with `400 Bad Request` because `resetToken` must
be a JSON string.

### Actual Result
The endpoint returns `200 OK` and performs the password reset with the
numeric token representation.

### Environment
- **Browser/OS:** Newman CLI on Windows (browser not applicable)
- **Version:** EShop requirements v2.0; Newman 6.2.2
````

`issues/bug_fr03-new-password-validation-bypass.md`

````markdown
`````

```markdown
---
labels: bug, fr-03, high
---

## Bug: Invalid new passwords bypass server-side validation

**Impacted Test Case ID(s):** API-016, API-017, API-018, API-020, API-021, API-022, API-023, API-024, API-046, API-047, API-050, API-052, API-055, API-057, API-060, API-061, API-062, API-067, API-080

### Description
The reset-password endpoint writes `newPassword` without validating that it is present, is a string, and satisfies the required minimum length and character classes. Invalid requests succeed and consume the one-time reset token; the object-valued case proves this by causing the subsequent valid retry to be rejected.

### Steps to Reproduce
1. Seed a registered account with a valid, unused six-digit reset token.
2. Send `POST /api/reset-password` using the valid email and token while omitting `newPassword`, supplying `null`, a number, an object, an empty string, or a password that violates one or more strength rules.
3. Observe the response and, where applicable, retry with a valid string password using the same token.

### Expected Result
Every invalid `newPassword` request is rejected with `400 Bad Request`; the account password remains unchanged and the reset token remains usable for a corrected request.

### Actual Result
The endpoint returns `200 OK` for the invalid values and updates the account while consuming the reset token. For API-080, the object-valued password receives `200 OK` and the subsequent valid retry receives `400 Bad Request` because the token was already consumed.

### Environment
```

`````
- **Browser/OS:** Newman CLI on Windows (browser not applicable)
- **Version:** EShop requirements v2.0; Newman 6.2.2
````

`issues/bug_fr03-unsupported-content-type-server-error.md`

````markdown
---
labels: bug, fr-03, medium
---

## Bug: Unsupported content type causes an internal server error

**Impacted Test Case ID(s):** API-078

### Description
The reset-password endpoint does not safely reject a JSON-shaped request
sent as `text/plain`. It attempts to destructure an unparsed body and
produces a server error instead of a client-error response.

### Steps to Reproduce
1. Seed a registered account with a valid, unused six-digit reset token.
2. Send otherwise valid JSON text to `POST /api/reset-password` with
`Content-Type: text/plain`.
3. Retry with the same email, token, and password using `Content-Type:
application/json`.

### Expected Result
The `text/plain` request receives a safe `4xx` response without
consuming the token, and the corrected JSON retry succeeds with `200
OK`.

### Actual Result
The `text/plain` request returns `500 Internal Server Error`. The
corrected JSON retry returns `200 OK`, confirming that the failed
request did not consume the token.
`````

`````
### Environment
- **Browser/OS:** Newman CLI on Windows (browser not applicable)
- **Version:** EShop requirements v2.0; Newman 6.2.2
````

`issues/bug_fr03-plaintext-password-storage.md`

````markdown
---
labels: bug, fr-03, critical
---

## Bug: Reset passwords are stored in plaintext

**Impacted Test Case ID(s):** API-068

### Description
The reset-password endpoint persists the submitted password value
directly rather than storing a secure password hash. This was confirmed
through the completed white-box database inspection required by the
reviewed test case.

### Steps to Reproduce
1. Seed a registered account with a valid, unused reset token and note
its existing password value.
2. Send a valid `POST /api/reset-password` request with a known
conforming new password.
3. After the request succeeds, inspect the account's password field
directly in the SQLite database.

### Expected Result
The new password is stored using a secure password-hashing mechanism and
the submitted plaintext value is not present in password storage.

### Actual Result
The request returns `200 OK`, and direct database inspection shows the
exact submitted reset password stored in plaintext.
`````

`````
### Environment
- **Browser/OS:** Newman CLI and direct SQLite inspection on Windows
(browser not applicable)
- **Version:** EShop requirements v2.0; Newman 6.2.2
````

Constraint check – the supplied `Bug_Report.md` path was missing, so I
created it and added only the required FR-03 feature section. The exact
text added is:

````markdown
# Bug Report

## Feature FR-03: Forgot Password / Password Reset

### Reset token accepts non-string JSON numbers
**Impacted Test Cases:** API-009, API-044

The endpoint accepts an issued `resetToken` represented as a JSON number
and completes the reset, instead of enforcing the documented string
representation and returning `400 Bad Request`.

### Invalid new passwords bypass server-side validation
**Impacted Test Cases:** API-016, API-017, API-018, API-020, API-021,
API-022, API-023, API-024, API-046, API-047, API-050, API-052, API-055,
API-057, API-060, API-061, API-062, API-067, API-080

The endpoint accepts missing, null, non-string, empty, and weak
`newPassword` values, updates the account, and consumes the reset token.
API-080 further shows that an object-valued password is accepted and
prevents the subsequent corrected retry because the token has already
been consumed.

### Unsupported content type causes an internal server error
**Impacted Test Cases:** API-078
`````

~~~
Sending otherwise valid JSON as `text/plain` causes `500 Internal Server Error` instead of a safe `4xx` rejection. The same token remains usable when the request is retried correctly as `application/json`.

### Reset passwords are stored in plaintext
**Impacted Test Cases:** API-068

Completed manual database verification confirmed that a successful reset stores the exact submitted password in plaintext instead of a secure hash.
````


The `append-bug` skill now requires an audit entry. Its referenced `$ai-export-audit` name is not installed, so I’m using the available equivalent `ai-audit` skill to append the required interaction and file-modification record to `AI_Audit.md`.

The `append-bug` workflow explicitly calls for a separate audit agent after the edits and constraint checks. I’m handing the completed file-change record to that audit agent now; it will append the next numbered artifact without altering the bug files.

Validation now confirms all 23 final failures in Section A.8—including the manually failed `API-068`—appear exactly once across the four bug groups. No passed or blocked case was reported as a defect.
~~~

- **File Modifications:**
  - `File:` `Bug_Report.md`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Complete FR-03 bug-report section consolidating 23 final failed cases into four root-cause groups: reset-token type coercion, invalid `newPassword` acceptance and token consumption, unsafe `text/plain` handling, and plaintext password storage.
  - `File:` `issues/bug_fr03-reset-token-type-validation.md`.

        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Complete medium-severity GitHub issue for API-009 and API-044, including description, reproduction steps, expected and actual results, environment, and labels.
    - `File:` `issues/bug_fr03-new-password-validation-bypass.md`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Complete high-severity GitHub issue for API-016, API-017, API-018, API-020, API-021, API-022, API-023, API-024, API-046, API-047, API-050, API-052, API-055, API-057, API-060, API-061, API-062, API-067, and API-080, including invalid-value coverage and reset-token consumption evidence.
    - `File:` `issues/bug_fr03-unsupported-content-type-server-error.md`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Complete medium-severity GitHub issue for API-078, documenting the `text/plain` request's `500 Internal Server Error`, the expected safe `4xx`, and successful corrected retry.
    - `File:` `issues/bug_fr03-plaintext-password-storage.md`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Complete critical-severity GitHub issue for manually verified API-068, documenting direct SQLite evidence that the submitted reset password is stored in plaintext.
- **Verdict:** INCOMPLETE
- **Reasoning:** The bug grouping is appropriate, but the new-password validation issue overstates the execution evidence. The Newman results show that the invalid

`newPassword` cases returned `200 OK`, but only API-080 directly demonstrates reset-token consumption through the failed corrected retry.

- **Student Fixes:** Revised the new-password validation bug summary, description, and actual result to report only confirmed behavior. Kept token-consumption evidence specific to API-080.

Artifact 17

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 15:12:56 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
  - `api-test-generator`
  - `contract-test-generator`
  - `domain-test-generator`
  - `state-transition-test-generator`
  - `security-test-generator`
  - `ai-audit`
- **Prompt:**

```
Use [$api-test-generator](D:\GitHub\SoftwareTesting-
HW06\\.agents\skills\api-test-generator\SKILL.md) for Pool B: `POST
/api/apply-coupon` (FR-09).
Write the reviewed analysis reports under `review/pool-b/reports/` and
the final candidate test cases to `test-cases/b-discount-coupons.csv`.
Use the API specification and system requirements as authoritative
sources.
```

- **AI Output:**

I’m using the `api-test-generator` skill because this request needs coordinated contract, domain, state-transition, and security analysis. I’ll follow its required review checkpoints and write only to the requested Pool B report and CSV paths.

The workflow’s first gate is analysis-only. I’m extracting one shared endpoint context from the authoritative specification and requirements now; four specialists will then write separate review reports in parallel. No test-case CSV will be generated until those reports are explicitly approved.

I’ve confirmed the endpoint is uniquely identified and the source contract is intentionally sparse: it documents the JSON fields and output shape but no HTTP status/error schemas. The shared context will preserve those gaps instead of inventing behavior, while FR-09 supplies the coupon rules and authentication requirement.

The shared context and orchestration status are now written. Per the skill, I’m starting all four Phase-1 specialists concurrently; each owns only its category report and cannot touch the shared context, status file, final CSV, or another specialist’s files.

Three specialists are active, which fills this runtime’s four-agent concurrency limit including me. The security specialist is queued to start as soon as one slot frees; this affects timing only, not the required report boundaries or review gate.

All four Phase-1 reports are complete. State testing is applicable because coupon eligibility depends on stored active/expiry/usage states, but the report correctly avoids assuming this endpoint consumes a use. I’m validating report structure, traceability, and file ownership before updating the review status.

Phase 1 analysis is complete and awaiting human review:

- [Contract report](D:\GitHub\SoftwareTesting-HW06\review\pool-b\reports\contract-report.md)

```
- [Domain report](D:\GitHub\SoftwareTesting-HW06\review\pool-
b\reports\domain-report.md)
- [State report](D:\GitHub\SoftwareTesting-HW06\review\pool-
b\reports\state-report.md)
- [Security report](D:\GitHub\SoftwareTesting-HW06\review\pool-
b\reports\security-report.md)
- [Shared API context](D:\GitHub\SoftwareTesting-HW06\review\pool-
b\shared-api-context.md)
- [Orchestration status](D:\GitHub\SoftwareTesting-HW06\review\pool-
b\orchestration-status.md)

Coverage includes 12 contract rules, 37 domain partitions, 7 boundaries,
5 state evaluations, and 7 security scenarios. State testing is
applicable, although no coupon-consumption side effect was inferred.

The reports remain `PENDING`. Per the required review gate, `test-
cases/b-discount-coupons.csv` has not been generated. Please review and
explicitly approve these exact report versions—or provide revision notes
—so I can proceed with Phase 2.
```

- **File Modifications:**
  - `File:` `review/pool-b/shared-api-context.md`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Complete normalized context for `POST /api/apply-coupon`, including authoritative source hashes and line references, request and response inventories, FR-09 coupon rules, SEC-01–SEC-07 applicability evidence, state cues, baseline request, and explicit specification gaps.
  - `File:` `review/pool-b/orchestration-status.md`.
    - `Action:` created, then edited after all Phase-1 reports completed.
    - `Deleted:` Initial `ANALYSIS_IN_PROGRESS` status and not-yet-produced category rows.
    - `Replaced With:` `AWAITING_HUMAN_REVIEW` status, four completed Phase-1 category rows, state applicability, report-version provenance, and the explicit

generation gate.
    - `Appended:` None.
  - `File:` `review/pool-b/reports/contract-report.md`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Complete Phase-1 CONTRACT analysis with request/response inventories, CR-001–CR-012, gaps, assumptions, and a PENDING human-review block.
  - `File:` `review/pool-b/reports/domain-report.md`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Complete Phase-1 DOMAIN analysis with baseline, parameter inventory, DP-001–DP-037, DB-001–DB-007, cross-parameter constraints, gaps, and a PENDING human-review block.
  - `File:` `review/pool-b/reports/state-report.md`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Complete Phase-1 STATE analysis marking state applicability as APPLICABLE, defining stored coupon and per-user states, modeling TR-001–TR-005, preserving the absence of documented endpoint mutations, and including a PENDING review block.
  - `File:` `review/pool-b/reports/security-report.md`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Complete Phase-1 SECURITY analysis with the SEC-01–SEC-07 applicability matrix, SS-001–SS-007, security gaps, and a PENDING human-review block.

- **Verdict:** INCOMPLETE
- **Reasoning:**
  - CONTRACT wrongly treats sample coupons as already active.
  - DOMAIN uses `unsupported syntax` without a documented syntax rule.
  - STATE should be `NOT_APPLICABLE`.
  - SECURITY misses SEC-05 coverage for `user_id`.

- **Student Fixes:**
  - Manually fixed the CONTRACT fixture wording.
  - Manually changed `unsupported syntax` to `unusual/arbitrary string`.
  - Asked the agent to change STATE to `NOT_APPLICABLE`.
  - Asked the agent to add SEC-05 coverage for `user_id`.
  - Asked the agent to keep cross-report summaries consistent.

Artifact 18

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 15:26:08 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
  - `api-test-generator`
  - `contract-test-generator`
  - `domain-test-generator`
  - `state-transition-test-generator`
  - `security-test-generator`
  - `ai-audit`

- **Prompt:**

```
Revise Pool B Phase-1 analysis based on the review.

- Replace the current transition model with a concise justification that
`POST /api/apply-coupon` has state-dependent preconditions but no
documented endpoint-driven state transition. Keep active/inactive,
expiry, and usage-limit coverage under DOMAIN.
- Add SEC-05 coverage for client-controlled `user_id` using a controlled
inert SQL/metacharacter input. Do not invent an HTTP status oracle.
- Preserve the rest of the approved analysis and update cross-report
summaries for consistency.

Do not generate Phase-2 test cases yet.
```

- **AI Output:**

I’m using the API test-generation workflow again for this Phase-1 revision. The review keeps CONTRACT and DOMAIN analysis substantively approved, while STATE must be recast as non-applicable to transition testing and SECURITY must gain controlled `user_id` coverage. I’ll keep the reports at the human-review gate and will not generate candidate JSON or the final CSV.

The review changes the classification, not the underlying FR-09 facts: active/inactive, expiry, and usage count remain testable partitions and boundaries, but they are DOMAIN setup conditions because this operation documents calculation only—no source state, trigger-driven mutation, or destination state. I’m updating that shared interpretation before the specialists revise their reports.

Three revision specialists are active, filling the runtime’s concurrency limit with me included. SECURITY is queued next; its required addition will use a controlled inert metacharacter value in client-supplied `user_id`, asserting only query-semantic safety—not a fabricated status code or error body.

The revisions are in place: STATE is now a minimal `NOT_APPLICABLE` record, DOMAIN retains all 37 partitions and 7 boundaries, and SECURITY adds `SS-008` for inert `user_id` metacharacter handling. I’m doing the final cross-report consistency check and updating the parent-owned status; Phase 2 remains untouched.

Pool B Phase-1 analysis has been revised:

- [State report](D:\GitHub\SoftwareTesting-HW06\review\pool-b\reports\state-report.md) now records `State Applicability: NOT_APPLICABLE`, with no transition model or review requirement.
- [Domain report](D:\GitHub\SoftwareTesting-HW06\review\pool-b\reports\domain-report.md) retains all 37 partitions and 7 boundaries, including active/inactive, expiry, and usage-limit coverage.
- [Security report](D:\GitHub\SoftwareTesting-HW06\review\pool-b\reports\security-report.md) adds `SS-008` for controlled inert `user_id` SQL/metacharacter input without inventing an HTTP response

```
oracle.
- [Contract report](D:\GitHub\SoftwareTesting-HW06\review\pool-
b\reports\contract-report.md), [shared context]
(D:\GitHub\SoftwareTesting-HW06\review\pool-b\shared-api-context.md),
and [orchestration status](D:\GitHub\SoftwareTesting-HW06\review\pool-
b\orchestration-status.md) were updated for consistency.

CONTRACT, DOMAIN, and SECURITY remain `PENDING` for the revised
versions. No candidate fragments or Phase-2 CSV were generated.
```

- **File Modifications:**
    - `File:` `review/pool-b/shared-api-context.md`.
        - `Action:` edited.
        - `Deleted:` Prior classification of active/inactive, expiry, and per-user usage conditions as applicable STATE dimensions for transition-specialist review.
        - `Replaced With:` State-dependent-precondition classification; exact `State Applicability: NOT_APPLICABLE` basis; DOMAIN ownership of active/inactive, expiry, and usage-limit coverage; and SEC-05 treatment of both `code` and client-controlled `user_id` as controlled inert database-query inputs without an HTTP status oracle.
        - `Appended:` None.
    - `File:` `review/pool-b/reports/contract-report.md`.
        - `Action:` edited.
        - `Deleted:` Prior v1 review metadata and the absence of an explicit cross-category disposition.
        - `Replaced With:` Preserved CR-001–CR-012 and substantive contract analysis plus a concise cross-category disposition assigning state-dependent eligibility coverage to DOMAIN, transition testing to `NOT_APPLICABLE`, and SEC-05 `code` / `user_id` coverage to SECURITY; version advanced to `contract-report-v2 — 2026-08-22` with `PENDING` review status.
        - `Appended:` None.
    - `File:` `review/pool-b/reports/domain-report.md`.
        - `Action:` edited.

    - `Deleted:` Prior wording that did not explicitly distinguish state-dependent eligibility preconditions from endpoint-driven transitions and prior draft review metadata.
    - `Replaced With:` Preserved DP-001–DP-037 and DB-001–DB-007 plus explicit DOMAIN ownership of active/inactive, expiry, and usage-limit partitions/boundaries, `STATE NOT_APPLICABLE` cross-report wording, revised DC-007, and version `POOL-B-DOMAIN-v2` with `PENDING` review status.
    - `Appended:` None.
  - `File:` `review/pool-b/reports/state-report.md`.
    - `Action:` replaced content.
    - `Deleted:` The prior APPLICABLE state model, state definitions, guards, TR-001–TR-005 transition table, invalid-transition discussion, gaps list, and PENDING review block.
    - `Replaced With:` Concise `STATE-PHASE1-v2` applicability record marking `State Applicability: NOT_APPLICABLE` and `Review Required: NO`, with exact source identity and justification that C1, C2, and C5 are state-dependent preconditions but the endpoint documents no state mutation or destination; active/inactive, expiry, and usage-limit coverage is routed to DOMAIN.
    - `Appended:` None.
  - `File:` `review/pool-b/reports/security-report.md`.
    - `Action:` edited.
    - `Deleted:` Prior SEC-05 coverage summary limited to controlled `code` inputs and v1 review metadata.
    - `Replaced With:` Existing SS-001–SS-007 preserved, new SS-008 for controlled inert SQL/metacharacter input in client-controlled `user_id`, updated SEC-05 coverage/gaps and cross-category consistency, and `security-report-v2` with `PENDING` review status. SS-008 prescribes no HTTP status, error schema, or message.
    - `Appended:` None.
  - `File:` `review/pool-b/orchestration-status.md`.
    - `Action:` edited.
    - `Deleted:` Prior STATE APPLICABLE/PENDING disposition, v1 category versions, and the requirement that all four categories receive approval.

        - `Replaced With:` `AWAITING_HUMAN_REVIEW` revision-complete status; v2 provenance; STATE `NOT_APPLICABLE` with no review required; DOMAIN ownership of state-dependent eligibility coverage; SECURITY SS-008 summary; and a generation gate requiring revised CONTRACT, DOMAIN, and SECURITY approval only.
        - `Appended:` None.

- **Verdict:** VALID
- **Reasoning:** The revision version of these documents look good and ready to be used in next step.
- **Student Fixes:** None

Artifact 19

- **User:** Vương Ngữ Tín Thành
- **Date and Time:** 2026-08-22 15:40:33 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
    - `api-test-generator`
    - `contract-test-generator`
    - `domain-test-generator`
    - `state-transition-test-generator`
    - `security-test-generator`
    - `ai-audit`
- **Prompt:**

```
Continue to phase 2
```

- **AI Output:**

Your instruction explicitly approves the current revised Phase-1 reports for generation. I’m entering Phase 2 with CONTRACT v2, DOMAIN v2, SECURITY v2, and the STATE `NOT_APPLICABLE` record; specialists will generate category-owned JSON fragments, then I’ll validate coverage, deduplicate within categories, and write the requested CSV only if the suite passes.

Phase 2 is now recorded as in progress. The state worker will materialize the required empty `[]` fragment from the preserved `NOT_APPLICABLE` basis; the other specialists will generate only from their reviewed rule, partition/boundary, and security-scenario models.

Generation produced 81 applicable candidates before parent validation: 15 CONTRACT, 58 DOMAIN, 0 STATE, and 8 SECURITY. The count exceeds the 35-case target because the reviewed DOMAIN model contains 37 partitions plus 21 atomic boundary representatives; I’m now checking whether any same-category cases are semantically duplicate while preserving every reviewed ID and boundary value.

One genuine same-category duplicate group was found: seven DOMAIN entries used the identical baseline preconditions, request, and expected result while separately labeling DP-001, DP-005, DP-008, DP-012, DP-019, DP-029, and DP-032. I’m consolidating them into one retained baseline case with all seven reviewed IDs and specification bases preserved, removing six duplicates without losing coverage.

After consolidation, the validated suite contains 75 distinct candidates: 15 CONTRACT, 52 DOMAIN, 0 STATE, and 8 SECURITY. All CR-001–CR-012, DP-001–DP-037, DB-001–DB-007, and SS-001–SS-008 remain covered, with no duplicate provisional IDs or remaining same-category semantic duplicates. I’m assigning final sequential IDs and writing the CSV now.

Phase 2 is complete.

Created [b-discount-coupons.csv](D:\GitHub\SoftwareTesting-HW06\test-cases\b-discount-coupons.csv) with 75 validated candidates:

```
- CONTRACT: 15
- DOMAIN: 52
- SECURITY: 8
- STATE: 0 (`NOT_APPLICABLE`)

All reviewed CR, DP, DB, and SS identifiers are covered. Six duplicate
DOMAIN baseline cases were consolidated without losing traceability. IDs
are unique and sequential from `API-001` to `API-075`.

Updated [orchestration-status.md](D:\GitHub\SoftwareTesting-
HW06\review\pool-b\orchestration-status.md) with coverage,
deduplication, assumptions, and validation results. No tests were
executed.
```

- **File Modifications:**
  - `File:` `review/pool-b/orchestration-status.md`.
    - `Action:` edited.
    - `Deleted:` The Phase-1 `AWAITING_HUMAN_REVIEW` workflow state and the pre-generation gate wording.
    - `Replaced With:` Phase-2 authorization provenance followed by final `COMPLETE` status, 75-case category counts, reviewed trace-ID coverage, STATE empty-output handling, six-case DOMAIN deduplication summary, validation results, preserved specification gaps, and confirmation that no tests were executed.
    - `Appended:` None.
  - `File:` `review/pool-b/candidates/contract-tests.json`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` JSON array containing 15 CONTRACT candidates covering CR-001–CR-012 with exact specification bases and explicit assumptions for undocumented behavior.
  - `File:` `review/pool-b/candidates/domain-tests.json`.
    - `Action:` created, then edited by same-category semantic deduplication.

        - `Deleted:` Six duplicate baseline records with provisional IDs `DOMAIN-005`, `DOMAIN-008`, `DOMAIN-012`, `DOMAIN-019`, `DOMAIN-029`, and `DOMAIN-032` after confirming identical preconditions, request input, and expected result.
        - `Replaced With:` A 52-record DOMAIN JSON array. Retained `DOMAIN-001` consolidates DP-001, DP-005, DP-008, DP-012, DP-019, DP-029, and DP-032 and preserves every specification basis and provisional ID; all DP-001–DP-037 and DB-001–DB-007 remain covered.
        - `Appended:` None.
    - `File:` `review/pool-b/candidates/state-tests.json`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Exact empty JSON array `[]`, preserving the reviewed `State Applicability: NOT_APPLICABLE` result without placeholder tests.
    - `File:` `review/pool-b/candidates/security-tests.json`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` JSON array containing 8 SECURITY candidates covering SS-001–SS-008, SEC-02, and SEC-05, including controlled inert client `user_id` metacharacter coverage without an invented HTTP status oracle.
    - `File:` `test-cases/b-discount-coupons.csv`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Final nine-column CSV with 75 distinct candidates, sequential IDs `API-001`–`API-075`, category counts CONTRACT 15 / DOMAIN 52 / SECURITY 8 / STATE 0, exact traceability, and preserved provisional specialist IDs in notes.
- **Verdict:** INCOMPLETE
- **Reasoning:**
    - The generator produced enough test cases and kept STATE empty.

    - Some DOMAIN boundary labels are reversed.
    - API-049 uses an unsupported floating-point precision value.
    - Some test preconditions do not match their request data.
    - The deduplication check says no duplicates remain, but some DOMAIN cases still overlap.

- **Student Fixes:**
    - Asked the agent to correct the boundary labels.
    - Asked the agent to remove the floating-point artifact from API-049.
    - Asked the agent to fix inconsistent BIGBUY, EXPIRED, and nonexistent-coupon preconditions.
    - Asked the agent to rerun same-category deduplication and preserve all traceability.
    - Asked the agent to regenerate the CSV, IDs, counts, and orchestration status consistently.

Artifact 20

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 15:49:41 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
    - `api-test-generator`
    - `domain-test-generator`
    - `ai-audit`
- **Prompt:**

```
Revise Pool B Phase-2 output based on human review.

- Fix DOMAIN BVA labels for minimum thresholds: for `total_amount >= min_order_amount`, just-below is outside, boundary is inside, and just-above is inside.
- Fix API-049 so it does not assert the floating-point artifact `30000.100000000002`. Use the documented mathematical result without inventing precision/rounding behavior.
- Fix inconsistent DOMAIN preconditions:
  - BIGBUY cases must establish BIGBUY eligibility.
  - EXPIRED expiry testing must arrange the EXPIRED coupon active and satisfy all non-expiry conditions.
  - The nonexistent-code case must explicitly establish that the submitted code does not exist.
- Re-run same-category semantic deduplication. In particular review the overlaps represented by current API-016/API-062, API-022/API-051, API-031/API-048, and API-033/API-054. Preserve all DP/DB traceability on whichever case is retained.
- Regenerate the final CSV with sequential IDs and update candidate files, counts, deduplication notes, and orchestration status consistently.

Do not remove cross-category coverage and do not execute the API tests yet.
```

- **AI Output:**

I’m using the API/domain test-generation workflow for this reviewed Phase-2 revision. A DOMAIN specialist will correct the candidate content and preconditions from the approved model; I’ll then perform parent-owned semantic deduplication, preserve every DP/DB trace ID, regenerate sequential final IDs, and revalidate the full cross-category suite without executing it.

The reviewed overlaps map to DOMAIN provisional cases as follows: API-016/API-062, API-022/API-051, API-031/API-048, and API-033/API-054. I’m handing the existing DOMAIN fragment to the specialist for factual corrections only; the parent will decide consolidation afterward so content repair and deduplication remain separate.

The specialist corrected 52 DOMAIN records without deduplicating: threshold labels now follow the inclusive predicate, the 300001 oracle is `30000.1`/`270000.9`, and BIGBUY, EXPIRED, and nonexistent-code setups are explicit. I’m now comparing the four reviewed overlap pairs semantically, not merely by exact string equality.

The semantic review confirms exactly four removable overlaps. Other repeated requests have different expiry or usage-count preconditions and therefore remain distinct. I’m retaining DOMAIN-001, DOMAIN-009, DOMAIN-020, and DOMAIN-022, merging the corresponding DB bases and provisional IDs from DOMAIN-053, DOMAIN-042, DOMAIN-039, and DOMAIN-045.

Validation now passes with 71 distinct cases: 15 CONTRACT, 48 DOMAIN, and 8 SECURITY. All DP/DB coverage remains present, the four reviewed overlap pairs are consolidated, no same-category semantic duplicates remain, and the corrected setup/oracle checks pass. I’m regenerating the final CSV and synchronizing the orchestration record.

Pool B Phase-2 output is revised and validated.

- [Final CSV](D:\GitHub\SoftwareTesting-HW06\test-cases\b-discount-coupons.csv): 71 sequential cases, `API-001`–`API-071`
- CONTRACT: 15
- DOMAIN: 48

```
- SECURITY: 8
- STATE: 0 (`NOT_APPLICABLE`)

Corrections include proper inclusive-threshold BVA labels, explicit
BIGBUY/EXPIRED/nonexistent-code setups, and mathematical results
`30000.1`/`270000.9`. After deduplication, the former API-049 is now
API-048.

Four additional DOMAIN overlaps were consolidated while preserving every
DP/DB basis. All CR, DP, DB, and SS coverage remains intact, with zero
remaining semantic duplicates. [Orchestration status]
(D:\GitHub\SoftwareTesting-HW06\review\pool-b\orchestration-status.md)
and candidate files were updated consistently.

No API tests were executed.
```

- **File Modifications:**
  - `File:` `review/pool-b/candidates/domain-tests.json`.
    - `Action:` edited and deleted content.
    - `Deleted:` Four semantically overlapping DOMAIN records: `DOMAIN-053`, `DOMAIN-042`, `DOMAIN-039`, and `DOMAIN-045`; the prior incorrect minimum-threshold BVA labels; the floating-point artifact `30000.100000000002`; and inconsistent BIGBUY, EXPIRED, and nonexistent-code preconditions.
    - `Replaced With:` A validated 48-record DOMAIN fragment. Minimum-threshold cases label just-below as outside and boundary/just-above as inside; the SAVE10/300001 oracle uses mathematical results `30000.1` and `270000.9` without a precision/rounding assertion; BIGBUY cases establish BIGBUY eligibility; EXPIRED isolates expiry while all other conditions qualify; and the nonexistent-code case explicitly establishes database absence. DOMAIN-001, DOMAIN-009, DOMAIN-020, and DOMAIN-022 retain the merged DB bases, provisional IDs, and boundary objectives from the four removed overlaps.
    - `Appended:` None.
  - `File:` `test-cases/b-discount-coupons.csv`.
    - `Action:` replaced content.

        - `Deleted:` The prior 75-row CSV with DOMAIN count 52, IDs `API-001` – `API-075`, the reviewed duplicate pairs, incorrect BVA labels/setup wording, and the floating-point artifact.
        - `Replaced With:` A 71-row nine-column CSV with sequential IDs `API-001` – `API-071`, category counts CONTRACT 15 / DOMAIN 48 / SECURITY 8 / STATE 0, corrected DOMAIN cases, zero same-category semantic duplicates, and complete CR/DP/DB/SS traceability.
        - `Appended:` None.
    - `File:` `review/pool-b/orchestration-status.md`.
        - `Action:` edited.
        - `Deleted:` The prior 75-case / DOMAIN 52 completion summary and six-duplicate-only deduplication account.
        - `Replaced With:` Final 71-case / DOMAIN 48 completion status, human-review correction summary, ten-total DOMAIN deduplication record, retained traceability mappings, validation results through `API-071`, and confirmation that no API tests were executed.
        - `Appended:` None.
- **Verdict:** INCOMPLETE
- **Reasoning:**
    - The requested BVA labels, API-049 precision, and fixture preconditions were fixed correctly.
    - The four previously reported duplicate pairs were merged correctly.
    - Some DOMAIN duplicates still remain.
    - API-032 and API-047 both test `SAVE10` with `total_amount=299999`.
    - API-025 and API-054 both test the coupon exactly at the expiry boundary.
    - Therefore, `Remaining semantic duplicate groups: 0` is not yet correct.
- **Student Fixes:**
    - Asked the agent to merge the remaining same-category duplicates.
    - Asked the agent to preserve both DP and DB traceability in the retained cases.
    - Asked the agent to regenerate sequential API IDs, counts, CSV, and orchestration status.

Artifact 21

- **User:** Vương Ngữ Tín Thành
- **Date and Time:** 2026-08-22 15:59:49 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
  - `api-test-generator`
  - `domain-test-generator`
  - `ai-audit`
- **Prompt:**

```
Revise the Pool B Phase-2 output again.

Same-category semantic deduplication is still incomplete.

- Merge the case for SAVE10 `total_amount=299999` with the DB-001 just-below case. Preserve both DP-021 and DB-001 traceability.
- Merge the coupon-at-expiry-date partition case with the DB-004 equal-to-expiry boundary case. Preserve both DP-013 and DB-004 traceability.
- Recheck the remaining DOMAIN cases for the same kind of overlap.
- Regenerate sequential API IDs, category/total counts, final CSV, and orchestration status consistently.
- Do not remove cross-category coverage or execute tests.
```

- **AI Output:**

I’m using the API/domain workflow for this second Phase-2 deduplication review. I’ll compare DOMAIN cases by effective setup, request, and oracle—including equivalent wording—merge only genuine same-category overlaps, preserve every DP/DB basis on retained records, and then regenerate and validate the full CSV without executing tests.

The two requested merges are confirmed. A broader equivalence scan also flags another likely overlap: DP-031’s SAVE10 prior-use count `2` and DB-006’s above-limit count `2`. I’m having the DOMAIN specialist independently audit all remaining cases for this exact partition-versus-boundary pattern before the parent performs the final merges.

The independent audit found four remaining duplicates total: the two you named plus DP-030/DB-006 at-limit and DP-031/DB-006 above-limit. Cases with different concrete expiry timing, coupon minimums, or usage maxima/counts are materially distinct and will remain. I’m consolidating those four pairs now.

Post-merge validation yields 67 distinct cases: 15 CONTRACT, 44 DOMAIN, and 8 SECURITY. The four removed provisional records are absent, each retained case contains both partition and boundary trace IDs, and CR/DP/DB/SS coverage is complete with no exact semantic duplicate groups.

Pool B Phase-2 deduplication is complete.

- [Final CSV](D:\GitHub\SoftwareTesting-HW06\test-cases\b-discount-coupons.csv): 67 sequential cases, `API-001`–`API-067`
- CONTRACT: 15
- DOMAIN: 44
- SECURITY: 8
- STATE: 0

Merged DOMAIN coverage now includes:

- API-032: `DP-021` + `DB-001` just-below
- API-025: `DP-013` + `DB-004` equal-to-expiry

```
- API-040: `DP-030` + `DB-006` at-limit
- API-041: `DP-031` + `DB-006` above-limit

All CR/DP/DB/SS traceability remains complete. No remaining same-
category semantic duplicates were identified, and no tests were
executed. [Orchestration status](D:\GitHub\SoftwareTesting-
HW06\review\pool-b\orchestration-status.md) was updated consistently.
```

- **File Modifications:**
  - `File:` `review/pool-b/candidates/domain-tests.json`.
    - `Action:` edited and deleted content.
    - `Deleted:` Four remaining semantically overlapping DOMAIN records: `DOMAIN-038` (DB-001 just-below), `DOMAIN-048` (DB-004 equal-to-expiry), `DOMAIN-054` (DB-006 at-limit), and `DOMAIN-055` (DB-006 above-limit).
    - `Replaced With:` A validated 44-record DOMAIN fragment. DOMAIN-021 retains `DP-021 + DB-001`; DOMAIN-013 retains `DP-013 + DB-004`; DOMAIN-030 retains `DP-030 + DB-006`; and DOMAIN-031 retains `DP-031 + DB-006`. Each retained case includes the removed record's boundary objective, exact specification basis, and provisional ID. Consolidated objective wording was normalized without changing behavior.
    - `Appended:` None.
  - `File:` `test-cases/b-discount-coupons.csv`.
    - `Action:` replaced content.
    - `Deleted:` The prior 71-row CSV with DOMAIN count 48 and sequential IDs through `API-071`, including the four reviewed overlap records.
    - `Replaced With:` A 67-row nine-column CSV with sequential IDs `API-001`–`API-067`, category counts CONTRACT 15 / DOMAIN 44 / SECURITY 8 / STATE 0, four newly consolidated partition/boundary pairs, no exact same-category semantic duplicate groups, and complete CR/DP/DB/SS traceability.
    - `Appended:` None.
  - `File:` `review/pool-b/orchestration-status.md`.
    - `Action:` edited.

        - `Deleted:` The prior 71-case / DOMAIN 48 result and ten-total DOMAIN deduplication summary.
        - `Replaced With:` Final 67-case / DOMAIN 44 completion status, fourteen-total DOMAIN deduplication record, explicit mappings for the second review pass, validation through `API-067`, and confirmation that cross-category coverage was preserved and no tests were executed.
        - `Appended:` None.

- **Verdict:** VALID
- **Reasoning:**
    - The remaining DOMAIN duplicate cases were merged correctly.
    - The agent also found and merged two extra usage-limit overlaps.
    - Test IDs and category counts were regenerated consistently.
- **Student Fixes:** None

Artifact 22

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 16:16:52 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:** `ai-audit`
- **Prompt:**

```
I've reviewed [pool-b](review/pool-b/). Base on them, write B2 to B6 in
[Main_Report.md](Main_Report.md)
```

- **AI Output:**

```
I’ll inspect the existing B-section structure in `Main_Report.md` and
the reviewed Pool B reports/status, then write B.2–B.6 to match the
report’s established style and preserve the reviewed conclusions.

I’ve matched Pool A’s narrative style and extracted the final reviewed
Pool B facts: 12 contract rules, 37 domain partitions, 7 boundary
models, STATE not applicable, 8 security scenarios, and 67 final cases
after semantic deduplication. I’m writing B.2–B.6 now.

The write is complete, and verification found an older placeholder B.6
block still present immediately afterward. I’m removing that duplicate
placeholder so Pool B has one authoritative B.6 section before B.7.

Completed Sections B.2–B.6 in [Main_Report.md]
(D:\GitHub\SoftwareTesting-HW06\Main_Report.md:281).

Added:

- Contract, domain, state-applicability, and security analyses
- Reviewed constraints, partitions, and boundaries
- Links to all Pool B analysis reports
- Final 67-case category breakdown and traceability
- Human-review corrections and deduplication summary
- Explicit note that tests have not been executed

The obsolete duplicate B.6 placeholder was also removed.
```

- **File Modifications:**
  - `File:` `Main_Report.md`.
    - `Action:` edited and deleted content.
    - `Deleted:` The obsolete duplicate Pool B B.6 placeholder block containing a second test-case link, the incorrect statement that Pool B cases came from Sections A.2–A.5, and an empty category-count table.
    - `Replaced With:` Completed Sections B.2–B.6 in the established Pool A narrative style. B.2 documents CR-001–CR-012, inputs, response formulas, and contract gaps; B.3 documents DP-001–DP-037, DB-001–DB-007, baseline, partitions, corrected

inclusive boundaries, and domain assumptions; B.4 records STATE `NOT_APPLICABLE`; B.5 evaluates SEC-01–SEC-07 and summarizes SS-001–SS-008; B.6 records the final 67-case breakdown, sequential IDs, complete traceability, human-review corrections, fourteen-case semantic deduplication, validation, and non-execution status.

    - `Appended:` Links to the reviewed Pool B contract, domain, state, security, orchestration-status, and final CSV artifacts within Sections B.2–B.6.

- **Verdict:** VALID
- **Reasoning:** The AI correctly summarized the test cases analysis into Main Report.
- **Student Fixes:** None

Artifact 23

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 16:38:22 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
  - `api-test-generator`
  - `ai-audit`
- **Prompt:**

```
Fix the remaining Pool B issues before Postman generation.

- Copy current [b-discount-coupons.csv](test-cases/b-discount-coupons.csv) to [pool-b](review/pool-b/) named `candidate-api-tests.csv`.
- Add explicit `VALID / INVALID / INCOMPLETE` audit results with brief reasoning for every AI-generated case in the appropriate review artifact.
- Merge human-added cases `API-068`–`API-074` into `test-cases/b-discount-coupons.csv`, preserving the existing schema and intended content.
- Revalidate Pool B: 74 unique sequential IDs, no duplicates, required fields complete, and report/CSV counts consistent.
- Update only affected artifacts. Do not generate the Postman collection yet.

Finish with the files changed and validation results.
```

- **AI Output:**

I’m using the API test-generation workflow instructions to preserve Pool B’s review and traceability conventions. I’ll first inspect the repository guidance and current Pool B artifacts, then make only the requested CSV/review updates and run structural validation.

The current Pool B final CSV has 67 cases, while the report already contains the seven human-authored additions as `API-068`–`API-074`. I’m tracing the exact intended rows and the existing Pool A audit convention so the Pool B review copy can distinguish AI-generated cases from human-added ones without altering the final test-case schema.

The repository’s AI-audit skill also applies because this task explicitly requires audit documentation for AI-produced cases. I’m reading its append rules now; its output will be limited to the standard audit log entry plus the Pool B artifacts directly requested.

The audit disposition is clear: all 67 retained AI-generated rows are `VALID` in their current reviewed form. Earlier incomplete generations were corrected and deduplicated before this snapshot; no rejected case remains in the final AI subset. I’ll record a per-row reason tied to each case’s category and trace IDs, then append the seven human rows only to the nine-column final CSV.

The merge and report synchronization are complete. I’m running the final structural audit now: exact schemas, all required cells, ID sequence and uniqueness, duplicate row signatures, category totals, AI audit completeness, first-nine-field preservation, and report/status count consistency.

Pool B is ready for Postman generation; no collection was generated yet.

Files changed:

- [candidate-api-tests.csv](D:/GitHub/SoftwareTesting-HW06/review/pool-b/candidate-api-tests.csv) — copied 67 AI cases with `Audit Result` and case-specific `Audit Reason`; all 67 are `VALID`.
- [b-discount-coupons.csv](D:/GitHub/SoftwareTesting-HW06/test-cases/b-

```
discount-coupons.csv) — added human cases `API-068`–`API-074`.
- [orchestration-status.md](D:/GitHub/SoftwareTesting-HW06/review/pool-b/orchestration-status.md) — updated final counts and validation status.
- [Main_Report.md](D:/GitHub/SoftwareTesting-HW06/Main_Report.md) — synchronized AI/human/final counts.

Validation results:

- PASS — 74 sequential IDs, `API-001`–`API-074`
- PASS — 74 unique IDs; zero exact duplicate rows
- PASS — all nine required fields complete
- PASS — final categories: CONTRACT 17, DOMAIN 48, SECURITY 9, STATE 0
- PASS — AI subset: 67 cases with 67 audit results and reasons
- PASS — review copy preserves the original nine fields for all AI cases
- PASS — report and CSV counts are consistent
- PASS — `git diff --check`
- Postman collection: not generated
```

- **File Modifications:**
    - `File:` `review/pool-b/candidate-api-tests.csv`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` A 67-row review copy of the AI-generated Pool B cases preserving the original nine fields and adding `Audit Result` and `Audit Reason`; every retained case is marked `VALID` with case-level reasoning.
    - `File:` `test-cases/b-discount-coupons.csv`.
        - `Action:` appended.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Seven human-authored cases `API-068`–`API-074` in the existing nine-column schema, covering unsupported `text/plain`, a zero-byte body, an empty coupon code, VIP100 at its exact minimum, JWT/body `user_id` usage-limit bypass, and 1%/100% percent coupons.

    - `File:` `review/pool-b/orchestration-status.md`.
        - `Action:` edited.
        - `Deleted:` The prior Phase-2-only 67-case completion/count statements and execution wording that did not mention Postman generation.
        - `Replaced With:` Pre-Postman completion status for 74 cases, AI/human and category breakdowns, AI audit disposition, merged human coverage, updated validation through `API-074`, and explicit confirmation that no Postman collection was generated.
        - `Appended:` The AI review artifact path and audit summary.
    - `File:` `Main_Report.md`.
        - `Action:` edited and appended.
        - `Deleted:` The prior Pool B B.6 table and narrative reporting only 67 final cases through `API-067`.
        - `Replaced With:` An AI-generated/human-added/final count table totaling 74 cases, IDs through `API-074`, the review-artifact audit summary, updated validation, and the no-Postman-generation statement.
        - `Appended:` A B.7 statement confirming that the seven human-authored cases are merged into the final CSV.
- **Verdict:** VALID
- **Reasoning:** The agent merged human cases and human review into single csv file correctly.
- **Student Fixes:** None

Artifact 24

- **User:** Vương Ngữ Tín Thành
- **Date and Time:** 2026-08-22 16:51:40 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
    - `postman-test-generator`
    - `ai-audit`
- **Prompt:**

```
Use [$postman-test-generator](D:\GitHub\SoftwareTesting-HW06\\.agents\skills\postman-test-generator\SKILL.md) to generate the Postman artifacts for Pool B from the final reviewed test cases in [b-discount-coupons.csv](test-cases/b-discount-coupons.csv).



Use the repository [api\_specification.md](reference/api_specification.md) and existing Pool B review artifacts [pool-b](review/pool-b/) as supporting context. Do not redesign or add test cases.



After generation, validate the produced artifacts and summarize:

- files created or changed
- number of Postman requests generated
- validation results
- any unresolved issues



Finally, log the artifact using [$ai-audit](D:\GitHub\SoftwareTesting-HW06\\.agents\skills\ai-audit\SKILL.md)&#x20;
```

- **AI Output:**

I’ll use `postman-test-generator` first to convert the reviewed Pool B CSV without altering test design, validate its outputs, then use `ai-audit` to record the completed artifact and file changes. I’m reading both skill workflows before touching repository files.

The generator requires full Postman v2.1 schema validation—not just JSON parsing—and a separate traceability report. I’m now loading that validation procedure plus the audit’s exact templates, then I’ll inspect the Pool B inputs and existing conventions.

Pool B contains 74 final rows (`API-001`–`API-074`), with exactly one request intended per row and no STATE cases. Many reviewed outcomes deliberately omit HTTP-status oracles, so the generated scripts will assert only explicit machine-checkable calculations/safety properties and will preserve exploratory/manual observations in descriptions and the report.

The reviewed CSV supplies no success status codes and no fixture-management API. I’ll therefore keep all 74 requests runnable with variables (`baseUrl`, `studentId`, and token variants), preserve every database/date/usage prerequisite as setup text, and explicitly flag those fixture actions as unresolved runtime setup rather than inventing seed flows.

The repository’s `python` alias is inaccessible in this Windows environment, so generation did not run and no output was produced. I’m switching to the installed Python interpreter already used by this repository, then I’ll run the same schema gate.

Generation and the full official-schema gate both pass. Next I’m independently checking one-to-one traceability, category counts, headers, descriptions, bodies, and assertion coverage, then running all 74 requests through Newman against a local deterministic compatibility mock (not the SUT).

The validation inspection caught three token/header mapping defects before finalization: API-004 lacked its reviewed valid Authorization

```
header, while API-014 and API-063 were normalized back to the generic
valid-token variable. I’m correcting those mappings and rerunning every
validation gate.

All corrected gates now pass: 74/74 traceability, category totals
17/48/0/9, complete descriptions and `X-Student-Id` injection, correct
token variants, full v2.1 schema validation, and Newman execution of 74
requests/27 assertions with zero failures. I’m recording these results
in the conversion report, then removing only the two temporary
validation helpers.

The final collection and report remain valid after cleanup: schema PASS,
74 traceability rows, zero pending markers, and `git diff --check` PASS.
The `ai-audit` workflow now requires appending the exact prompt/output
pair as Artifact 24; I’m recording the final handoff and its file
modifications verbatim.

The audit block was inserted after Artifact 2 because the patch anchor
was not unique. I caught this in the sequence check; I’m moving the
unchanged block to the document end so Artifact 24 is truly appended
after Artifact 23, as the audit skill requires.

Audit validation now passes with Artifacts 1–24 in exact order and the
required TODO fields preserved. A temporary untracked `diff.patch`
containing only this task’s generated diff also appeared during
patching; I identified it by timestamp/content and am removing that non-
deliverable before the final status check.

Generated and validated the Pool B Postman artifacts without changing or
adding test cases.

Files created or changed:

- [pool-b-discount-coupons.postman_collection.json]
(D:/GitHub/SoftwareTesting-HW06/postman/pool-b-discount-
coupons.postman_collection.json) — created; 74 requests.
- [pool-b-conversion-report.md](D:/GitHub/SoftwareTesting-
```

```
HW06/postman/pool-b-conversion-report.md) — created; complete row-to-request traceability, assertions, setup notes, and unresolved items.
- [AI_Audit.md](D:/GitHub/SoftwareTesting-HW06/AI_Audit.md) — appended Artifact 24.

Validation results:

- PASS — 74 CSV rows map to 74 unique requests (`API-001`–`API-074`); categories are CONTRACT 17, DOMAIN 48, STATE 0, SECURITY 9.
- PASS — every request preserves reviewed context and sends `X-Student-Id: {{studentId}}`.
- PASS — full official Postman Collection v2.1 schema validation.
- PASS — Newman 6.2.2 compatibility smoke run: 74 requests, 74 test scripts, 27 assertions, zero failures.
- PASS — conversion report has 74 traceability rows, no pending markers; `git diff --check` passes.

Unresolved issues: runtime coupon fixtures, usage counts, date boundaries, and token variants must be supplied externally because no fixture-management API is specified. Cases with undocumented statuses, exploratory outcomes, rounding behavior, or database/security side effects retain manual oracles. Full SUT conformance execution was not performed.

Artifacts 24-24 successfully appended to `D:\GitHub\SoftwareTesting-HW06\AI_Audit.md`.
```

- **File Modifications:**
  - `File:` `postman/pool-b-discount-coupons.postman_collection.json`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` A Postman Collection v2.1 containing exactly 74 reviewed Pool B requests, organized as CONTRACT 17, DOMAIN 48, and SECURITY 9, with runtime variables, preserved setup descriptions, and 27 machine-checkable assertions.
  - `File:` `postman/pool-b-conversion-report.md`.

        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` A 74-row conversion traceability report documenting setup notes, generated assertions, manual or unresolved oracles, schema validation, Newman compatibility, and remaining runtime setup requirements.
    - `File:` `AI_Audit.md`.
        - `Action:` appended.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Artifact 24 containing this exact user prompt, AI output, and file-modification record.
    - `File:` `tmp/generate_pool_b_postman.py`.
        - `Action:` created and deleted file.
        - `Deleted:` The temporary mechanical CSV-to-Postman generation helper after successful generation and validation.
        - `Replaced With:` None.
        - `Appended:` None.
    - `File:` `tmp/pool_b_mock_server.js`.
        - `Action:` created and deleted file.
        - `Deleted:` The temporary deterministic local Newman compatibility mock after the smoke run passed.
        - `Replaced With:` None.
        - `Appended:` None.
    - `File:` `tmp/move_audit_artifact.py`.
        - `Action:` created and deleted file.
        - `Deleted:` The temporary helper used to move the unchanged Artifact 24 block to the audit document end.
        - `Replaced With:` None.
        - `Appended:` None.

    - `File:` `diff.patch`.
        - `Action:` created and deleted file.
        - `Deleted:` A temporary untracked patch containing only this task's generated repository diff.
        - `Replaced With:` None.
        - `Appended:` None.
- **Verdict:** INCOMPLETE
- **Reasoning:**
    - `X-Student-Id` is added directly to each request.
    - The required collection-level pre-request script is missing.
- **Student Fixes:**
    - Move `X-Student-Id: {{studentId}}` injection to a collection-level pre-request script.
    - Remove duplicated `X-Student-Id` headers from individual requests.

Artifact 25

- **User:** Vương Ngữ Tín Thành
- **Date and Time:** 2026-08-22 17:01:09 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
    - `postman-test-generator`
    - `ai-audit`
- **Prompt:**

```
Fix the Pool B Postman artifacts:

- Inject `X-Student-Id: {{studentId}}` through a collection-level pre-request script instead of duplicating it in each request.
- Add a console log suitable for the required execution evidence.
- Keep all 74 test cases unchanged.
- Re-run Postman v2.1 schema and Newman compatibility validation.
- Update the conversion report.

Do not run the real SUT yet.
```

- **AI Output:**

I’ll update only the collection-level header injection/evidence behavior and the conversion report, preserving all 74 request definitions and test intent. I’m reapplying the Postman generator validation workflow, then the repository’s AI-audit workflow because these are AI-produced file edits.

The current collection has 74 duplicated request headers and no collection-level event. I’ll replace only that duplication with one `prerequest` event that upserts the header and logs request name/Test ID, method, URL, and resolved student ID; all folders, names, methods, URLs, bodies, descriptions, auth headers, and test scripts will be invariant-checked.

The combined validation command was rejected by Windows process creation before it ran, so the collection has not yet been modified. I’m splitting mutation/static validation from the background Newman mock to avoid that launcher limitation.

The revised collection passes the schema gate and structural invariants: 74 unchanged IDs/requests, zero request-level student headers, one collection-level pre-request event, and the evidence log is present. Newman also passed with all 74 injected headers observed by the mock and 74 evidence-log entries; no real SUT was contacted.

The finalized artifacts now pass schema and structural checks after temporary helpers were removed: 74 unique cases, no duplicated student headers, one collection-level injection/evidence script, 74 report trace rows, and no pending markers. I’m appending this completed edit as AI Audit Artifact 25.

Updated the Pool B Postman artifacts while preserving all 74 test cases.

Files changed:

- [pool-b-discount-coupons.postman_collection.json](D:/GitHub/SoftwareTesting-HW06/postman/pool-b-discount-coupons.postman_collection.json) — removed 74 request-level `X-Student-

```
Id` headers; added one collection-level pre-request script that upserts
the header and logs request/Test ID, method, URL, and resolved student
ID.
- [pool-b-conversion-report.md](D:/GitHub/SoftwareTesting-
HW06/postman/pool-b-conversion-report.md) — updated injection, evidence,
validation, timestamp, and SHA-256 details.
- [AI_Audit.md](D:/GitHub/SoftwareTesting-HW06/AI_Audit.md) — appended
Artifact 25.

Validation results:

- PASS — all 74 unique requests and reviewed Test IDs remain unchanged.
- PASS — zero request-level student headers; one collection-level pre-
request injection/evidence event.
- PASS — full Postman Collection v2.1 schema validation.
- PASS — Newman 6.2.2 compatibility: 74 requests, 74 injected headers
observed, 74 evidence logs, 74 test scripts, 27 assertions, zero
failures.
- PASS — 74 conversion-report traceability rows, zero pending markers,
and `git diff --check`.

The Newman run used only a local deterministic compatibility mock. The
real SUT was not started or contacted. Existing runtime fixture/token
requirements and manual oracles remain unresolved and unchanged.

Artifacts 25-25 successfully appended to `D:\GitHub\SoftwareTesting-
HW06\AI_Audit.md`.
```

- **File Modifications:**
  - `File:` `postman/pool-b-discount-coupons.postman_collection.json`.
    - `Action:` edited.
    - `Deleted:` The duplicated `X-Student-Id: {{studentId}}` header entry from each of the 74 request definitions.
    - `Replaced With:` One schema-supported collection-level pre-request event that resolves `studentId`, upserts `X-Student-Id`, and logs execution evidence containing request name/Test ID, method, URL, and resolved student ID.

        - `Appended:` None.
    - `File:` `postman/pool-b-conversion-report.md`.
        - `Action:` edited.
        - `Deleted:` Statements describing `X-Student-Id` as a request-level header and the previous validation timestamp, collection SHA-256, and Newman result.
        - `Replaced With:` Collection-level injection/evidence documentation plus validation results dated 2026-08-22 17:00:21 GMT+7 and SHA-256 `2323D04DF15152C14FB2FCA92813D93BA2E46F9F6EA799CD509689F29A2B75A7`.
        - `Appended:` A traceability check for the execution-evidence console log.
    - `File:` `AI_Audit.md`.
        - `Action:` appended.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Artifact 25 containing this exact prompt, AI output, and file-modification record.
    - `File:` `tmp/update_pool_b_collection.py`.
        - `Action:` created and deleted file.
        - `Deleted:` The temporary invariant-checked collection mutation helper after use.
        - `Replaced With:` None.
        - `Appended:` None.
    - `File:` `tmp/pool_b_compat_mock.js`.
        - `Action:` created and deleted file.
        - `Deleted:` The temporary local Newman compatibility mock after validation.
        - `Replaced With:` None.
        - `Appended:` None.
- **Verdict:** VALID
- **Reasoning:** The agent correctly fixed all found defects.
- **Student Fixes:** None

Artifact 26

- **User:** Vương Ngữ Tín Thành

- **Date and Time:** 2026-08-22 17:20:17 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
  - `postman-test-generator`
  - `ai-audit`
- **Prompt:**

```
Prepare the runtime preconditions for Pool B before Newman execution.
Use the reviewed Pool B test cases to identify all required database
fixtures, user coupon-usage states, and JWT variants. Follow the
existing Pool A setup approach where applicable.
Create a deterministic setup/reset mechanism so every test runs with its
required preconditions, including cases that require different usage
counts for the same coupon/user.
Do not change the 74 reviewed test cases and do not run the full Newman
suite yet.
Validate the setup and document how to use it.
```

- **AI Output:**

I’ll use the Postman generation workflow to add runtime setup without altering the 74 reviewed requests, following Pool A’s deterministic fixture pattern. I’ll first inventory every Pool B fixture/token/usage-state requirement and inspect Pool A’s seeder/reset mechanics, then build and validate the setup without running the full suite; the repository audit workflow will record the resulting file edits.

Pool A confirms the intended pattern: start the SUT first, then seed owned SQLite rows directly with schema checks and post-write verification. Pool B additionally needs per-request resets because multiple reviewed cases intentionally require conflicting usage counts for the same coupon/user; I’m now mapping those states to the actual SUT schema before designing the reset hook.

The reviewed runtime matrix resolves to 11 controlled coupon fixtures, users 1 and 2 present, users 0 and 999999999 absent where required, five JWT variants, and nine nonzero usage-count cases (`API-040`, `041`, `054`–`059`, `072`). I’ll reset the isolated coupon tables before every request, while snapshotting/restoring their prior contents in the Newman wrapper so conflicting counts cannot leak between cases.

The runtime design now separates concerns cleanly: a reusable fixture manifest/reset library, a Pool A-style one-case seeder for inspection, and a canonical Newman wrapper that snapshots the database, serves authenticated per-request resets on localhost, injects deterministic JWT variables, and restores the original state in `finally`. Next I’m wiring the collection’s pre-request hook to that local controller and building a self-test against a temporary SQLite database.

The first self-test caught an assumption in the validator—not in the fixtures: the collection executes by category folders, so its 74 IDs are complete but not numerically ordered. I’m changing both the validator and runner completion check to require exactly one reset for every reviewed ID regardless of folder execution order.

The full fixture self-test now passes for all 74 IDs on a temporary database, including snapshot restoration and all conflicting counts; the live SUT database was opened read-only only to validate schema compatibility. The collection also passes the full v2.1 schema. I’m adding a targeted seven-request Newman check against temporary fixture and HTTP mocks to verify the asynchronous reset hook and JWT/header delivery without touching the real SUT or running the full suite.

The targeted Newman validation passed for seven representative cases covering baseline state, conflicting usage counts, relative expiry dates, invalid-signature/limit-reached JWTs, student-header injection, and snapshot restoration. I’m retaining that harness as a reproducible validation tool, then I’ll update the runtime guide, validation record, and conversion report.

The runtime guide and conversion report now document the exact fixture/JWT matrix, safe commands, local controller behavior, and the “full suite not executed” boundary. I’m doing the final handoff checks now: request-item byte-equivalence against the prior collection, CLI parsing, validation-record consistency, schema validation, and repository cleanliness.

Final validation is clean: the 74 request items are byte-equivalent to the prior collection, all 74 temporary-database states and restoration pass, the actual SUT schema passes read-only validation, seven targeted Newman mock requests pass, and the full Postman v2.1 schema passes. Neither the real SUT nor the full Newman suite was run. I’m recording this work as AI Audit Artifact 26.

Prepared deterministic Pool B runtime preconditions without running the real SUT or full Newman suite.

Created:

- [pool_b_fixtures.py](D:/GitHub/SoftwareTesting-HW06/postman/pool_b_fixtures.py) — 11 coupon fixtures, controlled users, five JWT variants, nine usage-count overrides, reset/verify, snapshot, and restore logic.
- [seed_pool_b_fixtures.py](D:/GitHub/SoftwareTesting-

HW06/postman/seed_pool_b_fixtures.py) — Pool A-style one-case seeder for manual inspection.
- [run_pool_b_with_fixtures.py](D:/GitHub/SoftwareTesting-HW06/postman/run_pool_b_with_fixtures.py) — canonical Newman wrapper with authenticated localhost resets and automatic restoration.
- [validate_pool_b_runtime.py](D:/GitHub/SoftwareTesting-HW06/postman/validate_pool_b_runtime.py) and [validate_pool_b_targeted_newman.py](D:/GitHub/SoftwareTesting-HW06/postman/validate_pool_b_targeted_newman.py) — reusable setup validation.
- [pool-b-runtime-setup.md](D:/GitHub/SoftwareTesting-HW06/postman/pool-b-runtime-setup.md) — complete fixture matrix and usage instructions.
- [pool-b-runtime-validation.json](D:/GitHub/SoftwareTesting-HW06/postman/pool-b-runtime-validation.json) — machine-readable validation evidence.

Updated:

- [pool-b-discount-coupons.postman_collection.json](D:/GitHub/SoftwareTesting-HW06/postman/pool-b-discount-coupons.postman_collection.json) — added only collection-level runtime-controller wiring and variables; all 74 reviewed request items are unchanged.
- [pool-b-conversion-report.md](D:/GitHub/SoftwareTesting-HW06/postman/pool-b-conversion-report.md) — documented fixture automation, commands, validation, and remaining manual oracles.
- [AI_Audit.md](D:/GitHub/SoftwareTesting-HW06/AI_Audit.md) — appended Artifact 26.

Validation results:

- PASS — all 74 reviewed request items are byte-equivalent to the prior collection.
- PASS — all 74 per-case resets and exact snapshot restoration on temporary SQLite.
- PASS — 11 coupon fixtures, users 1/2 present, users 0/999999999 absent, five JWT variants, and all nine conflicting usage overrides.

```
- PASS — actual SUT SQLite schema checked read-only; the SUT was not
contacted.
- PASS — targeted Newman 6.2.2 validation for seven representative cases
using local mocks.
- PASS — full Postman Collection v2.1 schema and `git diff --check`.

For the later authorized full run, start the SUT and execute:

```powershell
C:\Users\xing0\AppData\Local\Python\bin\python.exe
postman\run_pool_b_with_fixtures.py --sut-dir D:\GitHub\eshop-sut --
newman-command npx.cmd newman
```

The wrapper snapshots and restores the test database automatically. The
standalone one-case seeder intentionally leaves its selected fixture
state in place. Existing exploratory/security/manual oracles remain
unchanged.

Artifacts 26-26 successfully appended to `D:\GitHub\SoftwareTesting-
HW06\AI_Audit.md`.
```

- **File Modifications:**
  - `File:` `postman/pool_b_fixtures.py`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Deterministic coupon/JWT/user/usage manifests plus schema validation, per-case reset verification, snapshot, and restoration functions for all 74 reviewed IDs.
  - `File:` `postman/seed_pool_b_fixtures.py`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.

        - `Appended:` A Pool A-style CLI that requires the initialized SUT, applies one selected Pool B state, verifies it, and optionally prints runtime JWT variables.
    - `File:` `postman/run_pool_b_with_fixtures.py`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` A canonical full-run wrapper that snapshots database state, serves authenticated localhost per-request resets, supplies runtime variables to Newman, verifies 74 reset IDs, and restores state in `finally`.
    - `File:` `postman/validate_pool_b_runtime.py`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Validation for the reviewed mapping, all 74 temporary-database states, snapshot restoration, JWT semantics, collection wiring, and optional read-only SUT schema.
    - `File:` `postman/validate_pool_b_targeted_newman.py`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` A seven-request Newman compatibility harness using temporary SQLite and local HTTP mocks, with machine-readable result recording.
    - `File:` `postman/pool-b-runtime-setup.md`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Runtime fixture tables, JWT definitions, validation commands, one-case inspection instructions, canonical full-run command, and restoration behavior.
    - `File:` `postman/pool-b-runtime-validation.json`.
        - `Action:` created.

        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Passing all-74 fixture/reset/restore/JWT/schema results plus the seven-request Newman compatibility result, explicitly recording no real-SUT or full-suite execution.
    - `File:` `postman/pool-b-discount-coupons.postman_collection.json`.
        - `Action:` edited.
        - `Deleted:` The prior collection description and pre-request script that only injected/logged `X-Student-Id`.
        - `Replaced With:` A runtime-aware description, blank fixture-controller variables, and a collection-level hook that retains student evidence logging, requests an authenticated localhost reset by Test ID, and skips the SUT request on setup failure.
        - `Appended:` None.
    - `File:` `postman/pool-b-conversion-report.md`.
        - `Action:` edited.
        - `Deleted:` The prior statements that runtime coupon/user/token setup was unresolved and the earlier collection hash/validation results.
        - `Replaced With:` The deterministic runtime strategy, controller variables, resolved automated preconditions, SHA-256 `85B1F2563DD2DEDD19BC5DDED320A639F0CD5680324793BA74FA932F692AD7EF`, and current fixture/schema/targeted-Newman results.
        - `Appended:` Links to the runtime guide and validation evidence.
    - `File:` `AI_Audit.md`.
        - `Action:` appended.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Artifact 26 containing this exact prompt, AI output, and file-modification record.
    - `File:` `tmp/wire_pool_b_runtime.py`.
        - `Action:` created and deleted file.
        - `Deleted:` The temporary invariant-checked collection wiring helper after use.

        - `Replaced With:` None.
        - `Appended:` None.

- **Verdict:** VALID
- **Reasoning:** The precondition prepared correctly.
- **Student Fixes:** None

## Artifact 27

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 18:28:56 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:** newman-result-analyzer
- **Prompt:**

```
Use [$newman-result-analyzer](D:\GitHub\SoftwareTesting-HW06\\.agents\skills\newman-result-analyzer\SKILL.md) to analyze the real Pool B Newman execution result in [pool-b-run.json](reports/pool-b/pool-b-run.json)&#x20;

Use the Pool B reviewed test cases, Postman collection, and API specification as supporting context.

For every failed or suspicious case:

- identify the Test ID
- classify it as TEST\_DEFECT, SETUP\_DEFECT, SUT\_BUG, or NEEDS\_MANUAL\_REVIEW
- explain the cause briefly
- do not modify the test cases or Postman collection

Summarize:

- total requests/tests/assertions
- passed and failed cases
- bug candidates
- manual-review cases
- recommended next actions

The report should be written in [Main\_Report.md](Main_Report.md).
```

- **AI Output:**

```
Updated [Main_Report.md](D:/GitHub/SoftwareTesting-HW06/Main_Report.md:389) with the Pool B Newman analysis.

Key results:

- 148 requests: 74 SUT requests plus 74 fixture-reset calls
- 74 logical test cases executed
- 27 assertions: 7 passed, 20 failed
- Logical cases: 54 `PASS`, 20 `FAIL_ASSERTION`
- 24 SUT bug-candidate case observations
- 1 `TEST_DEFECT`: `API-067`
- 1 failed case needing requirement review: `API-072`
- 0 setup defects
- 50 explicit manual checks remain

The report includes per-case outcomes, triage for every failed or suspicious case, reconciliation of duplicated Newman execution records, bug candidates, and recommended actions. Test cases and the Postman collection were not modified.
```

- **File Modifications:**
  - `File:` `Main_Report.md`.
    - `Action:` appended.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Section `B.8. Newman Execution Analysis`, including the evidence scope, normalized run totals, all 74 logical test-case outcomes, individual classification and causes for every failed assertion, six additional suspicious manual-only observations, the 50-case manual-oracle summary, bug candidates, recommended actions, and Newman coverage/reconciliation notes.
- **Verdict:** INCOMPLETE
- **Reasoning:** B.6 still incorrectly says Postman was not generated and tests were not executed.
- **Student Fixes:**

    - Remove the outdated execution statement from B.6.
    - State that the Postman collection was generated and Pool B was executed with Newman.
    - Refer readers to Section B.8 for execution results.

Artifact 28

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 19:12:17 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
    - `newman-result-analyzer`
    - `ai-audit`
- **Prompt:**

```
Resolve the remaining Pool B triage cases before bug reporting.

Focus only on:

- `API-067` — currently `TEST_DEFECT`
- `API-072` — currently `NEEDS_MANUAL_REVIEW`

For `API-067`:

- Review the test oracle against the reviewed security analysis and SUT
behavior.
- Determine what can actually be verified from black-box execution.
- If needed, use database inspection or implementation evidence to
verify query/state effects.
- Correct only the faulty assertion/oracle if justified.
- Re-run only the affected case after correction.

For `API-072`:

- Review the API specification and SUT implementation to determine
whether coupon usage must be tied to the authenticated JWT user or the
body `user_id`.
- Decide whether the observed behavior is an SUT bug, valid behavior, or
an unsupported requirement assumption.
- Do not change the reviewed human test case unless its expectation is
unsupported.

Update Section B.8 with the final classifications and evidence.
```

- **AI Output:**

```
Resolved both Pool B triage cases and updated [Main_Report.md]
(D:/GitHub/SoftwareTesting-HW06/Main_Report.md:521).

- `API-067`: `TEST_DEFECT — resolved`
  - Replaced the over-strong assertion with a database-diagnostics
oracle.
  - Implementation inspection confirmed parameterized queries and no
write operations.
  - Targeted rerun: 1 request, 1 passed assertion, 0 failures.
  - Database state was unchanged and restored afterward.
  - Evidence: [api-067-rerun.json](D:/GitHub/SoftwareTesting-
HW06/reports/pool-b/api-067-rerun.json).

- `API-072`: `TEST_DEFECT — unsupported requirement assumption`
  - Neither the specification nor reviewed context binds body `user_id`
to the JWT subject.
  - The observed behavior is valid for the documented body-driven
identity model.
  - Converted the reviewed case to exploratory and removed its
unsupported automated oracle.
  - It is not an SUT bug candidate.

Updated the [Postman collection](D:/GitHub/SoftwareTesting-
HW06/postman/pool-b-discount-coupons.postman_collection.json) and
[reviewed CSV](D:/GitHub/SoftwareTesting-HW06/test-cases/b-discount-
coupons.csv:73) only as required for these two corrections. Collection
schema validation passes, all 74 IDs remain intact, and only `API-067`
was rerun.
```

- **File Modifications:**
  - `File:` `Main_Report.md`.
    - `Action:` edited.
    - `Deleted:` The unresolved `API-067` test-defect and `API-072` manual-review conclusions, obsolete manual-oracle wording, and recommendations to investigate these two cases later.

        - `Replaced With:` Final Section B.8 classifications, corrected manual-oracle accounting, targeted rerun statistics, implementation and database-state evidence, and final bug-reporting guidance. Section B.7's `API-072` row was synchronized with its exploratory disposition.
        - `Appended:` A targeted `API-067` rerun and implementation-evidence subsection linking the new Newman artifact.
    - `File:` `postman/pool-b-discount-coupons.postman_collection.json`.
        - `Action:` edited.
        - `Deleted:` `API-067`'s unsupported no-success assertion and `API-072`'s unsupported rejection assertion and identity-binding description.
        - `Replaced With:` An `API-067` database-diagnostics assertion and an exploratory `API-072` name, description, and observation-only script with no fixed outcome oracle.
        - `Appended:` None.
    - `File:` `test-cases/b-discount-coupons.csv`.
        - `Action:` edited.
        - `Deleted:` `API-072`'s unsupported expectation that body `user_id` must be bound to the JWT identity and the coupon must be rejected.
        - `Replaced With:` An exploratory `API-072` objective and expected result that record identity-scoping behavior without treating either outcome as an FR-09 conformance failure.
        - `Appended:` None.
    - `File:` `reports/pool-b/api-067-rerun.json`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Newman JSON evidence for exactly one `API-067` request, one passed assertion, zero failures, and no request or script errors.
    - `File:` `tmp/run_api067_targeted.py`.
        - `Action:` created and deleted file.
        - `Deleted:` The temporary one-case Newman/database comparison runner after successful use.

        - `Replaced With:` None.
        - `Appended:` None.

- **Verdict:** VALID
- **Reasoning:**
    - Resolved the `API-067` test defect.
    - Resolved `API-072` as an unsupported requirement assumption.
- **Student Fixes:** None

Artifact 29

- **User:** Vương Ngữ Tín Thành
- **Date and Time:** 2026-08-22 19:30:30 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:** `ai-audit`
- **Prompt:**

```
[$ai-audit](D:\GitHub\SoftwareTesting-HW06\\.agents\skills\ai-audit\SKILL.md)&#x20;
```

- **AI Output:**

```
Artifacts 28-28 successfully appended to `D:\GitHub\SoftwareTesting-HW06\AI_Audit.md`.
```

- **File Modifications:**
    - `File:` `AI_Audit.md`.
        - `Action:` appended.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Artifact 28 containing the exact preceding Pool B triage-resolution prompt, AI output, and file-modification record.
- **Verdict:** VALID
- **Reasoning:** The agent grouped and extracted issues correctly.

- **Student Fixes:** None

Artifact 30

- **User:** Vương Ngữ Tín Thành
- **Date and Time:** 2026-08-22 20:07:27 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
  - `api-test-generator`
  - `contract-test-generator`
  - `domain-test-generator`
  - `state-transition-test-generator`
  - `security-test-generator`
  - `ai-audit`
- **Prompt:**

```
Use [$api-test-generator](D:\GitHub\SoftwareTesting-HW06\\.agents\skills\api-test-generator\SKILL.md) for Pool C – Admin Order Management.

Inputs:

- [api\_specification.md](reference/api_specification.md)&#x20;
- [system\_requirements.md](reference/system_requirements.md)&#x20;
- endpoint: PUT /api/admin/orders/:id/status



Log with [$ai-audit](D:\GitHub\SoftwareTesting-HW06\\.agents\skills\ai-audit\SKILL.md)
```

- **AI Output:**

```
Phase 1 for Pool C is complete and paused at the required human-review
gate. No candidate tests were generated.

Reports:

- [Contract](D:/GitHub/SoftwareTesting-HW06/review/pool-
c/reports/contract-report.md)
- [Domain](D:/GitHub/SoftwareTesting-HW06/review/pool-c/reports/domain-
report.md)
- [State](D:/GitHub/SoftwareTesting-HW06/review/pool-c/reports/state-
report.md)
- [Security](D:/GitHub/SoftwareTesting-HW06/review/pool-
c/reports/security-report.md)
- [Orchestration status](D:/GitHub/SoftwareTesting-HW06/review/pool-
c/orchestration-status.md)

The main review decisions are `shipping → canceled`, whether the FR-10
diagram is exhaustive, same-state/idempotent updates, strict `status`
schema behavior, the `id` domain, and unspecified response contracts.

Artifacts 30-30 successfully appended to D:\GitHub\SoftwareTesting-
HW06\AI_Audit.md.

Approve these exact report versions—or revise their review blocks—when
you want Phase 2 to generate `candidate-api-tests.csv`.
```

- **File Modifications:**
  - `File:` `review/pool-c/shared-api-context.md`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Complete normalized endpoint context for `PUT /api/admin/orders/:id/status`, including source hashes, request and response inventories, authentication/authorization, SEC-01–SEC-07 extraction, FR-10 state model, gaps, and assumptions.

- `File:` `review/pool-c/orchestration-status.md`.
  - `Action:` created.
  - `Deleted:` None.
  - `Replaced With:` None.
  - `Appended:` Phase 1 status, exact report hashes, validation result, review focus, and closed human-review gate record.
- `File:` `review/pool-c/reports/contract-report.md`.
  - `Action:` created.
  - `Deleted:` None.
  - `Replaced With:` None.
  - `Appended:` CONTRACT Phase 1 analysis with CR-001–CR-013 and `Review Status: PENDING`.
- `File:` `review/pool-c/reports/domain-report.md`.
  - `Action:` created.
  - `Deleted:` None.
  - `Replaced With:` None.
  - `Appended:` DOMAIN Phase 1 analysis with parameter inventory, equivalence partitions, explicit boundary analysis, and `Review Status: PENDING`.
- `File:` `review/pool-c/reports/state-report.md`.
  - `Action:` created.
  - `Deleted:` None.
  - `Replaced With:` None.
  - `Appended:` STATE Phase 1 analysis with five states, supported transitions TR-001–TR-013, reviewer-resolution items PR-001–PR-007, and `Review Status: PENDING`.
- `File:` `review/pool-c/reports/security-report.md`.
  - `Action:` created.
  - `Deleted:` None.
  - `Replaced With:` None.
  - `Appended:` SECURITY Phase 1 analysis with the SEC-01–SEC-07 applicability matrix, SS-001–SS-007, and `Review Status: PENDING`.

- `File:` `AI_Audit.md`.
    - `Action:` appended.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Artifact 30 containing this exact prompt, AI output, and file-modification record.

- **Verdict:** INCOMPLETE
- **Reasoning:**
  - CONTRACT, DOMAIN, and SECURITY are mostly correct.
  - STATE leaves some FR-10 transitions unresolved.
  - FR-10 defines the valid state machine, and FR-18 requires Admin updates to follow it.
  - PR-001 to PR-007 should be invalid.

- **Student Fixes:**
  - Manually approve CONTRACT, DOMAIN, and SECURITY.
  - Ask AI to mark PR-001 to PR-007 as INVALID.
  - Ask AI to remove the shipping → canceled ambiguity.
  - Ask AI to update all Phase 1 reports consistently.

Artifact 31

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 20:21:54 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
  - `api-test-generator`
  - `contract-test-generator`
  - `domain-test-generator`
  - `state-transition-test-generator`
  - `security-test-generator`
  - `ai-audit`

- **Prompt:**

```
Revise Pool C Phase 1 from my human review:

- FR-10's diagram is the authoritative state machine.
- Classify PR-001 through PR-007 as INVALID, including shipping →
canceled.
- FR-10 line 161 does not authorize shipping → canceled; FR-18 requires
Admin changes to follow FR-10.
- Keep same-state updates UNSPECIFIED.
- Propagate this decision consistently across shared context, CONTRACT,
DOMAIN, STATE, and orchestration status.
- Preserve all other analysis.
- Do not generate Phase 2 yet.
- Log with [$ai-audit](D:\GitHub\SoftwareTesting-
HW06\\.agents\skills\ai-audit\SKILL.md)
```

- **AI Output:**

```
Pool C Phase 1 has been revised consistently. Phase 2 remains blocked
and no candidate tests were generated.

Updated files:

- [Shared context](D:/GitHub/SoftwareTesting-HW06/review/pool-c/shared-
api-context.md)
- [Contract v2](D:/GitHub/SoftwareTesting-HW06/review/pool-
c/reports/contract-report.md)
- [Domain v2](D:/GitHub/SoftwareTesting-HW06/review/pool-
c/reports/domain-report.md)
- [State v2](D:/GitHub/SoftwareTesting-HW06/review/pool-c/reports/state-
report.md)
- [Orchestration status](D:/GitHub/SoftwareTesting-HW06/review/pool-
c/orchestration-status.md)

PR-001 through PR-007 are now `INVALID`, including `shipping →
canceled`. FR-10 line 161 is explicitly recorded as not authorizing that
edge, while same-state updates remain `UNSPECIFIED`. The SECURITY report
was verified consistent and left byte-for-byte unchanged. All reports
remain `PENDING`.

Artifacts 31-31 successfully appended to D:\GitHub\SoftwareTesting-
HW06\AI_Audit.md.
```

- **File Modifications:**
  - `File:` `review/pool-c/shared-api-context.md`.
    - `Action:` edited.
    - `Deleted:` The prior unresolved interpretation of omitted FR-10 edges, including the `shipping` → `canceled` ambiguity and the corresponding gap entry.
    - `Replaced With:` A human-review authority overlay making the FR-10 diagram exhaustive for non-self transitions, classifying the seven omitted edges as invalid, recording the line-161/FR-18 rationale, and preserving same-state behavior as unspecified.
    - `Appended:` None.

- `File:` `review/pool-c/reports/contract-report.md`.
  - `Action:` edited.
  - `Deleted:` The prior `shipping` → `canceled` ambiguity and unresolved review prompt.
  - `Replaced With:` `CONTRACT-PHASE1-v2`, including revised CR-009, explicit PR-001–PR-007 `INVALID` classifications, preserved same-state uncertainty, and a pending review record.
  - `Appended:` None.
- `File:` `review/pool-c/reports/domain-report.md`.
  - `Action:` edited.
  - `Deleted:` The prior ambiguous classification of Admin cancellation from `shipping` and unresolved non-self transitions.
  - `Replaced With:` `DOMAIN-P1-v2`, consistently applying all seven invalid classifications while preserving all unrelated partitions, boundaries, gaps, and pending review status.
  - `Appended:` None.
- `File:` `review/pool-c/reports/state-report.md`.
  - `Action:` edited.
  - `Deleted:` The PR-001–PR-007 proposed/ambiguous classifications and their resolved review questions.
  - `Replaced With:` `STATE-PHASE1-v2`, preserving PR IDs and classifying every PR edge as `INVALID`, with same-state updates still `UNSPECIFIED` and review status `PENDING`.
  - `Appended:` None.
- `File:` `review/pool-c/orchestration-status.md`.
  - `Action:` edited.
  - `Deleted:` Superseded report hashes and the pending questions about diagram exhaustiveness and `shipping` → `canceled`.
  - `Replaced With:` Revised v2 report hashes, resolved decision records, preserved gaps, validation results, and an explicitly closed Phase 2 gate.
  - `Appended:` None.

    - `File:` `AI_Audit.md`.
        - `Action:` appended.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Artifact 31 containing this exact prompt, AI output, and file-modification record.

- **Verdict:** VALID
- **Reasoning:**
    - The FR-10 state machine is now applied correctly
    - PR-001 to PR-007 are consistently marked INVALID
    - `shipping` to `canceled` is correctly rejected under FR-10 and FR-18.The FR-10 state machine is now applied correctly.
    - PR-001 to PR-007 are consistently marked INVALID.

- **Student Fixes:** None

Artifact 32

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 20:43:50 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
    - `api-test-generator`
    - `contract-test-generator`
    - `domain-test-generator`
    - `state-transition-test-generator`
    - `security-test-generator`
    - `ai-audit`

- **Prompt:**

```
I have reviewed and approved Pool C Phase 1.

Continue with [$api-test-generator](D:\GitHub\SoftwareTesting-HW06\\.agents\skills\api-test-generator\SKILL.md) Phase 2.

- Generate candidate tests for `PUT /api/admin/orders/:id/status`.
- Use only the approved CONTRACT, DOMAIN, STATE, and SECURITY reports.
- Target at least 35 test cases.
- Preserve CR/DP/TR/SS traceability.
- Keep unspecified behavior unresolved; do not invent HTTP status codes or schemas.
- Write the final candidate suite to `review/pool-c/candidate-api-tests.csv`.
- Log with [$ai-audit](D:\GitHub\SoftwareTesting-HW06\\.agents\skills\ai-audit\SKILL.md).
```

- **AI Output:**

```
Phase 2 is complete. Generated [candidate-api-tests.csv]
(D:/GitHub/SoftwareTesting-HW06/review/pool-c/candidate-api-tests.csv)
with 81 validated cases:

- CONTRACT: 23
- DOMAIN: 28
- STATE: 20
- SECURITY: 10

Coverage includes `CR-001–CR-013`, `DP-001–DP-028`, `TR-001–TR-013`,
`PR-001–PR-007`, `SS-001–SS-007`, and applicable `SEC-02/03/05`. IDs are
sequential from `API-001` to `API-081`; unspecified behavior remains
unresolved, and no numeric HTTP status codes or response schemas were
invented. No tests were executed.

Full validation and hashes are recorded in [orchestration-status.md]
(D:/GitHub/SoftwareTesting-HW06/review/pool-c/orchestration-status.md).

Artifacts 32-32 successfully appended to D:\GitHub\SoftwareTesting-
HW06\AI_Audit.md.
```

- **File Modifications:**
  - `File:` `review/pool-c/shared-api-context.md`.
    - `Action:` edited.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Phase 2 approval authority with the exact approved report hashes, target, traceability requirement, and unresolved-behavior guardrails.
  - `File:` `review/pool-c/candidates/contract-tests.json`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` 23 CONTRACT candidates covering CR-001–CR-013.
  - `File:` `review/pool-c/candidates/domain-tests.json`.

        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` 28 DOMAIN candidates covering DP-001–DP-028; no DB cases because the approved report defines no supported boundaries.
    - `File:` `review/pool-c/candidates/state-tests.json`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` 20 STATE candidates covering TR-001–TR-013 and PR-001–PR-007, with no same-state cases.
    - `File:` `review/pool-c/candidates/security-tests.json`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` 10 SECURITY candidates covering SS-001–SS-007 and applicable SEC-02, SEC-03, and SEC-05.
    - `File:` `review/pool-c/candidate-api-tests.csv`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Aggregated 81-row candidate suite with exact nine-field schema, stable API-001–API-081 IDs, compact JSON request cells, preserved provisional IDs, and complete approved traceability.
    - `File:` `review/pool-c/orchestration-status.md`.
        - `Action:` edited.
        - `Deleted:` The Phase 1-only gate state and absent-candidate markers.
        - `Replaced With:` Phase 2 approval record, specialist fragment hashes/counts, parent validation, final CSV hash, complete traceability, and preserved gaps/limitations.
        - `Appended:` None.

- `File:` `AI_Audit.md`.
    - `Action:` appended.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Artifact 32 containing this exact prompt, AI output, and file-modification record.

- **Verdict:** VALID
- **Reasoning:** Base on the analysis, the agent correctly generated test cases.
- **Student Fixes:** None

Artifact 33

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 22:01:09 GMT+7
- **Model Used:** GPT-5.6 Sol (primary Codex agent)
- **Skill Used:** api-test-generator, ai-audit
- **Prompt:**

```
Revise the reviewed Pool C test suite.

- Merge current API-024, API-030, and API-036 into one DOMAIN case
because they use the same pending-to-confirmed request and semantic
outcome.
- Preserve DP-001, DP-007, and DP-013 traceability and all related
provisional specialist IDs in the retained case.
- Regenerate sequential API IDs in both:
  - `review/pool-c/candidate-api-tests.csv`
  - `test-cases/c-order-management.csv`
- Preserve the human audit columns and update the affected audit
reasoning.
- Update Pool C counts and deduplication notes in `orchestration-
status.md`.
- Do not modify the approved Phase-1 reports.
- Do not add human cases or generate Postman tests yet.
```

- **AI Output:**

```
Revised Pool C:

- Merged former `API-024`, `API-030`, and `API-036` into retained `API-024`, preserving `DP-001`, `DP-007`, `DP-013` and `DOMAIN-P001`, `DOMAIN-P007`, `DOMAIN-P013`.
- Renumbered both suites sequentially from `API-001` through `API-079`.
- Preserved all human audit columns and synchronized affected audit reasons.
- Updated counts, hashes, and deduplication notes in [orchestration-status.md](D:/GitHub/SoftwareTesting-HW06/review/pool-c/orchestration-status.md).

Updated files: [candidate-api-tests.csv](D:/GitHub/SoftwareTesting-HW06/review/pool-c/candidate-api-tests.csv) and [c-order-management.csv](D:/GitHub/SoftwareTesting-HW06/test-cases/c-order-management.csv).

Validation confirmed 79 matching rows, no remaining same-category request/outcome duplicates, and no changes to Phase-1 reports or Postman files.
```

- **File Modifications:**
  - `File:` `review/pool-c/candidate-api-tests.csv`.
    - `Action:` edited and deleted content.
    - `Deleted:` Former duplicate DOMAIN rows `API-030` (`DP-007`, `DOMAIN-P007`) and `API-036` (`DP-013`, `DOMAIN-P013`), plus superseded IDs and audit-reason ID references on subsequent rows.
    - `Replaced With:` Retained `API-024` expanded to cover `DP-001`, `DP-007`, and `DP-013`, preserve `DOMAIN-P001`, `DOMAIN-P007`, and `DOMAIN-P013`, and explain the semantic consolidation in `Audit Reason`; all remaining rows renumbered sequentially through `API-079` with synchronized audit-reason IDs.
    - `Appended:` None.
  - `File:` `test-cases/c-order-management.csv`.
    - `Action:` edited and deleted content.

    - `Deleted:` Former duplicate DOMAIN rows `API-030` and `API-036`, plus superseded IDs on subsequent rows.
    - `Replaced With:` Retained `API-024` expanded with the merged DOMAIN traceability and provisional specialist IDs; all remaining rows renumbered sequentially through `API-079`.
    - `Appended:` None.
  - `File:` `review/pool-c/orchestration-status.md`.
    - `Action:` edited.
    - `Deleted:` Superseded 81-case total, 28-case aggregated DOMAIN count, zero-removal deduplication note, `API-001` through `API-081` range, nine-field-only candidate schema statement, and prior candidate hash.
    - `Replaced With:` 79-case total; CONTRACT 23, DOMAIN 26, STATE 20, SECURITY 10 counts; two-record semantic deduplication note; retained traceability details; `API-001` through `API-079` range; preserved human audit schema; and updated candidate/reviewed-suite hashes.
    - `Appended:` Reviewed suite path and reviewed suite SHA-256 record.
  - `File:` `AI_Audit.md`.
    - `Action:` appended.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Artifact 33 containing the exact previous prompt, AI output, and file-modification record.
- **Verdict:** VALID
- **Reasoning:**
  - The three duplicate DOMAIN cases were merged into API-024.
  - DP-001, DP-007, and DP-013 traceability is preserved.
  - All three provisional specialist IDs are preserved.
  - IDs are regenerated sequentially from API-001 to API-079.
- **Student Fixes:** None

Artifact 34

- **User:** Vương Ngữ Tín Thành

- **Date and Time:** 2026-08-22 22:29:45 GMT+7
- **Model Used:** GPT-5.6 Sol (primary Codex agent)
- **Skill Used:** `ai-audit`
- **Prompt:**

```
Add the reviewed Pool C human cases API-080 to API-085 in
[Main\_Report.md](Main_Report.md).

- Append them to [c-order-management.csv](test-cases/c-order-
management.csv) after API-079.
- Keep the existing 79 AI-generated cases unchanged.
- Mark API-080 to API-085 as human-authored cases.
- Preserve the exact case IDs, categories, objectives, expected results,
and notes.
- Keep multi-step cases represented clearly so they can later be
converted into ordered Postman requests.
```

- **AI Output:**

```
Appended human-authored `API-080` through `API-085` to [c-order-
management.csv](D:/GitHub/SoftwareTesting-HW06/test-cases/c-order-
management.csv:81).

Verified:

- Existing 79 AI-generated cases are unchanged.
- IDs, categories, objectives, expected results, and notes exactly match
`Main_Report.md`.
- Ordered workflows contain explicit step numbering: five two-step cases
and one four-step case.
- CSV parses successfully with 85 sequential cases.
```

- **File Modifications:**
  - `File:` `test-cases/c-order-management.csv`.
    - `Action:` appended.
    - `Deleted:` None.

        - `Replaced With:` None.
        - `Appended:` Human-authored cases `API-080` through `API-085` from `Main_Report.md` Section C.7, preserving their exact IDs, categories, objectives, expected results, and notes. Each case has explicit ordered request steps: five two-step workflows and one four-step lifecycle workflow. The existing `API-001` through `API-079` rows were unchanged.
    - `File:` `AI_Audit.md`.
        - `Action:` appended.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Artifact 34 containing the exact previous prompt, AI output, and file-modification record.
- **Verdict:** VALID
- **Reasoning:** The agent added the test cases base on provided list correctly.
- **Student Fixes:** None

Artifact 35

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 22:41:57 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:** `ai-audit`
- **Prompt:**

```
Complete Pool C Sections C.2–C.6 in [Main\_Report.md](Main_Report.md)
using [pool-c](review/pool-c/) as the source of truth.
```

- **AI Output:**

```
Completed Sections C.2–C.6 in [Main_Report.md]
(D:/GitHub/SoftwareTesting-HW06/Main_Report.md:579).

Added:

- CONTRACT, DOMAIN, STATE, and SECURITY analysis from the approved Pool
C reports.
- Complete CR/DP/TR/PR/SS traceability and documented limitations.
- AI/human/final counts: 79 / 6 / 85.
- Deduplication history and retained `API-024` traceability.
- Validation and pending Postman/Newman status.

Verified all counts against the CSVs, sequential `API-001`–`API-085`,
and that the original 79 AI-generated rows remain unchanged. No Pool C
source reports were modified.
```

- **File Modifications:**
  - `File:` `Main_Report.md`.
    - `Action:` edited and replaced content.
    - `Deleted:` Empty Pool C Sections C.2–C.5 and the incomplete C.6 text, including the incorrect Sections A.2–A.5 reference and blank testing-type counts.
    - `Replaced With:` Completed Pool C CONTRACT, DOMAIN, STATE, and SECURITY analyses based on `review/pool-c/`; CR/DP/TR/PR/SS coverage and documented gaps; the 79 AI-generated plus 6 human-authored case breakdown; the `API-024` deduplication record; sequential `API-001`–`API-085` validation; and the pending Postman/Newman status.
    - `Appended:` None.
  - `File:` `AI_Audit.md`.
    - `Action:` appended.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Artifact 35 containing the exact previous prompt, AI output, and file-modification record.
- **Verdict:** VALID

- **Reasoning:**
  - C.2–C.6 are now completed from the reviewed Pool C analyses.
  - C.7 remains unchanged.
- **Student Fixes:** None

Artifact 36

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 23:03:23 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
  - `postman-test-generator`
  - `ai-audit`
- **Prompt:**

```
[$postman-test-generator](D:\GitHub\SoftwareTesting-HW06\\.agents\skills\postman-test-generator\SKILL.md) [$ai-audit](D:\GitHub\SoftwareTesting-HW06\\.agents\skills\ai-audit\SKILL.md)&#x20;



Generate the Pool C Postman collection using the reviewed cases in [c-order-management.csv](test-cases/c-order-management.csv) and [pool-c](review/pool-c/).

- Generate all 85 cases for `PUT /api/admin/orders/:id/status`.
- Preserve multi-step human cases API-080–API-085 as ordered flows.
- Create deterministic, isolated fixtures for each case.
- Include valid Admin/non-Admin JWT setup as required.
- Add the required `X-Student-Id` pre-request behavior.
- Keep reviewed expected results unchanged.
- Output the collection under `postman/` and any fixture scripts needed.
- Validate the collection, but do not run Newman yet.



Take [run\_pool\_b\_with\_fixtures.py](postman/run_pool_b_with_fixtures.py) as example
```

- **AI Output:**

```
Generated Pool C with all 85 reviewed cases, represented as 93 requests.
API-080–API-085 remain ordered 2/2/2/2/4/2-step flows.

Key outputs:

- [Collection](D:/GitHub/SoftwareTesting-HW06/postman/pool-c-order-management.postman_collection.json)
- [Fixture runner](D:/GitHub/SoftwareTesting-HW06/postman/run_pool_c_with_fixtures.py)
- [Fixture definitions](D:/GitHub/SoftwareTesting-HW06/postman/pool_c_fixtures.py)
- [Conversion report](D:/GitHub/SoftwareTesting-HW06/postman/pool-c-conversion-report.md)
- [Validation results](D:/GitHub/SoftwareTesting-HW06/postman/pool-c-validation-results.json)
- [Runtime instructions](D:/GitHub/SoftwareTesting-HW06/postman/pool-c-runtime-setup.md)
- [AI audit](D:/GitHub/SoftwareTesting-HW06/AI_Audit.md)

Validation passed: full Postman v2.1 schema, exact CSV traceability, deterministic generation, all fixture reset/restore checks, Admin/non-Admin and invalid JWT fixtures, live SUT schema compatibility (read-only), and compilation of all 94 Postman event scripts. `X-Student-Id` is upserted collection-wide before every SUT request.

Newman was not run, as requested.
```

- **File Modifications:**
    - `File:` `postman/generate_pool_c.py`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Deterministic generator mapping all 85 reviewed CSV cases to 93 Postman requests, including six ordered flows, source descriptions, reviewed assertions, variables, and conversion reporting.

- `File:` `postman/pool-c-order-management.postman_collection.json`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Postman Collection v2.1 containing all reviewed Pool C cases, collection-wide `X-Student-Id` and fixture-reset behavior, step-specific assertions, and fixture-backed state oracles.
- `File:` `postman/pool-c-conversion-report.md`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Complete row-to-request traceability for API-001 through API-085, preserved preconditions and expected results, generated assertions, manual-oracle notes, fixture strategy, schema status, and Newman-compatibility status.
- `File:` `postman/pool_c_fixtures.py`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Deterministic isolated order fixtures, initial-state matrix, Admin/non-Admin and invalid JWT construction, schema validation, snapshot/reset/inspect/restore logic, and fixture summary.
- `File:` `postman/run_pool_c_with_fixtures.py`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Pool B-style authenticated loopback fixture runner that resets once per reviewed case, preserves ordered flows, supplies runtime variables, invokes Newman only when explicitly run later, and restores orders in `finally`.
- `File:` `postman/validate_pool_c_runtime.py`.
    - `Action:` created.
    - `Deleted:` None.

    - `Replaced With:` None.
    - `Appended:` Non-Newman validation for full Collection v2.1 schema conformance, exact CSV traceability, 85-case/93-request mapping, flow counts, fixture isolation/restoration, JWT semantics, runtime hooks, JavaScript compatibility markers, and optional read-only live SUT schema compatibility.
  - `File:` `postman/pool-c-validation-results.json`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Passing validation record showing 85 reviewed IDs, 93 requests, all six ordered flows, full schema success, fixture/JWT/traceability/static compatibility success, read-only SUT schema success, and `newmanExecuted: false`.
  - `File:` `postman/pool-c-runtime-setup.md`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Fixture, JWT, validation, restoration, and later Newman execution instructions.
  - `File:` `AI_Audit.md`.
    - `Action:` appended.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Artifact 36 containing the exact prompt, final AI output, and file-modification record for Pool C Postman generation.
- **Verdict:** INCOMPLETE
- **Reasoning:**
  - The invalid transition cases API-055–API-069 do not verify that the order state remains unchanged.
  - Their current automated check accepts either an HTTP error or any JSON `message`.
  - A wrong transition could therefore update the order successfully and still pass because a successful response may also contain a `message`.

- **Student Fixes:**
  - Asked the agent to add fixture-backed unchanged-state assertions to API-055–API-069.
  - Asked the agent to regenerate and revalidate the Pool C collection.

Artifact 37

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 23:19:45 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
  - `postman-test-generator`
  - `ai-audit`
- **Prompt:**

```
API-055–API-069 are invalid-transition cases and must also assert that
the target order remains in its original state. Their current
error/message assertions alone can accept a successful mutation.

Add fixture-backed unchanged-state assertions for all API-055–API-069,
regenerate the collection/report, and rerun validation.

Do not run Newman yet.
```

- **AI Output:**

```
Added fixture-backed unchanged-state assertions for API-055–API-069 and
regenerated the Pool C collection and conversion report.

Original-state assertions now enforce:

- API-055–API-058: `delivered`
- API-059–API-062: `canceled`
- API-063–API-064: `pending`
- API-065–API-066: `confirmed`
- API-067–API-069: `shipping`

Validation passed for all 85 cases and 93 requests, including the new
dedicated API-055–API-069 state-oracle gate, full Postman v2.1 schema
validation, fixture reset/restore, exact traceability, deterministic
regeneration, and compilation of all 94 Postman scripts.

Updated files: [collection](D:/GitHub/SoftwareTesting-HW06/postman/pool-
c-order-management.postman_collection.json), [conversion report]
(D:/GitHub/SoftwareTesting-HW06/postman/pool-c-conversion-report.md),
[generator](D:/GitHub/SoftwareTesting-HW06/postman/generate_pool_c.py),
[validator](D:/GitHub/SoftwareTesting-
HW06/postman/validate_pool_c_runtime.py), and [validation results]
(D:/GitHub/SoftwareTesting-HW06/postman/pool-c-validation-results.json).

Newman was not run.

Artifacts 37-37 successfully appended to `AI_Audit.md`.
```

- **File Modifications:**
  - `File:` `postman/generate_pool_c.py`.
    - `Action:` edited.
    - `Deleted:` The unchanged-state case set omitted API-055 through API-069.
    - `Replaced With:` The unchanged-state case set now includes every ID from API-055 through API-069.
    - `Appended:` None.

- `File:` `postman/pool-c-order-management.postman_collection.json`.
  - `Action:` regenerated and replaced content.
  - `Deleted:` API-055 through API-069 test scripts containing only error/message assertions and no post-request target-state oracle.
  - `Replaced With:` API-055 through API-069 test scripts retaining their error/message assertions and adding authenticated fixture-controller state inspection plus exact original-state assertions.
  - `Appended:` None.
- `File:` `postman/pool-c-conversion-report.md`.
  - `Action:` regenerated and replaced content.
  - `Deleted:` API-055 through API-069 traceability rows that listed only error/message assertions.
  - `Replaced With:` API-055 through API-069 traceability rows that also record the fixture-backed target-state assertion.
  - `Appended:` None.
- `File:` `postman/validate_pool_c_runtime.py`.
  - `Action:` edited.
  - `Deleted:` None.
  - `Replaced With:` None.
  - `Appended:` A dedicated validation matrix requiring the correct original-state marker and equality assertion for every request from API-055 through API-069.
- `File:` `postman/pool-c-validation-results.json`.
  - `Action:` regenerated and replaced content.
  - `Deleted:` Prior validation timestamp and result serialization.
  - `Replaced With:` Passing validation results dated 2026-08-22 23:19:26 GMT+7 after the unchanged-state correction; `newmanExecuted` remains `false`.
  - `Appended:` None.
- `File:` `AI_Audit.md`.
  - `Action:` appended.
  - `Deleted:` None.
  - `Replaced With:` None.

        - `Appended:` Artifact 37 containing the exact correction prompt, AI output, and file-modification record.

- **Verdict:** VALID
- **Reasoning:**
    - API-055–API-069 now verify the order state after each invalid transition.
    - The expected original states are mapped correctly across delivered, canceled, pending, confirmed, and shipping cases.
    - The existing response checks are preserved, while the new fixture-backed assertion prevents a successful mutation from being accepted.
- **Student Fixes:** None

Artifact 38

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-22 23:49:33 GMT+7
- **Model Used:** GPT-5.6 Sol
- **Skill Used:**
    - `newman-result-analyzer`
    - `ai-audit`
- **Prompt:**

```
Use [$newman-result-analyzer](D:\GitHub\SoftwareTesting-
HW06\\.agents\skills\newman-result-analyzer\SKILL.md) to analyze the
Pool C Newman run.

Use:

- [pool-c.json](reports/pool-c/pool-c.json)
- [pool-c-order-management.postman\_collection.json](postman/pool-c-
order-management.postman_collection.json)
- [c-order-management.csv](test-cases/c-order-management.csv)

Write the result into Section C.8 of [Main\_Report.md](Main_Report.md)

Reconcile 93 Postman requests with 85 logical test cases, account for
fixture/state helper requests, and classify each failed assertion as
SUT\_BUG, TEST\_DEFECT, SETUP\_DEFECT, or manual-review case.

Preserve failing evidence and group related bug candidates by root cause
without losing Test ID traceability.

Do not modify tests or rerun Newman.

After all, log this artifact using [$ai-audit]
(D:\GitHub\SoftwareTesting-HW06\\.agents\skills\ai-audit\SKILL.md)
```

- **AI Output:**

```
Completed the Pool C Newman analysis in [Main_Report.md]
(/D:/GitHub/SoftwareTesting-HW06/Main_Report.md:689).

- Reconciled all 245 Newman requests as 93 SUT requests, 85 fixture
resets, and 67 state-oracle helpers across 85 logical cases.
- Normalized 78 `PASS` and 7 `FAIL_ASSERTION` logical outcomes,
preserving all 8 failed assertion messages.
- Classified all 8 failures as `SUT_BUG` evidence and consolidated them
into 4 root-cause candidates with full Test ID traceability. `API-080`
retains its human/external requirement-authority review note.
- Recorded 26 execution-only characterization cases requiring manual
review and the additional HTTP 500 observation from `API-011`.
- Verified exactly 85 per-case rows and 8 failed-assertion triage rows.
No tests were modified and Newman was not rerun.

Artifacts 38-38 successfully appended to [AI_Audit.md]
(/D:/GitHub/SoftwareTesting-HW06/AI_Audit.md:3335).
```

- **File Modifications:**
  - `File:` `D:\GitHub\SoftwareTesting-HW06\Main_Report.md`.
    - `Action:` edited.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Section C.8 Newman execution analysis, including the normalized run summary, 85 per-logical-case outcome records, eight failed-assertion classifications, four traceable root-cause bug candidates, and reconciliation/manual-oracle notes.
  - `File:` `D:\GitHub\SoftwareTesting-HW06\AI_Audit.md`.
    - `Action:` appended.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Artifact 38 containing the exact Pool C analysis prompt, AI output, and file-modification record.
- **Verdict:** VALID

- **Reasoning:** The generated report contains no found problem.
- **Student Fixes:** None

Artifact 39

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-23 00:05:15 GMT+7
- **Model Used:** GPT-5
- **Skill Used:**
  - `append-bug`
  - `ai-audit`
- **Prompt:**

~~~~~
Use [$append-bug](D:\GitHub\SoftwareTesting-HW06\\.agents\skills\append-
bug\SKILL.md) to document the confirmed Pool C defects from C.8
[Main\_Report.md](Main_Report.md).
````

- **AI Output:**

````markdown
Documented the four confirmed Pool C root-cause defects and added the
FR-18 summary to [Bug_Report.md](/D:/GitHub/SoftwareTesting-
HW06/Bug_Report.md:55).

Created these issue files with the following exact content:

[Authorization scheme validation](/D:/GitHub/SoftwareTesting-
HW06/issues/bug_fr18-authorization-scheme-validation.md)

````
---
labels: bug, fr-18, medium
---

## Bug: Authorization scheme is not validated

**Impacted Test Case ID(s):** API-006

### Description

The Admin order-status endpoint accepts a valid Admin JWT under a non-
Bearer `Authorization` scheme. Authentication extracts and verifies the
second whitespace-delimited value without first requiring the documented
`Bearer` scheme.

### Steps to Reproduce

1. Seed an existing order in `pending` state and obtain a valid JWT
whose role is `admin`.
~~~~~

~~~markdown
2. Send `PUT /api/admin/orders/<existing-pending-order-id>/status` with
`Authorization: Basic <valid-admin-JWT>`, `Content-Type:
application/json`, and body `{"status":"confirmed"}`.
3. Query the order state after the request.

### Expected Result

The non-Bearer request does not satisfy the documented authorization
prerequisite and must not update the order.

### Actual Result

The request is authenticated despite using the `Basic` scheme, and the
target order changes from `pending` to `confirmed`. The Pool C Newman
artifact contains the failed unchanged-state assertion: [`pool-c.json`]
(../reports/pool-c/pool-c.json).

### Environment

- **Browser/OS:** Newman CLI on Windows (browser not applicable)
- **Version:** EShop requirements v2.0; Newman 6.2.2
````

[Admin role enforcement](/D:/GitHub/SoftwareTesting-
HW06/issues/bug_fr18-admin-role-enforcement.md)

````
---
labels: bug, fr-18, high
---

## Bug: Admin role is not enforced for order status updates

**Impacted Test Case ID(s):** API-008, API-034, API-076, API-085

### Description
~~~

`````
The Admin order-status endpoint validates JWT authenticity but does not
enforce `role = admin`. A valid JWT for a non-Admin user can perform the
protected order mutation.

### Steps to Reproduce

1. Seed an existing order in `pending` state and obtain valid JWTs for a
non-Admin user and an Admin user.
2. Send `PUT /api/admin/orders/<existing-pending-order-id>/status` with
the non-Admin Bearer token, `Content-Type: application/json`, and body
`{"status":"confirmed"}`.
3. Observe the response and query the target order state.
4. For the ordered control flow, retry the same update with the valid
Admin Bearer token.

### Expected Result

The non-Admin request is denied and the target order remains `pending`;
only the Admin-authorized request may perform the update.

### Actual Result

The non-Admin request is accepted, returns `200 OK` in API-085, and
changes the target order from `pending` to `confirmed`. Three
independent single-request cases and the ordered two-step flow reproduce
the missing role check in the Pool C Newman artifact: [`pool-c.json`]
(../reports/pool-c/pool-c.json).

### Environment

- **Browser/OS:** Newman CLI on Windows (browser not applicable)
- **Version:** EShop requirements v2.0; Newman 6.2.2
````

[Canceled final-state transition](/D:/GitHub/SoftwareTesting-
HW06/issues/bug_fr18-canceled-final-state-transition.md)
`````
`````

`````
````
---
labels: bug, fr-18, high
---

## Bug: Canceled orders can transition to delivered

**Impacted Test Case ID(s):** API-062

### Description

The order-status transition logic explicitly permits `canceled → delivered`, violating the FR-10 rule that `canceled` is a final state.

### Steps to Reproduce

1. Seed an existing order in `canceled` state and obtain a valid Admin JWT.
2. Send `PUT /api/admin/orders/<canceled-order-id>/status` with `Authorization: Bearer <valid-admin-JWT>`, `Content-Type: application/json`, and body `{"status":"delivered"}`.
3. Query the order state after the request.

### Expected Result

The transition is rejected and the order remains in the final `canceled` state.

### Actual Result

The endpoint accepts the invalid transition and changes the target order from `canceled` to `delivered`. The Pool C Newman artifact contains the failed final-state assertion: [`pool-c.json`](../reports/pool-c/pool-c.json).

### Environment
`````
`````

`````
- **Browser/OS:** Newman CLI on Windows (browser not applicable)
- **Version:** EShop requirements v2.0; Newman 6.2.2
````

[Unsupported content type](/D:/GitHub/SoftwareTesting-HW06/issues/bug_fr18-unsupported-content-type-server-error.md)

````
---
labels: bug, fr-18, medium
---

## Bug: Unsupported content type causes an internal server error

**Impacted Test Case ID(s):** API-080

### Description

The Admin order-status endpoint destructures `req.body` without guarding against an undefined body. When valid JSON text is sent as `text/plain`, the JSON parser does not populate `req.body`, and the endpoint raises an unhandled `TypeError`.

### Steps to Reproduce

1. Seed an existing order in `pending` state and obtain a valid Admin JWT.
2. Send `PUT /api/admin/orders/<existing-pending-order-id>/status` with `Authorization: Bearer <valid-admin-JWT>`, `Content-Type: text/plain`, and raw body `{"status":"confirmed"}`.
3. Query the order state.
4. Retry the same update with `Content-Type: application/json` and the same JSON body.

### Expected Result

The `text/plain` request receives a safe `4xx` response and leaves the
`````

`````
order `pending`; the corrected JSON retry succeeds and changes the order
to `confirmed`.

### Actual Result

The `text/plain` request returns `500 Internal Server Error` with an
unhandled `TypeError`, although it leaves the order unchanged. The
corrected JSON retry succeeds. Exploratory API-011 independently records
the same HTTP 500 behavior when the body is absent, corroborating the
root cause. The Pool C Newman artifact preserves both observations:
[`pool-c.json`](../reports/pool-c/pool-c.json).

### Environment

- **Browser/OS:** Newman CLI on Windows (browser not applicable)
- **Version:** EShop requirements v2.0; Newman 6.2.2
````

Appended this exact FR-18 section to `Bug_Report.md`:

````
## Feature FR-18: Admin Order Management

### Authorization scheme is not validated

**Impacted Test Cases:** API-006

The Admin order-status endpoint accepts a valid Admin JWT under the
`Basic` scheme and updates the target order because authentication does
not require the documented `Bearer` scheme.

### Admin role is not enforced for order status updates

**Impacted Test Cases:** API-008, API-034, API-076, API-085

Valid non-Admin JWTs can update order status because the protected Admin
route never checks `role = admin`. Independent contract, domain,
`````

`````
security, and ordered-flow cases reproduce the unauthorized mutation.

### Canceled orders can transition to delivered

**Impacted Test Cases:** API-062

The transition logic accepts `canceled → delivered`, allowing an order
to leave the FR-10 final `canceled` state.

### Unsupported content type causes an internal server error

**Impacted Test Cases:** API-080

Sending otherwise valid JSON as `text/plain` causes an unhandled
`TypeError` and `500 Internal Server Error` instead of a safe `4xx`
rejection. The same request succeeds when retried as `application/json`,
and exploratory API-011 corroborates the undefined-body failure.
````

Artifacts 39-39 successfully appended to [AI_Audit.md]
(/D:/GitHub/SoftwareTesting-HW06/AI_Audit.md).
`````

- **File Modifications:**
  - `File:` `D:\GitHub\SoftwareTesting-HW06\issues\bug_fr18-authorization-scheme-validation.md`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
    - `Appended:` Complete FR-18 issue for API-006, including metadata, reproduction steps, expected and actual results, evidence link, and environment.
  - `File:` `D:\GitHub\SoftwareTesting-HW06\issues\bug_fr18-admin-role-enforcement.md`.
    - `Action:` created.
    - `Deleted:` None.
    - `Replaced With:` None.
`````

        - `Appended:` Complete FR-18 issue consolidating API-008, API-034, API-076, and API-085 under the missing Admin-role check.
    - `File:` `D:\GitHub\SoftwareTesting-HW06\issues\bug_fr18-canceled-final-state-transition.md`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Complete FR-18 issue for API-062 and the invalid `canceled → delivered` transition.
    - `File:` `D:\GitHub\SoftwareTesting-HW06\issues\bug_fr18-unsupported-content-type-server-error.md`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Complete FR-18 issue for API-080, with API-011 retained as corroborating exploratory evidence.
    - `File:` `D:\GitHub\SoftwareTesting-HW06\Bug_Report.md`.
        - `Action:` appended.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Feature FR-18 section with four consolidated bug summaries and complete impacted-test traceability.
    - `File:` `D:\GitHub\SoftwareTesting-HW06\AI_Audit.md`.
        - `Action:` appended.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Artifact 39 containing the exact request, AI output, and file-modification record.
- **Verdict:** VALID
- **Reasoning:** The agent correctly grouped issues and create reports.
- **Student Fixes:** None

Artifact 40

- **User:** Vương Ngữ Tín Thành
- **Date and Time:** 2026-08-23 01:00:00 GMT+7
- **Model Used:** GPT-5
- **Skill Used:** newman-result-analyzer, ai-audit
- **Prompt:**

```
Resolve all remaining manual/execution-only test reviews in Pools A, B,
and C using the specification and execution evidence. Mark each case
PASS, FAIL, BLOCKED, or exploratory as appropriate, with short
reasoning. Do not invent results that require unavailable evidence.
```

- **AI Output:**

```
Resolved all 98 remaining manual/execution-only reviews.

- Final verdicts: **15 PASS, 15 FAIL, 14 BLOCKED, 54 exploratory**.
- Added a per-case evidence-based resolution report and linked it from
the main report.
- Preserved historical Newman execution statuses; `BLOCKED` is used
wherever required persistence, state, implementation, timing, expiry, or
configuration evidence is unavailable.
- No tests were rerun and no SUT results were invented.

Files: [resolution report](D:/GitHub/SoftwareTesting-
HW06/reports/manual-execution-review-resolution.md), [Main Report]
(D:/GitHub/SoftwareTesting-HW06/Main_Report.md), and [AI Audit]
(D:/GitHub/SoftwareTesting-HW06/AI_Audit.md).

Artifact 40 successfully appended to `D:\GitHub\SoftwareTesting-
HW06\AI_Audit.md`.
```

- **File Modifications:**
  - **File:** `reports/manual-execution-review-resolution.md`
    - **Action:** created

        - **Deleted:** None
        - **Replaced With:** None
        - **Appended:** Complete per-case final review for 98 remaining Pool A, B, and C manual/execution-only cases, including decision rules, short evidence reasoning, pool totals, and consolidated totals.
    - **File:** `Main_Report.md`
        - **Action:** edited and appended
        - **Deleted:** Statements that Pool A API-030, Pool B manual cases, and Pool C execution-only cases still had pending reviews.
        - **Replaced With:** Final adjudication language distinguishing historical Newman status from PASS, FAIL, BLOCKED, and exploratory review verdicts.
        - **Appended:** Section C.9 linking the authoritative resolution report and recording consolidated totals.
    - **File:** `AI_Audit.md`
        - **Action:** appended
        - **Deleted:** None
        - **Replaced With:** None
        - **Appended:** Artifact 40 for this interaction.
- **Verdict:** VALID
- **Reasoning:** Human review confirmed that the resolution report assigns all 98 previously unresolved manual/execution-only cases an evidence-bounded disposition. The per-pool totals reconcile to 15 PASS, 15 FAIL, 14 BLOCKED, and 54 exploratory, and the main report links the authoritative resolution without rewriting historical Newman results.
- **Student Fixes:** None required for this artifact.

Artifact 41

- **User:** Vương Ngữ Tín Thành
- **Date and Time:** 2026-08-23 00:59:21 GMT+7
- **Model Used:** GPT-5.6 Sol (Codex)
- **Skill Used:** `ai-audit`
- **Prompt:**

```
Implement the missing HW06 CI/CD requirement: add a GitHub Actions
Newman workflow, a short CI/CD report, and instructions/evidence
placeholders for one all-pass run and one intentional single-failure
run. Reuse the existing collections/scripts and avoid changing test
logic unnecessarily.
```

- **AI Output:**

```
Implemented the missing HW06 CI/CD requirement.

- Added [newman.yml](/D:/GitHub/SoftwareTesting-
HW06/.github/workflows/newman.yml:1) for push, pull-request, and manual
Newman runs. It pins the documented SUT commit, uploads JSON/HTML/log
evidence, defaults to green `API-001`, and offers an intentional one-
failure `API-006` scenario.
- Added [CI_CD_Report.md](/D:/GitHub/SoftwareTesting-
HW06/CI_CD_Report.md:1) with setup notes, reproduction instructions,
expected counts, and evidence placeholders.
- Extended [run_pool_c_with_fixtures.py](/D:/GitHub/SoftwareTesting-
HW06/postman/run_pool_c_with_fixtures.py:29) with backward-compatible `-
-test-id` filtering through a temporary collection. Existing Postman
assertions remain unchanged.

Validation passed: Python compilation, CLI help, temporary `API-001`
filtering (one leaf plus retained collection setup), and `git diff --
check`. Actual GitHub-hosted evidence remains intentionally pending
until the workflow is committed and run.

I preserved unrelated concurrent changes in `Main_Report.md` and
`reports/manual-execution-review-resolution.md`.

Artifacts 41-41 successfully appended to [AI_Audit.md]
(/D:/GitHub/SoftwareTesting-HW06/AI_Audit.md).
```

- **File Modifications:**
  - `File:` `D:\GitHub\SoftwareTesting-HW06\.github\workflows\newman.yml`.

        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` GitHub Actions Newman workflow with automatic all-pass CI, manual all-pass or intentional-single-failure selection, pinned SUT setup, and evidence artifact upload.
    - `File:` `D:\GitHub\SoftwareTesting-HW06\CI_CD_Report.md`.
        - `Action:` created.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Short CI/CD implementation report, run instructions, expected outcomes, and two evidence placeholder sections.
    - `File:` `D:\GitHub\SoftwareTesting-HW06\postman\run_pool_c_with_fixtures.py`.
        - `Action:` edited.
        - `Deleted:` The unconditional full-collection command construction and full-suite-only fixture reset comparison.
        - `Replaced With:` Backward-compatible optional Test-ID filtering into a temporary collection and selected-case-aware fixture verification; default full-suite behavior remains intact.
        - `Appended:` `filtered_collection` helper and repeatable `--test-id` CLI option.
    - `File:` `D:\GitHub\SoftwareTesting-HW06\AI_Audit.md`.
        - `Action:` appended.
        - `Deleted:` None.
        - `Replaced With:` None.
        - `Appended:` Artifact 41 containing the exact request, AI output, and file-modification record.
- **Verdict:** INVALID
- **Reasoning:** Human review found that the claimed all-pass mode executed only Pool C `API-001` and the intentional-failure mode executed only `API-006`; neither mode ran all

reviewed API cases, and two dispatch modes did not satisfy the requirement for two sample commits.

- **Student Fixes:** Superseded by Artifact 43: replace the single-case modes with full Pool A, B, and C execution, require two distinct pushed commits, and remove the temporary Pool C Test-ID filtering.

Artifact 42

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-23 10:31:30 GMT+7
- **Model Used:** GPT-5
- **Skill Used:** None
- **Prompt:**

```
Check for this repo against [2026.HW06.API Testing\_En.md]
(<reference/2026.HW06.API Testing_En.md>).

State out what need to fix. What is missing.
```

- **AI Output:**

```
The repository is not submission-ready yet. The API testing work is
substantial, but several mandatory deliverables are missing or
incomplete.

## Must fix before submission

1. **Complete CI/CD evidence.**
   [CI_CD_Report.md](D:/GitHub/SoftwareTesting-HW06/CI_CD_Report.md:39)
still contains `TODO` placeholders. You need:

   - One commit with the complete 241-case suite passing.
   - One separate commit with exactly one intentional failing test.
   - Commit links, Actions run links, screenshots, timestamps, totals,
and artifact names.
   - Because the pinned SUT contains real defects, create/fix a suitable
SUT revision for the green baseline without hiding failures or excluding
tests.

2. **Push the latest commits.**
   Local `main` is two commits ahead of `origin/main`:

   - `86aa381 Add GitHub Issues`
   - `ad3c2af Implement CI workflow and review pool A-C`

   The public repository currently referenced by [Main_Report.md]
(D:/GitHub/SoftwareTesting-HW06/Main_Report.md:13) therefore does not
contain all local work, based on the current tracking state.

3. **Add the mandatory AI Critique.**
   No 200–300-word critique was found. It must address AI
errors/incompleteness, why they happened, and lessons learned. Provide
it in Markdown and PDF as required by the [assignment]
(<D:/GitHub/SoftwareTesting-HW06/reference/2026.HW06.API
Testing_En.md:121>).

4. **Finish the AI Audit.**
   [AI_Audit.md](D:/GitHub/SoftwareTesting-HW06/AI_Audit.md:8) still has
```

```
an empty accuracy table and a TODO. It currently contains:

   - 23 `VALID`
   - 1 `INVALID`
   - 17 `INCOMPLETE`
   - 41 total artifacts

   Also, Artifact 41 says it was superseded by “Artifact 43,” but
Artifacts 42–43 do not exist. Append the missing interactions that
produced the current full-suite CI design, correct the reference, add
the explicit AI-use declaration, and fill the accuracy summary.

5. **Create the required `README.md`.**
   It must include:

   - Self-assessment table and total grade.
   - Number of APIs.
   - AI-generated, human-added, executed, passed, failed, blocked, and
exploratory cases.
   - Number of distinct bugs.
   - Public repository link.

6. **Create Excel deliverables.**
   The repo has CSV test cases but no `.xlsx` file. The assignment
explicitly requires Excel test cases and a test summary.

7. **Generate required PDFs.**
   No PDF files exist. At minimum, provide:

   - Main Report PDF.
   - AI Audit Report PDF.
   - AI Critique PDF.

8. **Add the Git commit log file.**
   The Git history is well structured, but the required text-based
exported commit log is missing—for example, `Git_Commit_Log.txt`.
```

```
9. **List the Postman features actually used.**
   The report does not contain the mandatory feature list. Document evidence for features such as workspace, collections, collection variables, pre-request scripts, tests, Newman reporters, multi-request flows, and CI. Clearly state which optional features were not used; consider adding an environment and a genuine data-driven run where reasonable.

10. **Create the final submission archive.**
    No correctly named ZIP exists. The required name is:

    `23127261_HW06_AI_API_<grade>.zip`

## Quality/compliance fixes

- Pool B executed 74 cases but the Newman artifact has only 27 assertions. Many cases are observational. Ensure every behavior with a defined oracle has a meaningful automated assertion; reserve assertion-free execution for genuinely unspecified behavior.
- Pool B checks `discount_amount` and `final_amount`, but does not appear to assert the exact response key set and types. Strengthen schema validation where the API specification defines the response structure.
- Add a short statement confirming the chosen three-API combination is not duplicated by another group member.
- The API-generator pseudocode and Mermaid diagram exist in [api-test-generator.md](D:/GitHub/SoftwareTesting-HW06/skills-design/api-test-generator/api-test-generator.md:34). Explicitly document that the submitted diagram was personally designed. If AI generated the diagram itself, it must be redrawn manually to satisfy the anti-cheat constraint.
- Local bug files contain screenshots, and [Bug_Report.md](D:/GitHub/SoftwareTesting-HW06/Bug_Report.md:1) links ten live issues. Verify manually that each live GitHub issue—not merely the local Markdown copy—has its screenshot attached.
- Consider linking `Bug_Report.md`, `CI_CD_Report.md`, AI Critique, and AI Audit clearly from the main report/README.
```

```
## Already in good shape

- Correct selection of one Pool A, B, and C API.
- Test-count target exceeded: 76/67/79 AI cases and 6/7/6 human additions.
- Every retained AI case has a case-level review verdict and reason.
- Domain, contract, security, and applicable state-transition analyses are present.
- Newman JSON and HTML reports exist for all three pools.
- `X-Student-Id` injection is implemented, and the cited Postman console screenshot visibly includes the header log and localhost execution.
- Ten consolidated bug reports and twelve evidence images exist.
- API-generator design, pseudocode, diagrams, and demonstration links exist.
- Phase-oriented Git commits exist for generation, review, extension, execution, and bug reporting.
```

- **File Modifications:** None
- **Verdict:** VALID
- **Reasoning:** The repository review correctly identified the mandatory deliverables that were missing or incomplete at that time. It also separated completed API-testing work from remaining submission requirements and supported the findings with repository evidence.
- **Student Fixes:** None

Artifact 43

- **User:** Vương Ngũ Tín Thành
- **Date and Time:** 2026-08-23 10:31:30 GMT+7
- **Model Used:** GPT-5
- **Skill Used:** None
- **Prompt:**

```
Tell me how to do step 1
```

- **AI Output:**

````
Use a fixed fork of the SUT to create the legitimate green baseline,
then introduce one deliberately wrong assertion in the test repository
for the red run.

## A. Produce the all-pass commit

1. Fork `ttbhanh/eshop-sut` into your GitHub account, for example:

   `https://github.com/vntthanh/eshop-sut`

2. In that fork, fix all defects currently detected by the suite. Use
the ten issues already listed in [Bug_Report.md]
(D:/GitHub/SoftwareTesting-HW06/Bug_Report.md:1), particularly:

   - Reset-token type and password validation.
   - Password hashing.
   - Unsupported content types returning controlled `4xx`.
   - Coupon percentage calculation and inclusive minimum.
   - JWT validation for coupons.
   - Authorization scheme and Admin-role enforcement.
   - Invalid canceled-order transitions.

3. Run all three collections locally against the fixed SUT. Do not
continue until every automated assertion passes.

4. Commit and push the SUT fixes. Record that SUT commit SHA.

5. In [.github/workflows/newman.yml](D:/GitHub/SoftwareTesting-
HW06/.github/workflows/newman.yml:20), change:

```yaml
repository: ttbhanh/eshop-sut
ref: 85af3ba875c88283615e22cb108f13e2fccaf0e9
```

to:
````

````
```yaml
repository: vntthanh/eshop-sut
ref: <fixed-sut-commit-sha>
```

Use an immutable commit SHA, not `main`.

6. Commit the green baseline:

```powershell
git add .github/workflows/newman.yml CI_CD_Report.md
git commit -m "ci: establish full Newman all-pass baseline"
git push origin main
```

7. Open the GitHub Actions run and confirm:

   - All 241 logical cases were included.
   - Pools A, B, and C all ran.
   - Zero failed assertions.
   - The job is green.
   - JSON/HTML artifacts were uploaded.

8. Save a screenshot under:

```text
reports/ci/evidence/all-pass-run.png
```

Record the test-repository commit SHA, SUT commit SHA, commit URL, run
URL, totals, timestamp, and artifact name in [CI_CD_Report.md]
(D:/GitHub/SoftwareTesting-HW06/CI_CD_Report.md:41).

## B. Produce the exactly-one-failure commit

1. Start immediately from the green test-repository commit.
````

````
2. Change exactly one reliable passing assertion. Pool C `API-001` is a
suitable choice: change its expected final state from `confirmed` to an
impossible value such as `intentional-ci-failure`.

Do not change the request, fixtures, or any other test.

3. Commit and push:

```powershell
git add postman/pool-c-order-management.postman_collection.json
git commit -m "test: demonstrate one intentional Newman failure"
git push origin main
```

4. Confirm the new Actions run:

   - All 241 cases still execute.
   - Exactly one assertion fails.
   - No request, fixture, or script error occurs.
   - The failure is the deliberately changed `API-001` assertion.

5. Save a screenshot as:

```text
reports/ci/evidence/single-failure-run.png
```

Fill the second evidence section in [CI_CD_Report.md]
(D:/GitHub/SoftwareTesting-HW06/CI_CD_Report.md:51). The red commit’s
parent SHA must be the green commit SHA.

## C. Restore the final branch

Revert only the intentional assertion change and commit the restoration:

```powershell
git revert <intentional-failure-commit-sha>
````
````

````
git push origin main
```

This preserves all three commits in history:

```text
green baseline → intentional one-failure → restore green
```

Finally, commit the screenshots and completed CI report. Do not use `--suppress-exit-code`, filter the suite, or weaken genuine assertions to manufacture the green run.
````

- **File Modifications:** None
- **Verdict:** VALID
- **Reasoning:** The response gave clear steps for creating the two CI examples requested by the assignment. It preserved full-suite execution, required real exit codes, and warned against hiding failures or filtering tests to create misleading evidence.
- **Student Fixes:** None

Artifact 44

- **User:** Vương Ngữ Tín Thành
- **Date and Time:** 2026-08-23 10:31:30 GMT+7
- **Model Used:** GPT-5
- **Skill Used:** None
- **Prompt:**

```
The [2026.HW06.API Testing_En.md](<reference/2026.HW06.API Testing_En.md>) does not require fixing bug. Remove the all test cases passing requirement
```

- **AI Output:**
```

```
You are correct that the assignment does not require repairing the SUT’s
bugs. However, it explicitly requires CI evidence of:

> “two sample commits: one whose pipeline run shows all API test cases
passing, and another whose pipeline run shows one test case failing.”

See [the CI/CD requirement](<D:/GitHub/SoftwareTesting-
HW06/reference/2026.HW06.API Testing_En.md:91>).

Therefore, I should remove the suggestion that you must fix the bugs,
but I cannot accurately remove the all-tests-passing requirement from
the checklist. The specification creates a practical conflict: genuine
defect-detection tests cannot all pass against the defective SUT.

The safest submission approach is:

- Keep the genuine failing tests and bug evidence unchanged.
- Keep the full CI workflow.
- State clearly in the CI report that the green-run requirement cannot
be produced against the assigned SUT without modifying the SUT,
weakening assertions, or concealing genuine defects.
- Do not fabricate a green run.
- Ask the lecturer/TA whether an alternate fixed SUT revision or a
demonstration-only test is expected for the green commit.

So: **fixing SUT bugs is not required; an all-passing sample pipeline is
still explicitly required by the provided assignment.**
```

- **File Modifications:** None
- **Verdict:** VALID
- **Reasoning:** The response correctly distinguished between fixing SUT bugs and providing the required CI evidence. It quoted the assignment requirement that one sample commit show all API tests passing and another show one failing test case.
- **Student Fixes:** None